SA HB 39:2015

# Installation code for metal roof and wall cladding

handbook

HB

AUSTRALIAN STEEL INSTITUTE

VICTORIAN BUILDING AUTHORITY

Handbook

# Installation code for metal roof and wall cladding

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# PREFACE

This Handbook was prepared by a steering committee and peer reviewed by the Standards Australia Committee WS-014, Plumbing and Drainage, from contributions, sketches and the like received from installers, manufacturers of roofing materials and components, TAFE institutes and industry training organisations, to supersede, HB 39—1997.

The development of this Handbook was made possible by arrangement between the Victorian Building Authority, the Australian Steel Institute Ltd and Standards Australia Limited.

Acknowledgement is made to the Master Plumbers & Mechanical Services Association of Australia, the Association of Hydraulic Services Consultants Australia Ltd, along with the support of the steel roof installation industry.

The intention of this Handbook is to provide guidelines and basic standards of good practice for use by industry training providers, the Australian metal roofing installation industry and roofing contractors in any State or Territory.

AS/NZS 3500.3:2003, *Plumbing and drainage,* Part 3: *Stormwater drainage*, AS/NZS 3500.5:2012, *Plumbing and drainage,* Part 5: *Housing installations,* and HB 114:1998, *Guidelines for the design of eaves and box gutters*, also include important criteria for roof drainage, roof flashings and cappings, which should be read in conjunction with this Handbook.

Requests for interpretations or suggestions for improvement should be forwarded in the first instance to Standards Australia.

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STANDARDS AUSTRALIA

# Handbook
# Installation code for metal roof and wall cladding

## SECTION 1 GENERAL

### 1.1 SCOPE

This Handbook provides information and basic guidelines on the selection, performance and installation of metal roofing and wall cladding. Although the fixing details for roof drainage systems relate mainly to steel, acknowledgment is made of the other materials in common use such as profiled fibreglass reinforced polyester (FRP) and polycarbonate.

When utilised, the design and installation measures contained in this Handbook will provide a weatherproof exterior and ensure that all rainwater is directed to the stormwater drainage system.

The recommendations and best practice methods set out in this Handbook will ensure that—

(a) the life of the roof sheeting, flashings and roof drainage systems including accessories, is optimised;

(b) the installation of roof coverings and flashings are weathertight;

(c) the installation of the roof drainage system is capable of discharging established rainfall intensities;

(d) in the event of a blockage or partial blockage of the stormwater system or downpipes rainfall is discharged external to the building with no detrimental effect to the building or its contents;

(e) the end effect of poor transportation, handling and storage is recognised and avoided;

(f) safe work practices are observed, with full awareness of solar radiation effects on exposed skin and sun glare on eyesight;

(g) the selection of materials reflects local environmental factors;

(h) the correct types of fasteners and recommended fixing patterns are utilised;

(i) the minimum and effective roof pitch recommendations for each profile are met;

(j) the minimum and effective grade for roof drainage systems is met;

(k) provision is made for expansion in roof coverings, gutters, flashings and downpipes;

(l) roof noise is minimised; and

(m) insulation is installed to ensure effective performance.

As the bulk of this Handbook refers to roofing, it is recommended that the user also source any appropriate additional information required for metal wall cladding installations from specific product manufacturers.

NOTE: The general design and sizing principles of roof drainage systems may also be utilised when installing drainage outlets, downpipes and overflow provision in above-ground external areas of buildings such as drains from balconies, patios and the like.

## 1.2 REFERENCED DOCUMENTS

The following documents are referred to in this Handbook.

| | |
|---|---|
| AS | |
| 1397 | Continuous hot-dip metallic coated steel sheet and strip—Coatings of zinc and zinc alloyed with aluminium and magnesium |
| 1562 | Design and installation of sheet roof and wall cladding |
| 1562.1 | Part 1: Metal |
| 1657 | Fixed platforms, walkways, stairways and ladders—Design, construction and installation |
| 3566 | Self-drilling screws for the building and construction industries |
| 3566.1 | Part 1: General requirements and mechanical properties |
| 3959 | Construction of buildings in bushfire-prone areas |
| 4055 | Wind loads for housing |
| 4256 | Plastic roof and wall cladding materials |
| 4256.2 | Part 2: Unplasticized polyvinyl chloride (uPVC) building sheets |
| 4256.5 | Part 5: Polycarbonate |
| AS/NZS | |
| 1170 | Structural design actions |
| 1170.0 | Part 0: General principles |
| 1170.1 | Part 1: Permanent, imposed and other actions |
| 1170.2 | Part 2: Wind actions |
| 1562 | Design and installation of sheet roof and wall cladding |
| 1562.3 | Part 3: Plastic |
| 1665 | Welding of aluminium structures |
| 1892 | Portable ladders |
| 1892.5 | Part 5: Selection, safe use and care |
| 2179 | Specifications for rainwater goods, accessories and fasteners |
| 2179.1 | Part 1: Metal shape or sheet rainwater goods, and metal accessories and fasteners |
| 2312 | Guide to the protection of structural steel against atmospheric corrosion by the use of protective coatings |
| 2728 | Prefinished/prepainted sheet metal products for interior/exterior building applications—Performance requirements |
| 3500 | Plumbing and drainage |
| 3500.3 | Part 3: Stormwater drainage |
| 3500.5 | Part 5: Housing installations |
| 4256 | Plastic roof and wall cladding materials |
| 4256.3 | Part 3: Glass fibre reinforced polyester (GRP) |
| 4389 | Safety mesh |
| ISO | |
| 9223 | Corrosion of metals and alloys—Corrosivity of atmospheres—Classification, determination and estimation |
| ABCB | |
| NCC | National Construction Code of Australia |
| | Condensation in Buildings Handbook, 2014 |

ASTM

A240 Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

D257 Standard Test Methods for DC Resistance or Conductance of Insulating Materials

IPCA Insulated Panel Council of Australasia
Code of Practice

Safe Work Australia
Preventing Falls in Housing Construction, Code of Practice, 2012

Workplace Health and Safety Queensland
Managing the risk of falls at workplaces, Code of Practice (December 2011)

## 1.3 TERMS AND DEFINITIONS

The following terms are used in this Handbook (see also Figure 1.3).

### 1.3.1 Accessories

Gutters for rainwater, ridge capping, valley gutters, flashings, downpipes, gutter brackets and the like.

### 1.3.2 Aluminium/zinc alloy-coated steel

Steel sheeting that is protected from corrosion by a hot dip coating of 50% to 60% aluminium, 1% to 2% silicon with the remainder zinc, and incorporating less than 1% of minor additions of control elements.

### 1.3.3 Aluminium/zinc/magnesium alloy-coated steel

Steel sheeting that is protected from corrosion by a hot dip coating of 47% to 57% aluminium, 1% to 3% magnesium, 1% to 2% silicon with the remainder zinc, and incorporating less than 1% of minor additions of control elements.

### 1.3.4 Barge

A finishing at the gable end of a roof, fixed parallel to the roof slope.

### 1.3.5 Concealed-fastened

A roof cover fixed by means of hidden, or secret, fixing clips or brackets.

### 1.3.6 Cross-sectional area of eaves gutter profile

The area beneath a line not less than 10 mm below the overflow provision of the gutter or with internal brackets less any allowance for the effects of the brackets, as determined by the manufacturer.

### 1.3.7 Downpipe

A pipe to carry roof water from gutters and roof catchments to drains or storage tanks.

### 1.3.8 Eaves gutter (spouting)

Internal (concealed) or external roof gutter attached to an eaves overhang.

### 1.3.9 Electrolytic (galvanic) corrosion

A type of corrosion that occurs when two dissimilar metals are in contact in a moist environment.

NOTE: Corrosion of the more reactive metal is accelerated by the presence of the less reactive metal.

### 1.3.10 Expansion joint

A joint in a long run of gutter, roof coverings or flashing designed to allow for thermal expansion and contraction.

### 1.3.11 Fall (slope)

The slope of the roof or gutter, usually expressed in degrees, or as a ratio of vertical height to horizontal distance (e.g. 1 in 20).

### 1.3.12 Fixings

Screws, nails, rivets or clouts used to fasten the roof sheeting or accessories to the building structure.

### 1.3.13 Flashing

A rigid or flexible material, usually metal, fixed over, against or built into an abutment to form a weathertight joint.

NOTE: When used to cover an inclined intersection against an abutment, the flashing may be described either as 'raking' when the top edge is secured into a chase cut parallel to the slope of the roof, or 'stepped' when the top edge is secured into the horizontal joints of a brick or masonry abutment and stepped up the slope from course to course.

### 1.3.14 Flashing, apron

An overflashing used to obtain a weathertight joint, usually where a roof abuts a vertical wall or penetration.

### 1.3.15 Flashing, counter (or over)

A flashing dressed down, as a cover only, over a separate upstand.

### 1.3.16 Flashing, hanging

Side, front or back cover piece used to prevent entry of water between abutting surfaces and other gutters, flashings and soakers.

### 1.3.17 Flashing, pressure

A flashing, fixed and sealed to a smooth finished wall, that covers the upstand to an apron or soaker flashing; *also* referred to as a 'K' flashing.

### 1.3.18 Flashing, soaker

A side or end flashing extended under roof coverings (which have the ribs closed) upstream of a penetration or at an abutment overflashed with a hanging flashing, or pressure flashing against the upstand.

### 1.3.19 Flashing, transverse

A flashing fixed across the ends of roof coverings at the upper end of the sheets such as at a ridge.

### 1.3.20 Freeboard

The vertical distance between a design maximum water level and the top of gutter to prevent wind driven spillages.

### 1.3.21 Galvanised steel

Steel sheeting protected against corrosion by a zinc coating applied by the continuous hot-dip process.

### 1.3.22 High capacity overflow

An overflow device fitted to a box gutter sump to relieve excess flow during high rainfall events.

### 1.3.23 Insulated panel

An insulated laminated roofing panel that is manufactured from different materials permanently bonded together so that they act as a single structural element.

### 1.3.24 Mansard

A roof built at two pitches, the steeper pitch commencing at the eaves and the flatter pitch finishing at the ridge.

NOTE: Sometimes used with reference to a steeply pitched roof finishing at the edge of a flat roof.

### 1.3.25 Monel metals

An alloy of nickel and copper and other metals (such as iron and/or manganese and/or aluminium).

### 1.3.26 Oil-canning

Variation from flatness of sheet metal, creating undulations along the surface.

NOTES:

1 The result is poor appearance and potential ponding.

2 Oil-canning is usually controlled by forming stiffening ribs along or across the pans of a profile.

### 1.3.27 Pan (trays)

The flat portion between the ribs in a pan-type preformed sheet.

### 1.3.28 Penetration

A projection through the roof (e.g. vent pipe, roof light or duct).

### 1.3.29 Pierce-fastened

A roof cover drilled and fixed by means of a screw or nail.

### 1.3.30 Ponding

Undrained water on a roof.

### 1.3.31 Pop

Gutter outlet point; also referred to as downpipe outlet and spout.

### 1.3.32 Preformed sheet

A roofing sheet with longitudinal ribs, which increase its resistance to vertical loads.

### 1.3.33 Rainhead

A collector connected to a downpipe at the end of a gutter, external to the building that permits a free flow from the end of the gutter.

### 1.3.34 R-value

The thermal resistance ($m^2$.K/W) of a component calculated by dividing its thickness by its thermal conductivity.

### 1.3.35 Rib

A longitudinal upstand in a preformed sheet produced by bending, folding or crimping.

### 1.3.36 Ridge capping

Formed metal designed to weatherproof the junction at the apex of opposing roof slopes.

### 1.3.37 Sarking

A flexible membrane designed to provide thermal reflectance and to collect condensation and moisture that may form on or penetrate a roof covering.

### 1.3.38 Side overflow device

An overflow device fitted to a box gutter to discharge all roof water clear of a building due to total or partial blockage of outlets, downpipes or stormwater drains.

### 1.3.39 Siphonic drainage system

A site-specific roofing stormwater drainage system (outside the scope of this document) specifically designed by a manufacturer of the system for a particular application.

### 1.3.40 Soaker gutter (chimney gutter)

A small gutter located on the upper side of a chimney stack.

### 1.3.41 Sole

The internal, bottom surface (base) of a roof gutter.

### 1.3.42 Spreader

A device fitted to the foot of a downpipe to distribute evenly rainwater onto a roof at a lower level in the direction of flow.

### 1.3.43 Sump

An internal collector of water from a gutter system for discharge to a connecting downpipe.

### 1.3.44 Swarf

Metallic filings and debris resulting from installation and fixing practices.

### 1.3.45 Thermal stress

Stress or movement due to expansion and contraction.

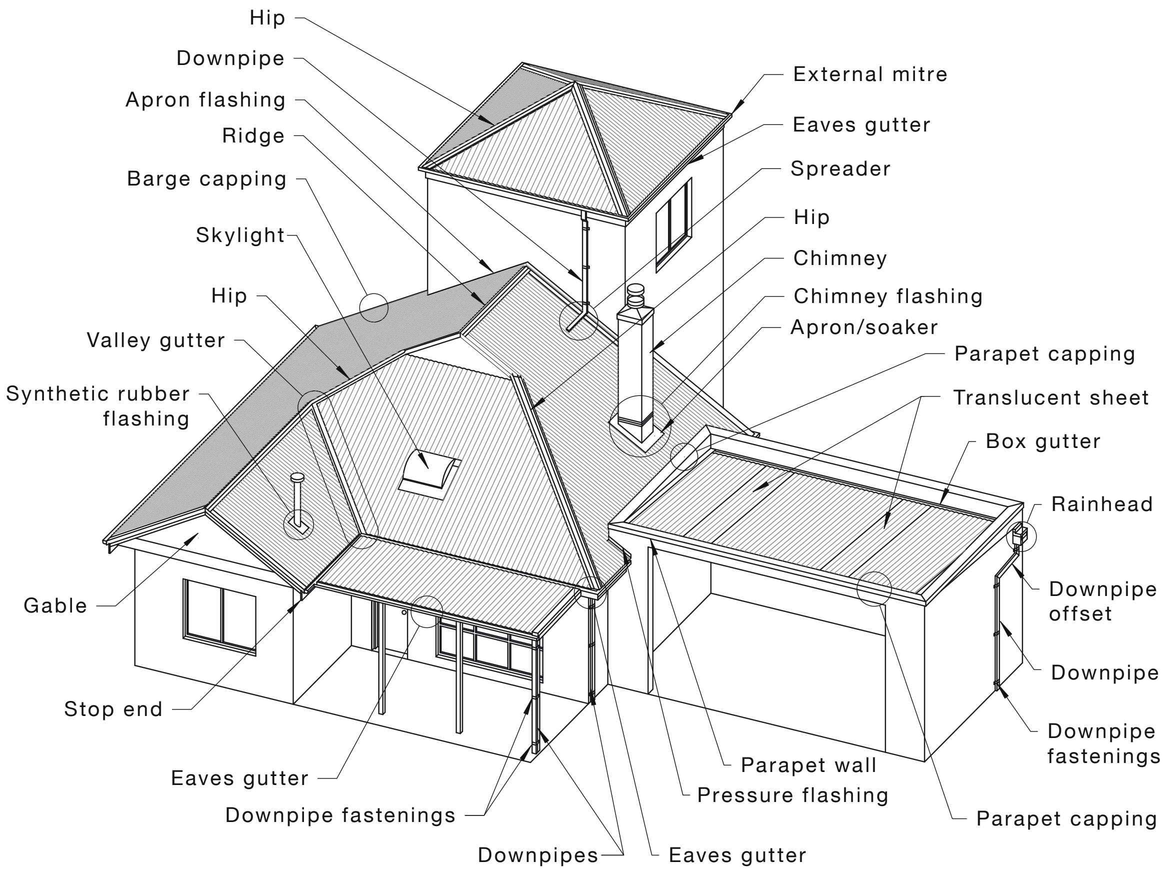


FIGURE 1.3 COMMON TERMS

# SECTION 2 MATERIAL SELECTION

## 2.1 ENVIRONMENTAL EFFECTS

The performance of sheet roofing is affected by many factors, some of which are controllable and some not.

The main uncontrollable factor is the environment, the effect of which can be minimised by choosing a material appropriate to the conditions under which the sheet is to perform.

NOTE: Material selection is highly dependent upon environmental conditions. Appendix A gives a description of atmospheric environments and atmospheric classifications that are to be taken into account, in combination with manufacturers' literature, when choosing a particular material for a particular application.

## 2.2 METAL THICKNESS

The thickness of metal is specified in AS/NZS 2179.1 and is shown in Tables 2.2(A), 2.2(B), 2.2(C), 2.2(D) and 2.2(E).

### TABLE 2.2(A)

### MINIMUM THICKNESS OF BOX GUTTERS

| Minimum base metal thickness, mm | | | | | |
|---|---|---|---|---|---|
| **Zinc-coated steel** | **Aluminium/zinc or aluminium/zinc/ magnesium alloy-coated steel** | **Copper and copper alloy** | **Aluminium and aluminium alloy** | **Stainless steel** | **Zinc** |
| 0.50 (G300) or 0.42 (G550) | 0.50 (G300) or 0.42 (G550) | 0.55 | 0.90 | 0.45 | 0.70 |

NOTE: The designations in parentheses are grades according to AS 1397.

### TABLE 2.2(B)

### MINIMUM THICKNESS OF EAVES GUTTERS WITH SOLE WIDTH EQUAL TO OR LESS THAN 200 mm

| Effective cross-sectional area $mm^2$ | Minimum base metal thickness, mm | | | | | |
|---|---|---|---|---|---|---|
| | **Zinc-coated steel** | **Aluminium/zinc or aluminium/zinc/ magnesium alloy-coated steel** | **Copper and copper alloys** | **Aluminium alloys** | **Stainless steels** | **Zinc** |
| ≤ 10 000 | 0.50 (G300) or 0.42 (G550) | 0.50 (G300) or 0.42 (G550) | 0.55 | 0.70 | 0.45 | 0.70 |
| > 10 000 ≤ 20 000 | 0.60 (G300) or 0.48 (G550) | 0.60 (G300) or 0.48 (G550) | 0.70 | 0.70 | 0.55 | 0.90 |

NOTE: The designations in parentheses are grades according to AS 1397.

## TABLE 2.2(C)

## MINIMUM THICKNESS OF VALLEY GUTTERS WITH WIDTH EQUAL TO OR LESS THAN 600 mm AND CONTINUOUS SUPPORT SYSTEM

| Minimum base metal thickness, mm | | | | | |
|---|---|---|---|---|---|
| **Zinc-coated steel** | **Aluminium/zinc or aluminium/zinc/ magnesium alloy-coated steel** | **Copper and copper alloys** | **Aluminium alloys** | **Stainless steel** | **Zinc** |
| 0.50 (G300) or 0.42 (G550) | 0.50 (G300) or 0.42 (G550) | 0.55 | 0.55 | 0.45 | 0.70 |

NOTE: The designations in parentheses are grades according to AS 1397.

## TABLE 2.2(D)

## MINIMUM THICKNESS OF CIRCULAR VERTICAL OR GRADED DOWNPIPES

| Nominal size, diameter* mm | | Minimum base metal thickness mm | | | | | |
|---|---|---|---|---|---|---|---|
| **Vertical†** | **Graded‡** | **Zinc-coated steel (G300)** | **Aluminium/zinc or aluminium/zinc/ magnesium alloy-coated steel (G300)** | **Copper and copper alloy** | **Aluminium alloys§** | **Stainless steel** | **Zinc** |
| 50 | 50 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 65 | 65 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 75 | 75 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 100 | 100 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 125 | | 0.50 | 0.50 | 0.70 | 0.70 | 0.50 | 0.90 |
| 140 | | 0.50 | 0.50 | 0.70 | 0.70 | 0.50 | 0.90 |
| 150 | | 0.50 | 0.50 | 0.70 | 0.70 | 0.50 | 0.90 |
| 200 | | 0.70 | 0.70 | 0.90 | 0.80 | 0.70 | 1.0 |
| 250 | | 0.70 | 0.70 | 1.0 | 0.90 | 0.70 | 1.2 |
| 300 | | 0.70 | 0.70 | 1.0 | 1.0 | 0.70 | 1.2 |

* Dimension may be less at one end to permit telescoping.

† Cross-sectional area less than 75 000 mm$^2$.

‡ Cross-sectional area less than 10 000 mm$^2$.

§ For yield strength of not less than 180 MPa.

NOTE: The designation in brackets is the grade according to AS 1397.

## TABLE 2.2(E)
## MINIMUM THICKNESS OF RECTANGULAR OR SQUARE—VERTICAL OR GRADED DOWNPIPES

| Vertical† | Graded‡ | Minimum base metal thickness mm | | | | | |
|---|---|---|---|---|---|---|---|
| | | Zinc-coated steel (G300) | Aluminium/zinc or aluminium/zinc/magnesium alloy-coated steel (G300) | Copper and copper alloy | Aluminium alloys§ | Stainless steel | Zinc |
| 75 × 50 | 75 × 50 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 75 × 70 | 75 × 70 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 100 × 50 | 100 × 50 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 100 × 75 | 100 × 75 | 0.40 | 0.40 | 0.55 | 0.55 | 0.40 | 0.80 |
| 100 × 100 | | 0.50 | 0.50 | 0.70 | 0.70 | 0.50 | 0.90 |
| 150 × 100 | | 0.50 | 0.50 | 0.70 | 0.70 | 0.50 | 0.90 |
| 150 × 150 | | 0.70 | 0.70 | 0.80 | 0.80 | 0.70 | 1.0 |
| 200 × 200 | | 0.70 | 0.70 | 1.0 | 1.0 | 0.70 | 1.2 |
| 250 × 250 | | 0.70 | 0.70 | 1.2 | 1.2 | 0.70 | 1.2 |

* Dimension may be less at one end to permit telescoping.

† Cross-sectional area less than 75 000 $mm^2$.

‡ Cross-sectional area less than 10 000 $mm^2$.

§ For yield strength of not less than 180 MPa.

NOTE: The designation in brackets is the grade according to AS 1397.

## 2.3 DURABILITY OF STEEL

To gain maximum benefit and maximum life from steel used in roofing it requires some form of protective coating to prevent corrosion (rusting). There are numerous ways of eliminating corrosion but none as effective, practical and economical as metallic coatings using zinc or other alloys. Depending on the intended application, the recommended coating may be a thin layer of electroplated zinc, a heavy layer of zinc applied by hot-dipping (galvanising), aluminium/zinc or aluminium/zinc/magnesium alloy coatings.

In the case of plain zinc coating, the thicker the coating, the longer the life of the product, while the coating on aluminium/zinc alloy-coated steel lasts up to four times longer than galvanised steel with the same coating thickness, under the same conditions of atmospheric exposure.

The metallic coatings protect the base steel in two ways—

(a) by providing a physical barrier between the steel and the atmosphere; and

(b) by galvanic action, an electrochemical process in which one metal corrodes preferentially to another.

Steel may be painted for protection, but any scratched or uncoated area will still allow the exposed steel to rust. Zinc-coated and aluminium/zinc alloy-coated steel will not corrode even if there are scratches or other slight abrasions in the coating because the coating sacrifices itself, corroding instead of the steel (see Clause 2.5). Thus, the steel is left structurally sound to perform the job for which it was intended.

As the environment becomes more aggressive, the coating thickness for a given life also needs to increase.

The sheet manufacturer will be able to assist with the correct material specification.

Tables 2.3(A) and 2.3(B) provide information on direct contact between various metals and on drainage from one surface over another.

NOTE: Appendix A provides information on atmospheric environments.

## TABLE 2.3(A)

## ACCEPTABILITY OF DIRECT CONTACT BETWEEN METALS

| **Roof drainage system components and any cladding material** | **Accessory or fastener material** | | | | | | | | | | | | **Fastener material** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Aluminium alloys** | | **Copper and copper alloys*** | | **Stainless steel (300 series)** | | **Zinc-coated steel and zinc** | | **Aluminium/zinc alloy-coated and aluminium/zinc/magnesium alloy-coated steel** | | **Lead** | | **Ceramic or organic coated** |
| | **Atmospheric classification** | | | | | | | | | | | | |
| | **SI and VS** | **Mild** | **SI and VS** | **Mild** | **SI and VS** | **Mild** | **SI and VS** | **Mild** | **SI and VS** | **Mild** | **SI and VS** | **Mild** | **SI and VS and mild** |
| Aluminium alloys | Yes | Yes | No | No | † | Yes | ‡ | ‡ | Yes | Yes | No | No | Yes |
| Copper and copper alloys | No | No | Yes | Yes | No | Yes | No | No | No | No | No | Yes | Yes |
| Stainless steel (300 series) | No | No | No | No | Yes | Yes | No | No | No | No | No | Yes | Yes |
| Zinc-coated steel and zinc | Yes | Yes | No | No | No | Yes | Yes | Yes | Yes | Yes | No | Yes | Yes |
| Aluminium/zinc and aluminium/zinc/magnesium alloy-coated steel | Yes | Yes | No | No | No | Yes | ‡ | ‡ | Yes | Yes | No | No | Yes |
| Lead§ | No | No | Yes | Yes | Yes | Yes | No | Yes | No | No | Yes | Yes | Yes |

* Includes Monel metal rivets.

† Grade 316 in accordance with ASTM A240 is suitable.

‡ Unpainted zinc-coated steel and zinc are suitable for direct contact but are not to receive drainage from an inert catchment (see Clause 2.6).

§ Due to its toxicity, lead is not recommended for rainwater goods.

LEGEND:

SI, VS, Mild = severe industrial, very severe and mild classifications (see AS/NZS 2312)

Yes = acceptable—as a result of bimetallic contact, either no additional corrosion of rainwater goods will take place or, at worst, only very slight additional corrosion; also implies that the degree of corrosion would not significantly shorten the service life

No = not acceptable—moderate to severe corrosion of rainwater goods will occur, a condition that may result in a significant reduction in the service life

NOTES:

1 Unless adequate separation can be assured, prepainted rainwater goods are to be considered in terms of the base or coated steel.

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## TABLE 2.3(B)

## ACCEPTABILITY OF DRAINAGE FROM AN UPPER SURFACE TO A LOWER METAL SURFACE

| Lower roof drainage system material | Upper cladding or rainwater goods material | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Aluminium alloys | Copper and copper alloys | Stainless steel (300 series) | Zinc-coated steel and zinc | Aluminium/zinc alloy-coated and aluminium/zinc/ magnesium alloy-coated steel | Lead | Prepainted metal | Roof tiles | | Plastic | Glass |
| | | | | | | | | All terracotta glazed concrete | Unglazed concrete | | |
| Aluminium alloys | Yes | No | * | Yes | Yes | * | Yes | Yes | Yes | Yes | Yes |
| Copper and copper alloys | * | Yes | * | * | * | Yes | * | Yes | Yes | Yes | Yes |
| Stainless steel (300 series) | * | * | Yes | * | * | Yes | * | Yes | Yes | Yes | Yes |
| Zinc-coated steel and zinc | No | No | No | Yes | No | | No | No | Yes | No | No |
| Aluminium/zinc and aluminium/zinc/ magnesium alloy-coated steel | Yes | No | * | Yes | Yes | No | Yes | Yes | Yes | Yes | Yes |
| Lead§ | * | * | * | * | * | Yes | * | Yes | | Yes | Yes |

* Whilst drainage between the materials shown would be acceptable, direction material contact is to be avoided [see Table 2.3(A)].

LEGEND:
Yes = acceptable
No = not acceptable

NOTES:

1 'Acceptable' and 'not acceptable' imply similar performances to those noted in Table 2.3(A).

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## 2.4 COATING MASS

The life of any given piece of metallic-coated steel depends not on the base metal thickness itself, but on thickness of metallic coating, be it zinc (galvanised), aluminium/zinc or aluminium/zinc/magnesium.

AS 1562.1 and AS/NZS 2179.1 specify the relevant coating mass for various roofing and rainwater goods applications. Roofing applications require a minimum Z450 galvanised, AZ150 aluminium/zinc coating, or AM125 aluminium/zinc/magnesium-coated steel complying with AS1397 for bare metallic-coated steel (see Figure 2.4). Prepainted materials are to comply with AS/NZS 2728.

*Examples*:

(a) Z450 galvanised steel:

where

Z = zinc

450 = a minimum coating of 450 g of zinc applied to each square metre of steel, or a nominal coating of 225 g/m$^2$ per side.

(b) AM125 aluminium/zinc/magnesium alloy-coated steel:

where

AM = aluminium/zinc/magnesium

125 = a minimum coating of 125 g of zinc applied to each square metre of steel, or a nominal coating of 62.5 g/m$^2$ per side.

For rainwater goods such as gutters or downpipes, AS/NZS 2179.1 requires a minimum coating of Z275, AZ150 or AM125 complying with AS 1397 for bare metallic-coated steel. Prepainted materials are to comply with AS/NZS 2728. Severe environmental conditions require AZ200 or AM150.

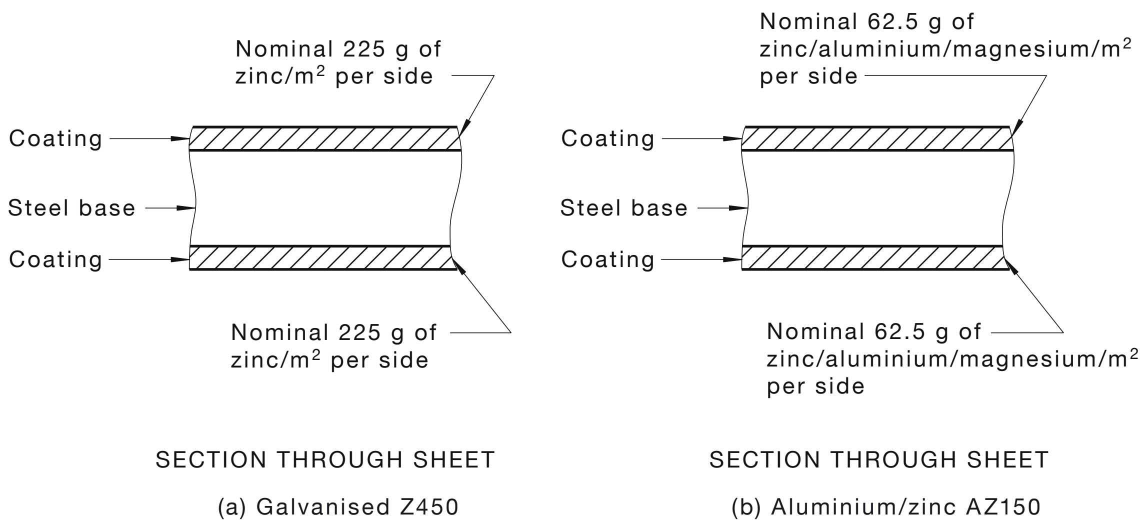


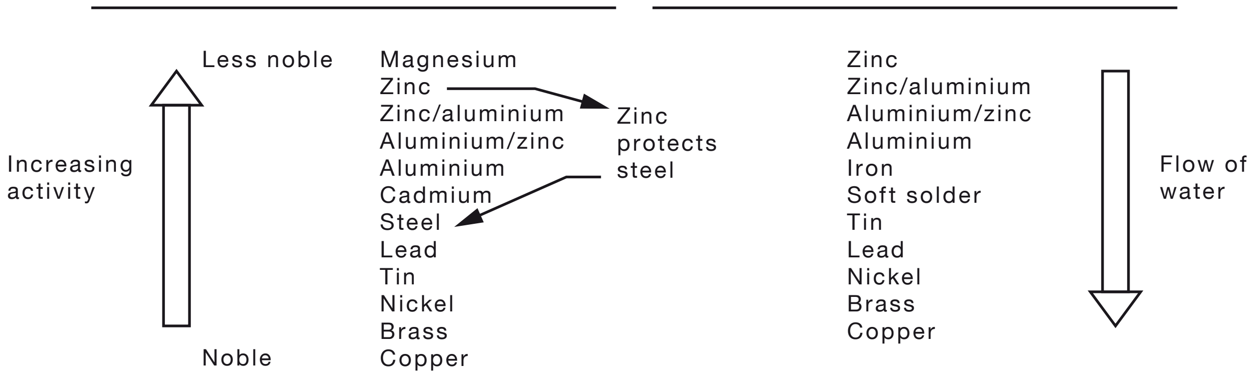

FIGURE 2.4 COATING MASS

## 2.5 DISSIMILAR METALS—SACRIFICIAL PROTECTION

When two or more dissimilar metals are in contact and moisture is present, one of the metals is relatively protected while the other suffers accelerated corrosion. This is commonly known as galvanic or electrolytic corrosion.

Figure 2.5 shows an abridged galvanic series of metals and includes only those commonly used for building applications. This series is laid out in descending order of activity. Metals near the top of the series will sacrificially protect those metals below it. From this position, it can be seen that zinc is more active than steel and will, therefore, protect it sacrificially.

NOTE: Monel metal rivets are not to be used with metal roofing, flashings and rainwater goods.




NOTE: An exception to the need for direct or electrical contact for corrosion to occur is when a more noble metal corrodes and dissolves in water, then flows over the less noble one. The more noble metal will deposit on the less noble one with similar results to direct contact.

(a) Sacrificial protection

NOTE: Dissimilar metals may be used if the flow of water over the metals is as shown above.

(b) Material compatibility

FIGURE 2.5 DISSIMILAR METALS

## 2.6 INERT CATCHMENT

An inert catchment exists when roofing materials such as glazed tiles, prepainted metals, glass, acrylic, fibreglass, aluminium/zinc steel, aluminium/zinc/magnesium alloy-coated steel, etc., are used. Rainfall flowing down these materials does not take metal salts or minerals into solution so the quality of the rainwater is virtually unchanged. High purity rainwater does not allow downstream galvanised gutters and flashings to form an erosion-resistant protective coating, as would be the case with non-inert catchments such as unglazed concrete tiles, galvanised steel or fibre cement [see Figures 2.6(A) and 2.6(B)]. This results in unsatisfactory performance and early replacement.

Aluminium/zinc alloy-coated steel, factory prepainted steel and aluminium/zinc/magnesium alloy-coated steel are not affected by inert catchments.

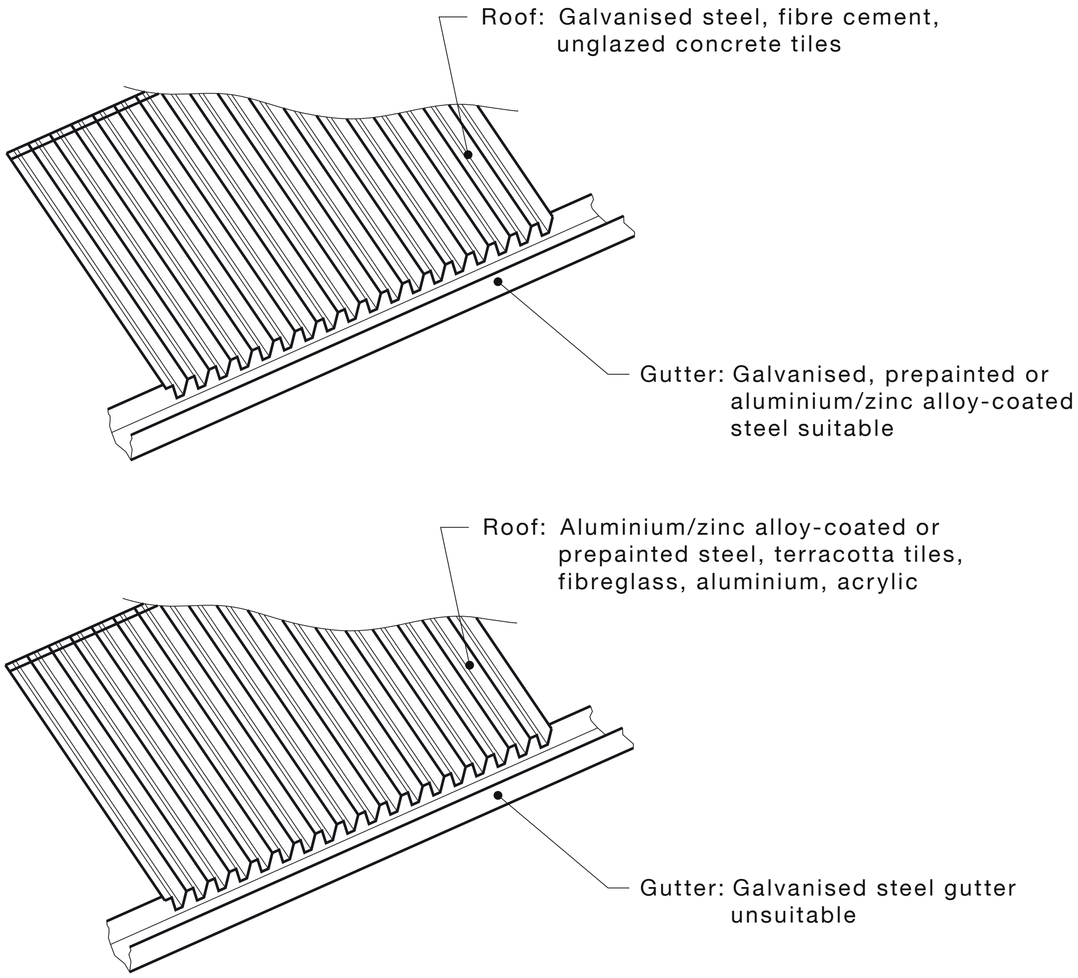
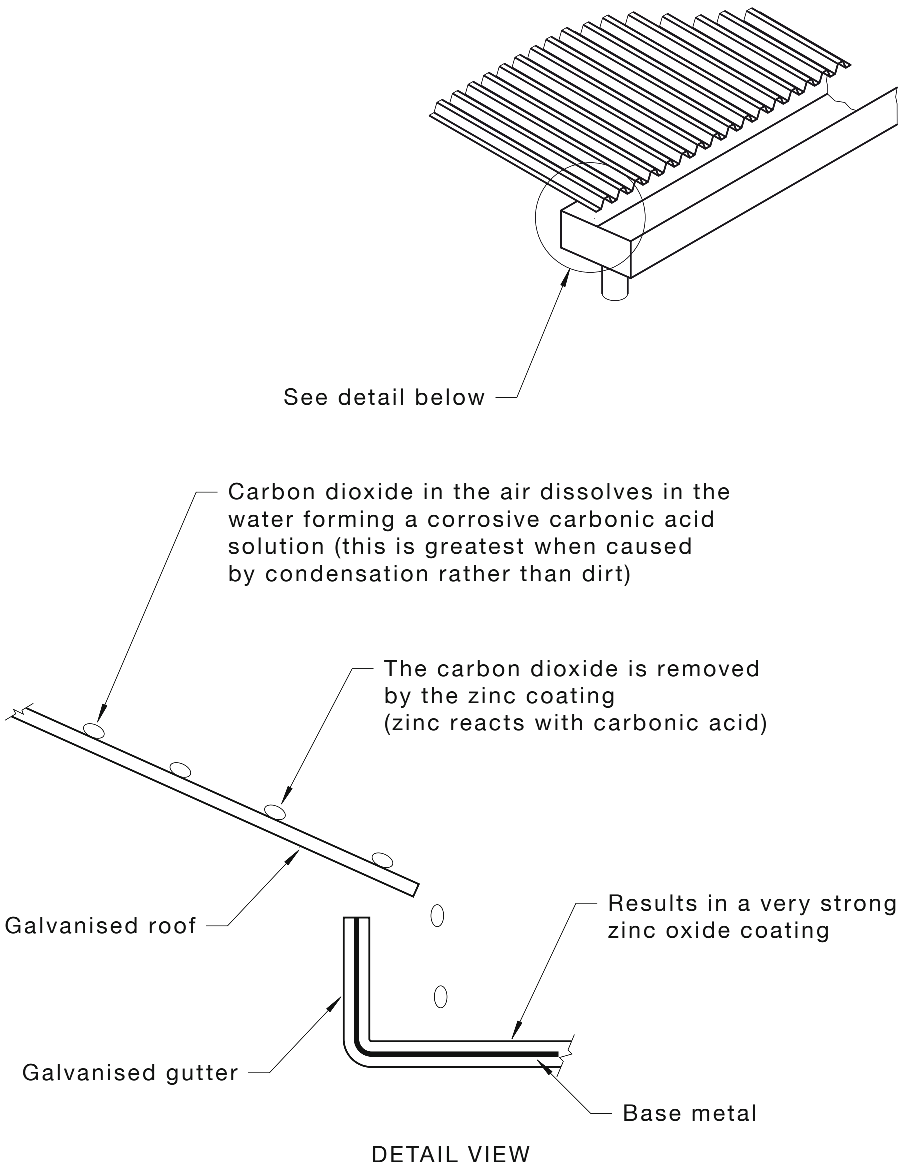
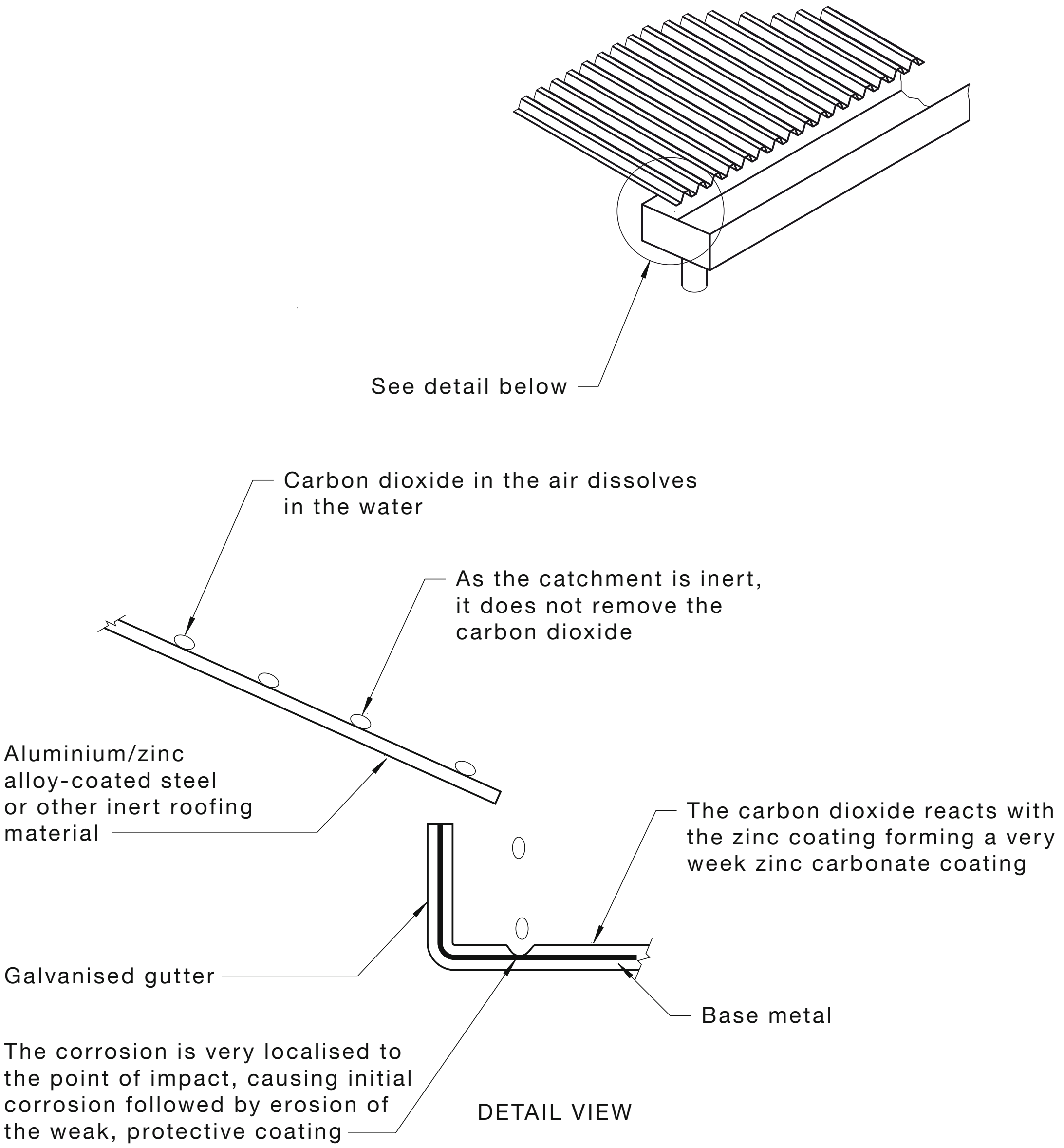



FIGURE 2.6(A) DRIP SPOT CORROSION




(a) No corrosion

FIGURE 2.6(B) (in part) INERT CATCHMENT MECHANISMS




(b) Corrosion

FIGURE 2.6(B) (in part) INERT CATCHMENT MECHANISMS

## 2.7 CUT EDGE PROTECTION

### 2.7.1 General

The sacrificial protection afforded the steel will delay corrosion of cut edges while there is zinc, aluminium/zinc or aluminium/zinc/magnesium alloy coating in the vicinity of the cut edges.

Almost every metal-coated steel product has the edges trimmed or cut and, when piercing occurs within the sheet surface, a further 'cut edge' is generated.

NOTE: Prime examples of such products are roofing and guttering, the total perimeter of which have trimmed edges.

These items are first slit to width and then cropped to length. Holes are often pierced to accommodate fasteners and yet corrosion does not affect the steel's useful life.

*Regardless of the environment, the more coating present, the more protection afforded to the steel, both on flat unmarked surfaces and at cut edges.*

Galvanic action causes zinc compounds to build up automatically at cut edges and scratches by an electrolytic reaction when water or moisture is present.

### 2.7.2 Cutting sheet

A clean sharp cut will protect itself; a burred and burnt edge will corrode and stain the downstream roof cover.

*Never use carborundum discs to cut metallic roof sheets or accessories. Power saws with metal-cutting blades will produce fewer damaging particles and burring. Alternative cutting tools such as electric shears and nibblers are also suitable but care is to be taken to ensure the straightness of cuts.*

## 2.8 MARKING OF ROOFING MATERIALS

Any colour pencil, except black pencils, felt-tipped pens or a string line and chalk dust may be used to mark roofing materials for cutting.

NOTES:

1 Black pencil marks can be a problem because the 'lead' is made from graphite and clay. Graphite is carbon which, when placed in contact with most metals, creates an electric cell when wet. This cell acts like a battery and chews at the metal surface leaving an indelible mark.

2 Caution should be taken when utilising chalk lines as some chalks are known to leave permanent marks.

## 2.9 SILICONE RUBBER SEALANT SELECTION AND APPLICATION

The sealant industry in Australia produces a wide variety of building sealants, which together embrace a multitude of end use applications and an even greater range of conditions. The sealant providing the optimum properties relating to metal roofing is neutral-cure silicone rubber. Sealants are generally not classed as adhesives.

Different metals require different sealants, but the same installation techniques are used.

All surfaces to be bonded are to be clean and dry and free from contaminates such as old sealant or oil. Mineral turpentine is generally suitable, but care is necessary to ensure the complete removal of residual solvent.

In the event that solvents such as xylol or toluol are used, gloves are to be worn for skin protection and adequate ventilation is to be provided.

Surface preparation is to be carried out immediately before sealant application. The joint is to be finished as soon as practicable after sealant extrusion to prevent premature curing, which may cause poor bonding to the second surface.

Sealant manufacturers specify the maximum open time for sealants.

For best practice, the application of sealant is to be as follows (see Figure 2.9):

(a) Apply the sealant in a continuous steady flow to achieve a fully filled void free joint.

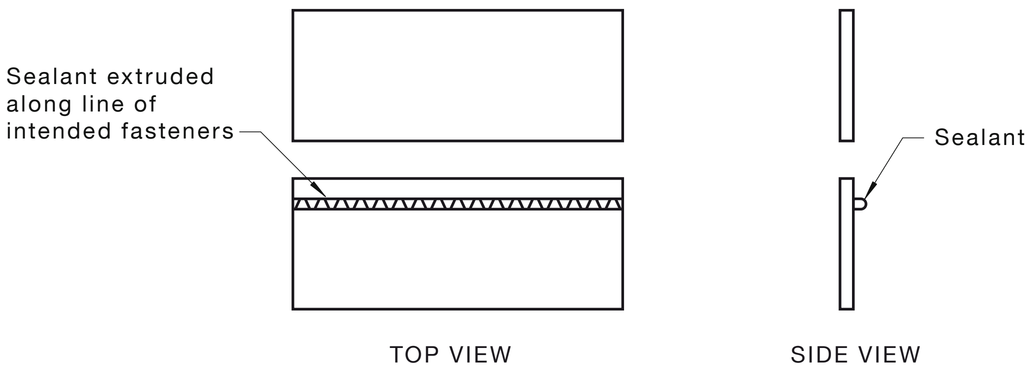
(b) Apply the sealant throughout the line of intended fasteners for complete bonding to avoid entrapment of air and to ensure that the fasteners are thoroughly sealed within the lap.

    The sealed lap is to be 25 mm.

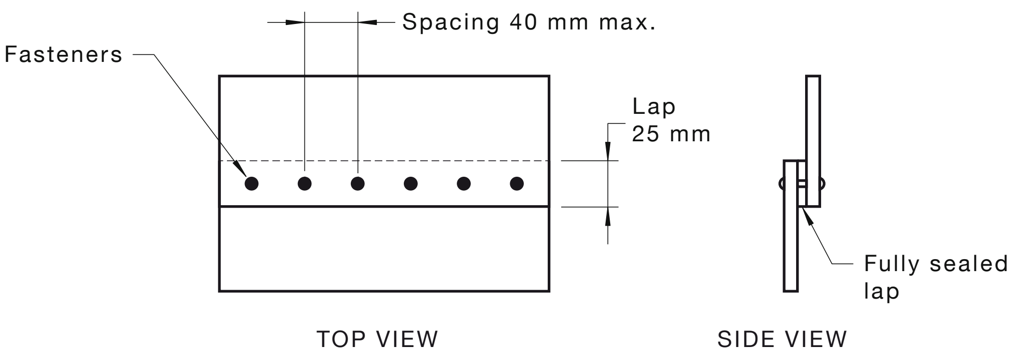
Sealed aluminium rivets or solid fasteners are preferred and are to be spaced at no more than 40 mm centres.

(c) When complete, remove any excess sealant extruded from the lap by means of a plastic spatula.

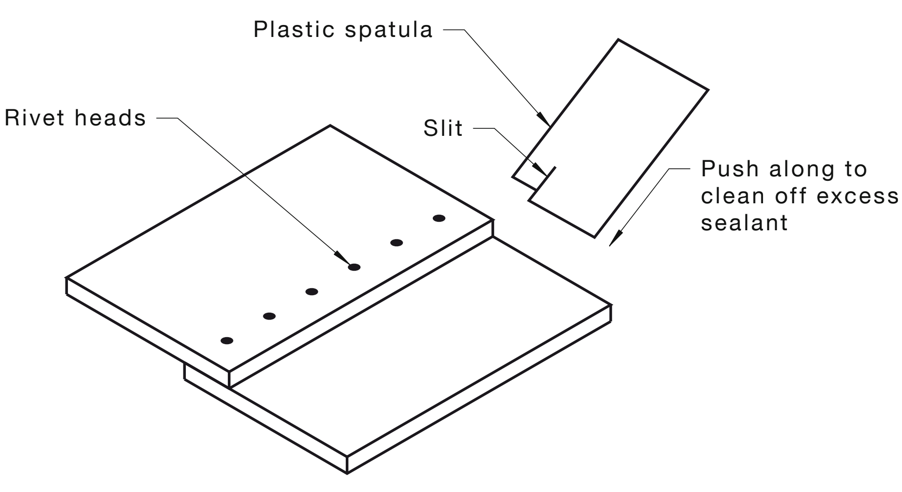
NOTE: This is preferable to wiping the surface with a cloth, which tends to contaminate the area with residual sealant and looks untidy when dust adheres to it.




(a) Placement of sealant




(b) Fasteners




(c) Completed joint

FIGURE 2.9 SEALANT JOINTS

## 2.10 SYNTHETIC RUBBER FLASHINGS

Like many other organic materials, synthetic rubber flashings such as ethylene propylene diene monomer (EPDM) and silicone are liable to undergo gradual degradation and change with the passage of time.

The main cause is degradation by atmospheric oxygen, which may be aggravated by ultraviolet light (UV) or high temperature, resulting in loss of strength flexibility and other properties.

The effect of light is confined to the surface and if the colour is black, the effect is generally negligible. In the case of non-black compounds, the type of processing oil has a significant effect on properties such as ageing, stain and UV resistance besides colour-fastness.

It is generally accepted that carbon black is the best possible UV absorber for a rubber compound. Non-black compounds used for flashings utilise special UV absorbers, which produce comparable protection to carbon black.

A more serious surface effect is the cracking produced by traces of ozone in rubber that is slightly stretched, say 10% to 20%. Even the minute amount of ozone present in the normal atmosphere (about 1 part per 10 million) can produce serious cracking if the rubber is not specially formulated to resist the effect.

## 2.11 MATERIAL COMPOSITION OF COMPOUNDS

### 2.11.1 Temperature range

The operating temperature ranges are as follows:

(a) *EPDM*: −30°C to 115°C: continuous

116°C to 150°C: intermittent

(b) *Silicone*: −60°C to 200°C: continuous

201°C to 250°C: intermittent

### 2.11.2 EPDM for normal conditions

EPDM is specifically formulated to give exceptional resistance to ozone, UV and water. Synthetic rubbers will effectively seal and remain flexible over a wide range of temperature extremes.

Under normal weather and environmental conditions, life expectancy and effectiveness is approximately 20 y.

### 2.11.3 Silicone for special conditions

Where flashings are required to withstand a greater range of temperatures, such as from steampipes, solid fuel heater flues and some water heater flues, special silicone units are available. The normal life of silicone units is approximately 20 y; however, continuous high temperature applications may affect the service life.

### 2.11.4 Carbon black and corrosion

Corrosion between dissimilar metals is a well-understood phenomenon. Carbon black can be found in a range of items, is highly conductive and can initiate corrosion of metallic-coated steels, particularly in marine environments. By way of example:

(a) The use of black 'lead' pencils on bare galvanised, aluminium/zinc alloy-coated, aluminium/zinc/magnesium alloy-coated and other coated steels allows the formation of corrosion in roofing products where the steel has been marked. In severe environments, this can result in severe knife-like corrosion of the metallic coating.

(b) Carbon black washers can also initiate corrosion; hence, they are not to make contact with bare, prepainted metallic-coated steel or prepainted stainless steels.

NOTE: Washers free of carbon black are readily available.

(c) Carbon black can also be found in gutter mesh applications. These applications should not be used unless the manufacturer can provide evidence of testing against ASTM D257, which measures the current flow through a material. If the material has a reading greater than $0.5 \times 10^{-6}$ A, the material is not to be used in contact with prepainted or bare metallic-coated steels.

# SECTION 3 MAINTENANCE AND CARE

## 3.1 SITE STORAGE

Sheets that have become wet during transport should be unpacked immediately and each sheet dried thoroughly on both sides before stacking. Sheets should be stored clear of the ground under cover or, for outside storage, fully wrapped with tarpaulins or plastic sheeting with the sheet and wraps clear of the ground (see Figure 3.1).

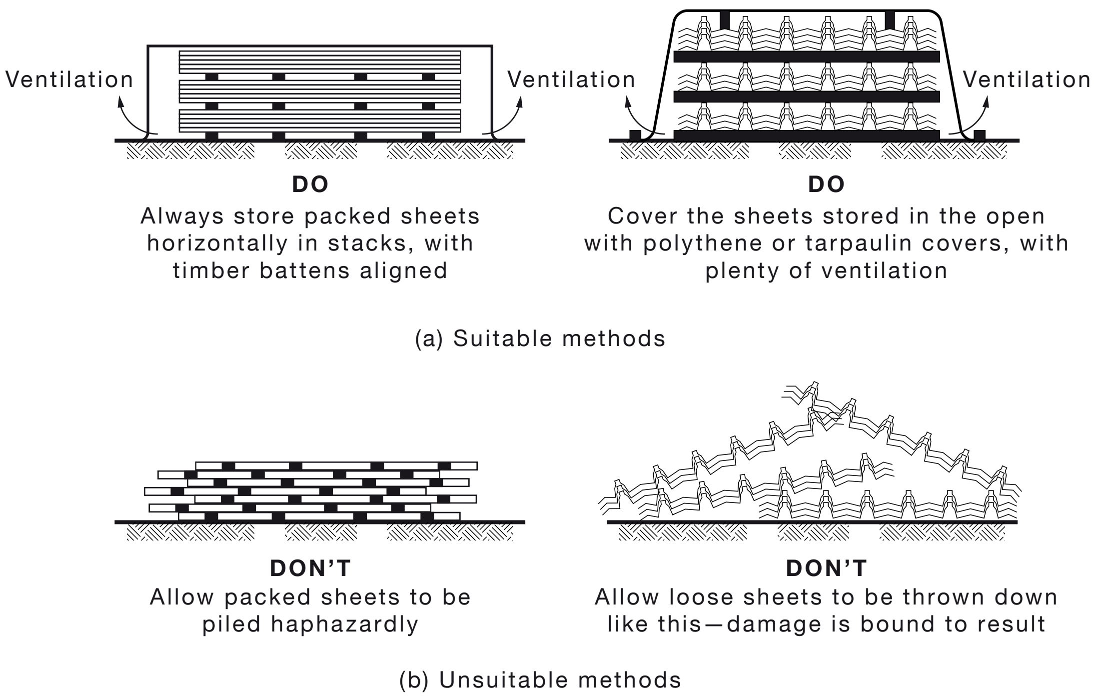



FIGURE 3.1 STORAGE

## 3.2 WALKING ON ROOFS

Where a risk of fall exists, protection against falling is to be provided by way of safety mesh, harness or perimeter protection (see Clause 4.2).

The surfaces of aluminium/zinc, aluminium/zinc/magnesium-coated steel and aluminium are prone to ‘scuffing’ from footwear or heavy equipment dragged across the surface. As this affects the appearance if not the performance of the material, preventive measures to protect the surfaces are to be taken (see Clause 3.4).

When walking on a roof, for best practice the following is to be observed (see Figure 3.2):

(a) Always wear soft-soled shoes on the roof avoiding the ribbed types, which pick up and hold small stones and swarf.

(b) On metal decking—

  (i) walk in the trays or on at least two ribs, as close as possible to the roofing support;

  (ii) distribute body weight evenly over the soles of the feet, trying not to concentrate weight on heels or toes; and

(iii) do not walk in the pan immediately adjacent to flashings or translucent sheeting—walk at least one pan away.

(c) As sheets freshly taken from packs may have an oily surface, take care when walking over newly laid sheets before any oil has evaporated.

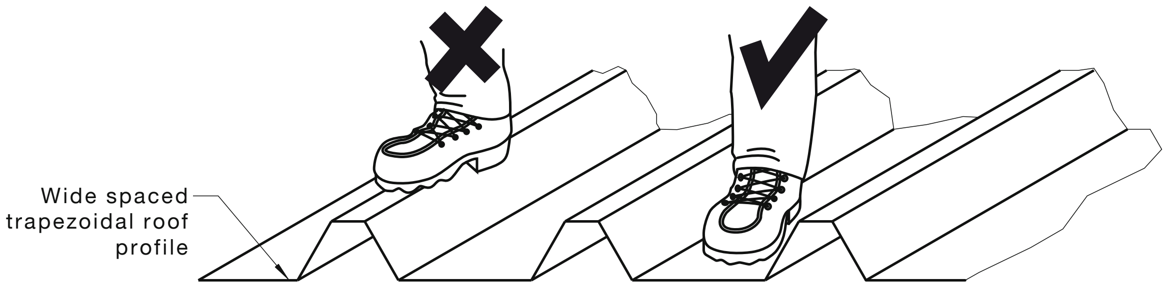



FIGURE 3.2 WALKING ON ROOFS

## 3.3 PROTECTION OF HEAVY TRAFFIC AREAS

On commercial and industrial installations, where constant foot traffic is required for the ongoing maintenance of plant or equipment installed on the roof, permanent walkways compatible with the roofing material should be installed (see Clauses 2.4 and 2.5).

## 3.4 TEMPORARY WALKWAYS

If heavy roof traffic cannot be avoided during the installation process, it is important to protect the roof coverings by the installation of temporary walkways (see Figure 3.4).

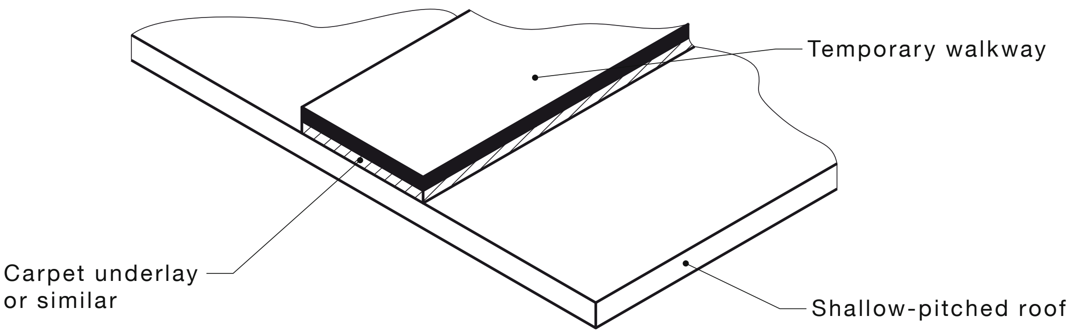



FIGURE 3.4 TEMPORARY WALKWAYS

## 3.5 CUTTING SHEETS ON SITE

Sheets should be placed face down on padded protection to reduce potential damage to the surface. Abrasive discs are to be avoided as they create copious amounts of swarf and burred edges, both of which can cause unsightly rust stains. Steel cutting blades in a minimum 4000 rpm saw, power shears, nibblers and slot-shears produce relatively clean sharp edges with a minimum of swarf (see Figure 3.5).

*Whenever possible, cutting should be done on the ground, and never over unprotected materials.*

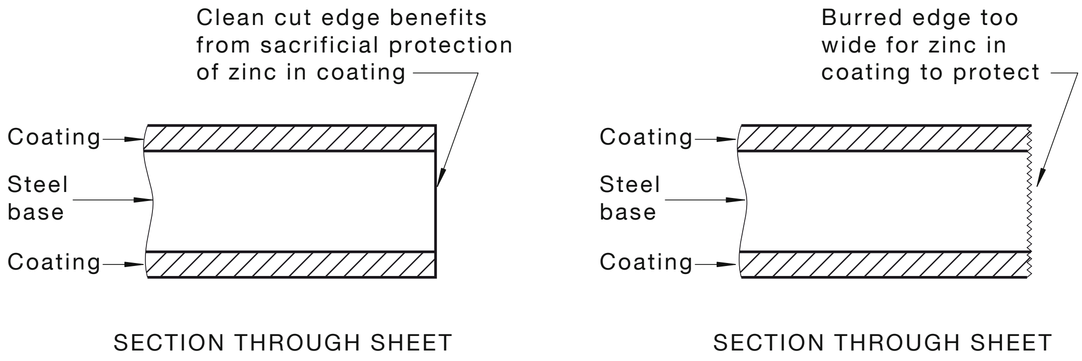



FIGURE 3.5 CUTTING STEEL SHEETS

## 3.6 CLEANING UP

Normal installation practices such as drilling and cutting usually leave offcuts and metallic swarf on or around the roof area. These materials and all other debris, including blind rivet shanks, nails and screws are to be cleaned from the roof area and gutter regularly during the installation process as unsightly staining of the surface due to oxidation of the metal particles will result, leading to corrosion and possible failure of the roofing material or guttering. Where practicable, the entire installation should be cleaned down with a blower vac, swept or, alternatively, if a water supply is available, hosed down at the completion of the work.

## 3.7 UNWASHED AREAS

Where exposed to the atmosphere in a sheltered situation such as under eaves, under solar hot water collectors or under photovoltaic panels, metal roofing will be more adversely affected than normally exposed sheeting (see Figure 3.7).

Accelerated corrosion in these instances is caused by particles of dust (frequently acid-laden), salt and industrial fallout, which are deposited by the wind onto the metal surface. Because of the sheltered location, rain cannot wash them from the surface, allowing them to build up and react aggressively with the metal when damp.

Metal roofing in such areas should be regularly washed with fresh water and mild detergent solution to remove grime, which would otherwise generate early corrosion.

NOTE: Any chemicals used for roof cleaning should be carefully selected to ensure they do not pose a risk to human health, and consideration should be given to the temporary disconnection of rainwater storage tanks utilised for drinking water during the cleaning process.

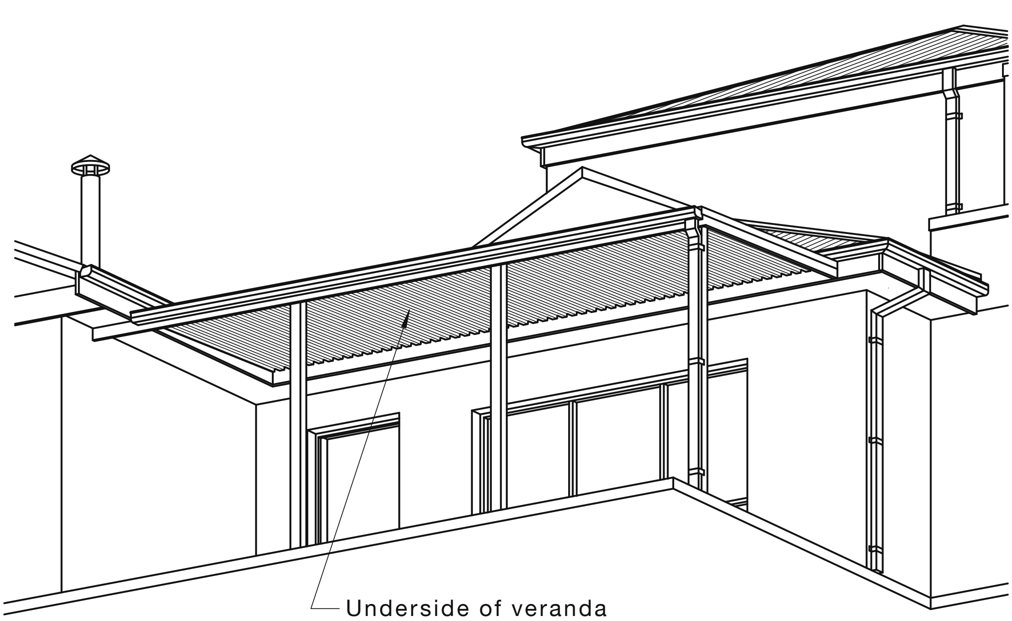



FIGURE 3.7 TYPICAL UNWASHED AREAS

## 3.8 SURFACE DAMAGE

There are occasions when a prepainted surface can be scratched or slightly damaged. The general rules are:

(a) Do not use paint spray cans to touch up scratches.

(b) If you cannot see the scratch from the ground, balcony, window, leave it alone.

(c) If damage is severe, the sheets in question are to be replaced.

## 3.9 STRIPPABLE COATINGS

Prepainted metal roofing components are often supplied with a protective plastic film. The protection is provided to prevent damage to the painted surface during forming, delivery and fabrication.

It is essential that the film be removed either prior to or during actual installation or immediately after.

The film is to be removed from areas to be lapped so that metal to metal joints can be achieved rather than have the film sandwiched within the laps.

**WARNING: SOME TYPES OF PROTECTIVE PLASTIC FILM ON ROOF COMPONENTS ARE AFFECTED BY SUNLIGHT AND ARE TO BE STORED IN THE SHADE. IF ANY SIGNIFICANT EXPOSURE TO SUNLIGHT HAS OCCURRED, REMOVAL CAN BE DIFFICULT IF NOT IMPOSSIBLE.**

## 3.10 ROOF NOISE—RAIN AND HAIL

The use of insulating blanketing placed over a foil laminate will reduce noise levels. The laminate is to be pulled taut enough to hold the blanket hard against the underside of the roofing (see Figure 3.10).

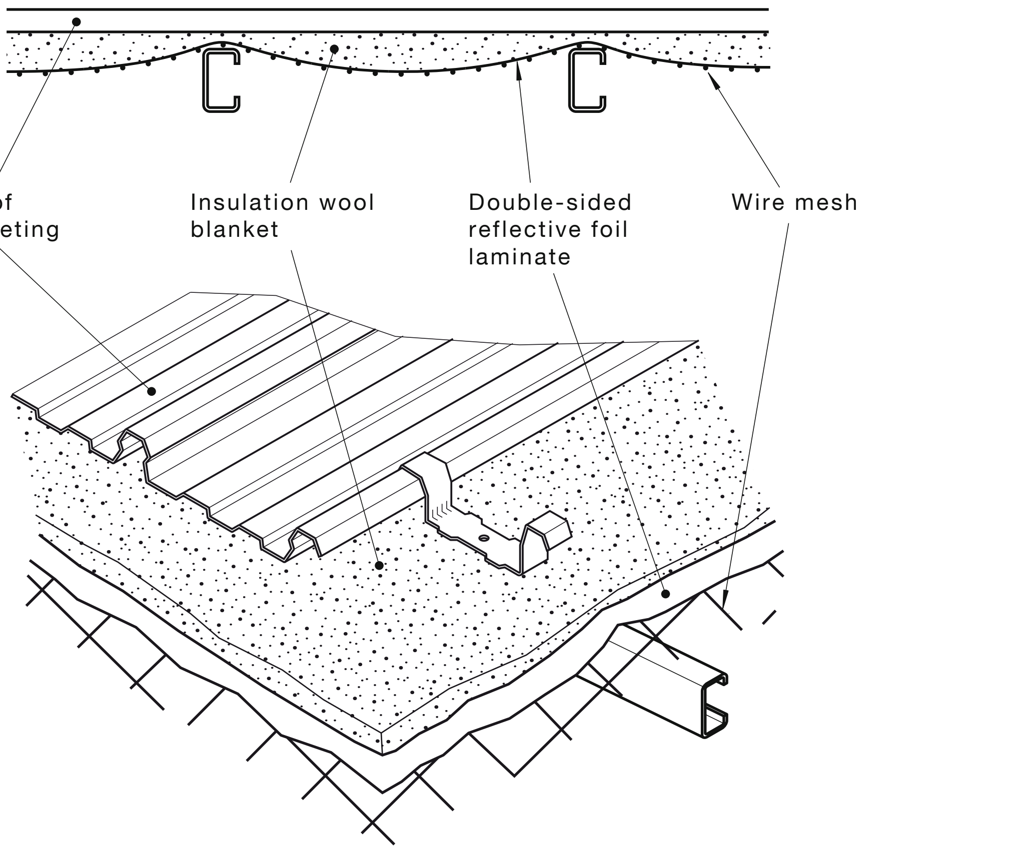
NOTE: Appropriate protective clothing should be worn when working with fibreglass insulation.

Actual contact dampens rain-induced vibration at the point of impact and marked noise reduction is achieved.

For purlin spacing of over 1200 mm, wire mesh is needed to support the blanket and to ensure contact is maintained. It has to be ensured that the mesh is not over-tightened, as the roof deck may be difficult to install.

Insulation wool blankets up to a thickness of 125 mm for pierced decking and 50 mm for concealed fastened profiles will compress sufficiently over the supports to allow normal sheet fixing procedures to proceed; however, the length of the fasteners is to be increased slightly to allow for the thickness of the compressed blanket between the sheeting and support (see Figure 3.10).

In addition to noise reduction, the wool blanket will improve the thermal insulation value of the roof system.




NOTE: This Figure is reproduced in adapted form with permission from BlueScope Steel Limited trading as BlueScope Lysaght.

FIGURE 3.10 ROOF NOISE

## 3.11 THERMAL STRESS

Expansion and contraction of pierce-fastened steel roof sheeting due to temperature changes within the sheeting, particularly caused by passing clouds or a strong breeze, can sometimes result in noise from the roof.

This noise is generally minor, but it is possible for louder noises to be generated by friction between the sheeting and roof structure. Expansion, if suppressed until the stress build-up exceeds the frictional resistance, results in sudden release, producing a series of sharp cracks.

This problem, although not common, is usually associated with high solar absorptance dark coloured prepainted sheeting fastened directly to timber supports, which have a high coefficient of friction, due to particular sap content or painted surface. This, combined with the tendency of the underside of the sheeting to stick, results in additional friction.

The situation is more pronounced with high solar absorptance dark colours due to their higher heat absorption, which increases the temperature and, therefore, the thermal movement. The risk of this occurring may be reduced by using lower solar absorptance dark colours, which are now widely available, or by inserting a separation membrane between the sheets and the supports to lower the resistance to movement (see Figure 3.11).

Reflective foil laminate, often used to provide a vapour and heat barrier, can operate as a separation membrane and assist in reducing the friction. Insulation blankets will perform a similar function.

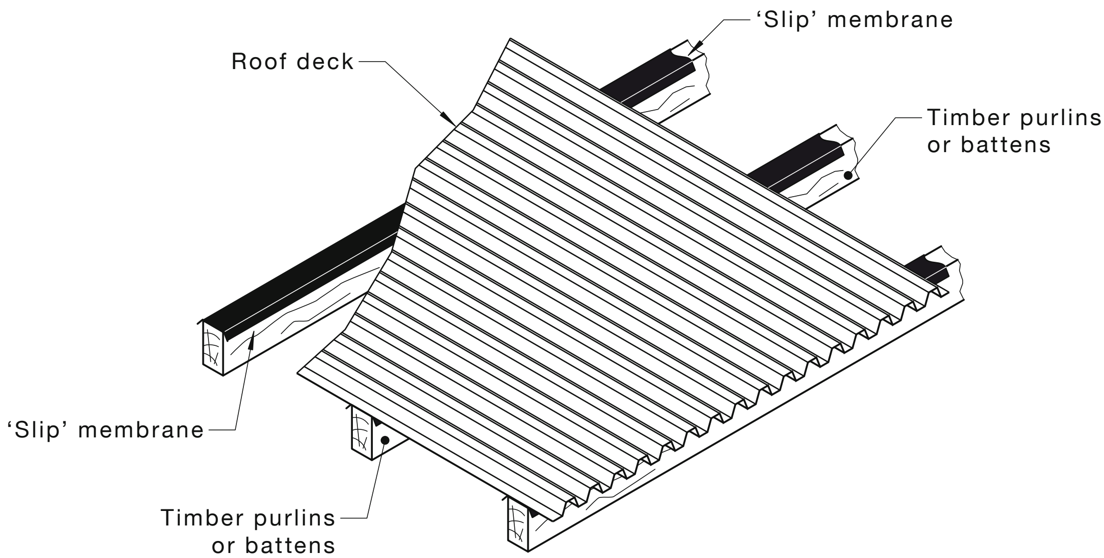



FIGURE 3.11 SEPARATION MEMBRANE

## 3.12 SUNSCREEN AND PREPAINTED METAL SHEETS

Given the outdoor nature of roofing and walling installation, suitable precautions should be taken to prevent personal sun damage. It has been found that certain sunscreens containing semi-conducting metal oxides such as titanium dioxide ($TiO_2$) and zinc oxide (ZnO) can accelerate the degradation of organic materials, including paint systems.

For personal safety, and to protect the surface of prepainted steel, it is recommended to—

(a) wear clean, dry, cut-resistant gloves that are suitable for the task;

(b) take suitable protection against personal sun damage, and

(c) prevent contact of the painted surface with titanium dioxide ($TiO_2$) and zinc oxide (ZnO) containing sunscreens.

# SECTION 4 ROOF SAFETY

## 4.1 EYE AND SKIN PROTECTION

Codes of practice and guidance material have been developed in each State and Territory. They contain specific guidance for persons working in conditions subject to UV radiation, including information for adopting a risk management approach for sun protection at work. Following the guidance will ensure standards of safety and welfare, as required under the Work Health and Safety Act and Work Health and Safety Regulations, are achieved.

Persons who work on roofs need to take extra care of eyes and skin, because ultraviolet radiation reflects off roof coverings, roof insulation, concrete and all light covered surfaces, creating increased levels of radiation.

Damage is permanent and irreversible and increases with each exposure.

Wind-burn can only dry the skin but, often, what is considered to be wind-burn may in fact be a severe ultraviolet radiation burn.

Cloudy days, where ultraviolet light is scattered in all directions, will also create conditions for very high doses of radiation.

Some substances can cause photosensitivity. Photosensitivity is an abnormally high reactivity in the skin or eyes to UV radiation or natural sunlight. It may be induced by ingestion, inhalation or skin contact with certain substances known as photosensitisers. Symptoms will vary with the amount of UV radiation, type and amount of photosensitiser, skin type, age and sex of the person exposed.

Ingestion or topical application of particular medications may cause photosensitivity in some individuals.

Photosensitivity may occur in every person. It is usually dose related and may not happen the first time the medication is taken.

NOTE: It should be stressed that administration of the medication should not stop until medical advice has been sought.

Avoiding exposure to direct sunlight will control the photosensitivity. A doctor or pharmacist should be consulted about the availability of alternative medications.

It is the ultraviolet (UV) radiation component of sunlight that is harmful; and the level of UV radiation is not directly related to temperature or brightness of sunlight. This means that harm can still occur on cool or cloudy days during the peak UV periods of the year. Exposure to high UV Index levels can contribute to skin cancer.

NOTE: The UV index is a measure of UV radiation and the higher the index value the greater the potential for damage to skin.

Under Australian Occupational Health and Safety legislation, employers are to consider steps to reduce this risk and protect employees from ongoing exposure to solar UV radiation, which can lead to skin cancer.

Implementing a comprehensive sun protection program, which includes a range of simple protective measures, can prevent sun-related injuries and reduce the suffering and costs associated with skin cancer, including reduced productivity, morale and financial returns.

Conduct a risk assessment to identify employees performing roof work, who have a high risk of exposure to solar UV radiation, and work situations where exposure to solar UV radiation occurs and consider the following protective measures:

(a) Engineering controls, which are measures that reduce exposure to solar UV radiation by a physical change to the work environment.

(b) Administrative controls, which are measures that reduce exposure to solar UV radiation by a change in work procedure and the way work is organised, such as—

  (i) considering changes to outdoor work schedules to minimise exposure;

  (ii) providing training to roof workers to enable them to work safely in the sun; and

  (iii) issuing SunSmart UV alerts and informing employees when it is necessary to use sun protection control measures.

(c) Personal protective equipment and clothing, which are measures that reduce exposure to solar UV radiation by providing a personal barrier between roof workers' eyes and skin and the hazard.

Employees also have a duty to take care of their own health and safety and cooperate with employers' efforts to improve health and safety. To work safely in the sun, employees are to follow workplace sun protection policies and procedures, attend training and follow instructions and advice provided, and use supplied protective equipment as instructed (see Figure 4.1).

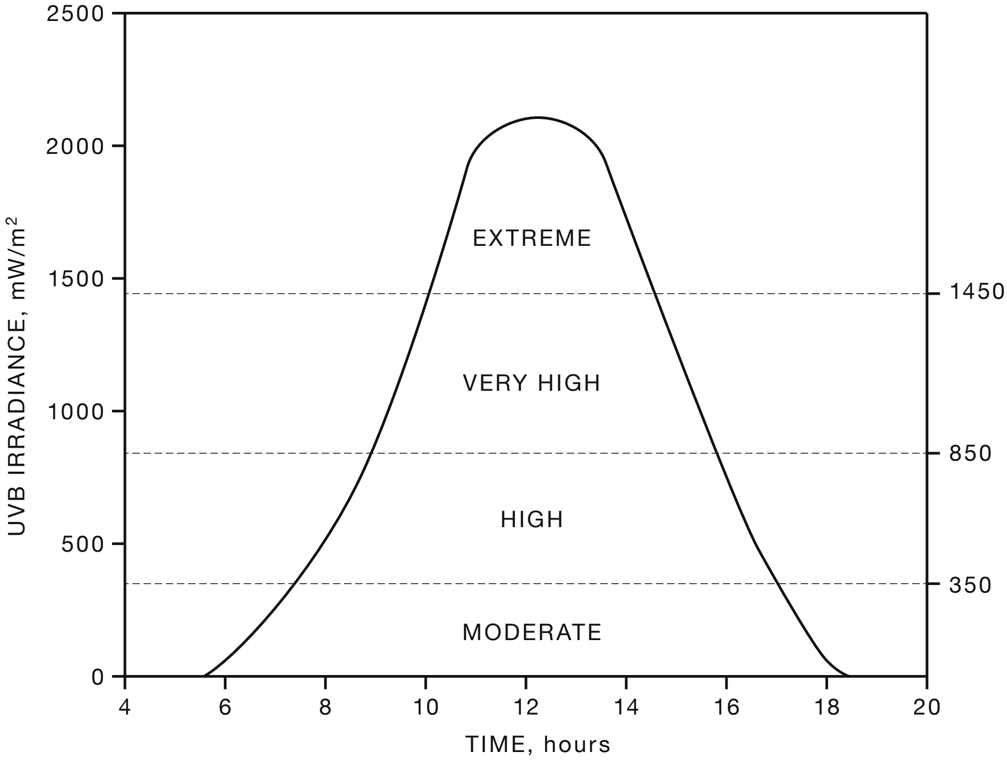



FIGURE 4.1 AMOUNT OF UV RECEIVED ON A SUMMER DAY

## 4.2 FALL PROTECTION

Various publications, regulations, codes of practice and guidance materials that deal with the risk of falls from heights are available from States and Territories. These include:

(a) *Managing the risk of falls at workplaces Code of Practice 2011* (December 2011), Workplace Health and Safety Queensland.

(b) *Preventing falls in housing construction, Code of Practice* (July 2012), Safe Work Australia.

NOTE: *Preventing falls in housing construction, Code of Practice,* provides practical guidance for employers and employees on ways to effectively manage the risks associated with working at height in the housing/construction sector and prescribes circumstances in which the provision of physical fall prevention measures are required. Preventing falls in housing, Code of Practice is an advisory document.

Each of these regulations, codes of practice and guidance materials on fall protection have specific criteria for how to manage the risk associated with falls from heights and have been developed or adopted in each State and Territory. They contain practical guidance for persons exposed to a risk of a fall and others in the vicinity, including information for adopting a risk management approach for all work at height, emergency and rescue procedures as well as detailed guidance when working at 2 m and above and they are to be followed. Each jurisdiction will make a decision on the adoption of the National Code of practice or any other code of practice or guidance material.

The preferred fall protection method to control a risk of a fall over 2 m requires the minimisation of the fall risk by ensuring that a passive fall prevention device, a work positioning system or a fall arrest system is used before implementing a lesser control under regulation. The protection measures are to include:

(i) Ensuring that a fall prevention system used in the business or undertaking is designed, manufactured and installed so as to be capable of bearing the weight of a person who (or object that) could fall.

(ii) Ensuring that an anchorage point in a fall prevention system used in the business or undertaking is designed, manufactured and installed so as to be capable of bearing the weight of a person who (or object that) could fall.

(iii) Providing guardrailing as a passive fall prevention device in the vicinity of any opening through which (or any edge over which) a person or object could fall.

(iv) Ensuring that any ladder used to minimise risk is fit for purpose, appropriate for the duration of the work and set up correctly.

(v) Ensuring emergency and rescue procedures are developed to provide emergency assistance to any person using a fall prevention system or ladder.

A risk assessment is to be conducted to identify employees performing roof work who have a high risk of exposure to a fall and consideration given to the following in the order of priority or hierarchy of control:

(A) Any work that may involve a fall hazard is to be carried out on the ground or on a solid construction.

(B) A passive fall prevention device is to be used, that is plant, equipment or material that is designed to prevent a fall and after installation does not require ongoing adjustment, alteration or operation to maintain its ability to prevent a fall such as a temporary work platform, roof safety mesh or guardrailing. Persons are to avoid walking or standing on the mesh except in recovery from a fall (see Clause 4.4).

NOTES:

1 Temporary work platform means a fixed, mobile or suspended scaffold, elevated work platform, mast climbing work platform, a work box support by a crane, hoist, forklift truck or other mechanical plant, building maintenance equipment, portable or mobile fabricated platform or temporary platform that provides a work area and is designed to prevent a fall.

2 Safety mesh is not intended as provision for access or a working platform.

(C) A work positioning system is to be used, that is any plant, equipment or structure, other than a temporary work platform, that enables a person or thing to be positioned and safely supported at a location for the duration of the relevant work being carried out (e.g. an industrial rope access system or travel restraint system).

A travel restraint system means plant or equipment that is worn or attached to a person and designed to prevent the person from reaching an edge or elevated surface from which the person could fall; for example a harness or belt that is attached to one or more lanyards, each attached in turn to a static line or anchorage point.

(D) A fall arrest system is to be used, that is plant, equipment or material designed to arrest a fall to prevent injury to a person [e.g. an industrial safety net, catch platform or safety harness system (not a travel restraint system)].

(E) A ladder is used, which is fit for purpose, appropriate for the duration of the work and set up correctly.

## 4.3 WORK ON EXISTING ROOFS

### 4.3.1 General

When existing roofs are to be repaired or maintenance is to be undertaken, particular attention is to be paid to the fall protection controls given in Clause 4.2 and the following, as required:

(a) Identifying the existing material on the roof.

(b) Inspecting the material to determine its condition.

(c) Determining whether safety mesh is in existence and its condition.

(d) Inspecting the support structure of the roof to determine its soundness.

(e) Identifying any penetrations, skylights, etc., and securing.

*Always remember the following:*

*(i) If the material on the roof is not immediately identifiable or in sound condition, assume the roof is fragile.*

*(ii) Place warning signs at any opening or point wherefrom persons can gain access to the roof, to warn them of the dangerous and hazardous situation.*

### 4.3.2 Fragile roofs

Regulations, codes of practice and guidance material for working on a fragile surface have been developed in each State and Territory. They contain specific guidance for those who carry out work on, or in the vicinity of, any fragile surface. Following the guidance will ensure standards of safety and welfare, as required under the Work Health and Safety (WHS) Act and Regulations, are achieved.

Although preparation and planning is of vital importance in any area of roofing work, it is particularly important to take special care when working on, or in the vicinity of, a fragile roof.

For best practice, consideration and provision for the following is to be taken, given the site conditions:

(a) Ensuring that a passive fall prevention device is used to safeguard the surface against persons walking or placing items on the surface.

(b) Ensuring the surface, including the underside, is inspected to determine the extent of its fragility, and warning signs are in place.

(c) Implementing the fall protection controls measures of Clause 4.2.

### 4.3.3 General handling

Regulations, codes of practice and guidance material for manual handling and the use of plant have been developed in each State and Territory. They contain specific guidance for those who carry out manual handling tasks and use plant such as access and lifting equipment. Following the guidance will ensure standards of safety and welfare, as required under the Work Health and Safety (WHS) Act and Regulations, are achieved.

### 4.3.4 On-site storage

Particular attention is to be paid to the following:

(a) Roof sheeting is normally delivered to the site in strapped bundles if intended for direct lifting on to a roof, or delivered loose if being stored for uplifting to the roof as required. Bundles of sheets being stored on site are to be stacked clear of the ground and protected from rain with waterproof covers (see Clause 3.1).

(b) Moisture can penetrate between the surface of stacked sheets by capillary action or wind and can cause deterioration of the surface coating, resulting in poor appearance or a shortened life expectancy. If sheets become wet, they are to be wiped with a clean cloth and stacked to dry (see Clause 3.1).

### 4.3.5 Lifting to roof

When craning long lengths from truck to roof, a spreader bar with a fabric sling is to be used. Otherwise, sheets may be unloaded and passed up to the roof singly. Gloves should be worn to protect the hands.

### 4.3.6 Roof traffic

Roof lights installed on roof areas are not to be walked on.

## 4.4 SAFETY MESH

### 4.4.1 General

Where required by AS 1562.3, safety mesh is to be fitted in roof construction under plastic roof cladding materials for Class 1 and Class 10, and Class 2 and Class 9 buildings, as defined in the NCC.

### 4.4.2 Fall protection

Whenever work is to be carried out within 2 m of any edge on a new or existing roof from which any person could fall, provision is to be made to prevent persons falling in accordance with the relevant regulations, codes of practice and guidance material on fall protection that have been developed and or adopted in each State and Territory in conjunction with appropriate edge and perimeter protection.

One of the preferred systems for protection against injury through falling in the sheet laying process is the provision of a permanent wire safety mesh over the area to be roofed.

NOTE: Safety mesh installed beneath metallic roof coverings is a common installation fall protection measure only.

### 4.4.3 Fixing methods

For best practice, the method of fixing the mesh to the purlins is as follows:

(a) Pass each longitudinal wire through a hole drilled in the top of the purlin and tie off with at least four full turns around the wire, as shown in Figure 4.4.3(A). Where the mesh is to be fixed to timber purlins, use 40 mm × 3.5 mm staples.

(b) Side lap the runs of mesh by 150 mm (one opening width). Where the purlin spacing exceeds 1.7 m, provide intermediate fixing with 40 mm × 2 mm staples. Carry out intermediate stapling of the mesh from underneath.

(c) Where it is necessary to make longitudinal joins, the knot and tie is to be the full length of the tail wire, which is 300 mm in length. Tie the tail wire at least four times around the knot. Then place the other wire under the wire and tie. The 300 mm tail wire may be achieved by cutting the longitudinal wire close to a join. The join is to be the full width of the mesh with every longitudinal wire joined.

    Any variation to the recommended method of tying is to be avoided. Figure 4.4.3(B) illustrates the tying procedure.

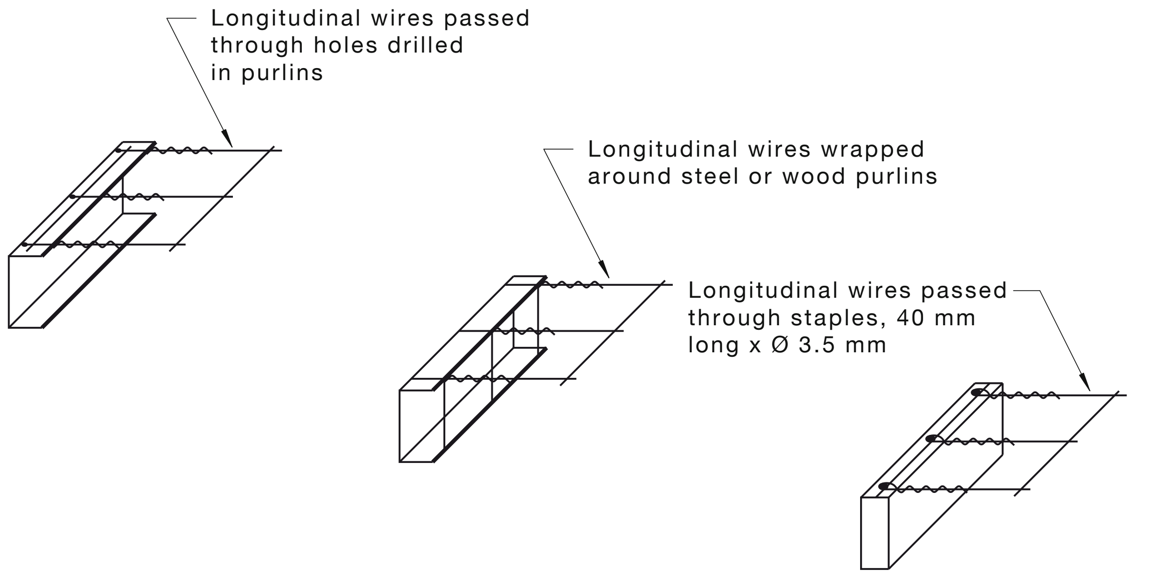
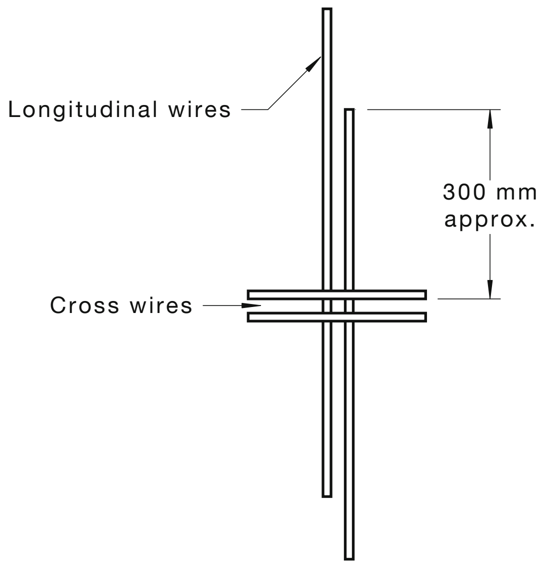
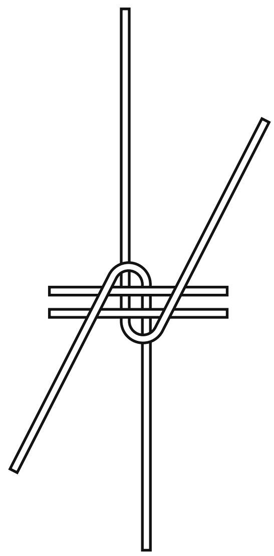


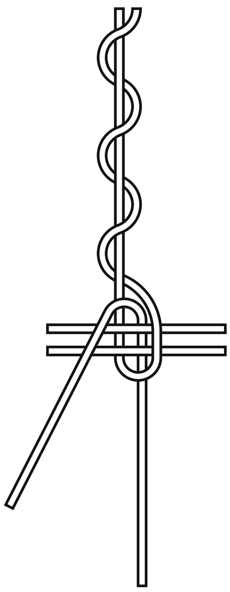

FIGURE 4.4.3(A) WIRE FIXING



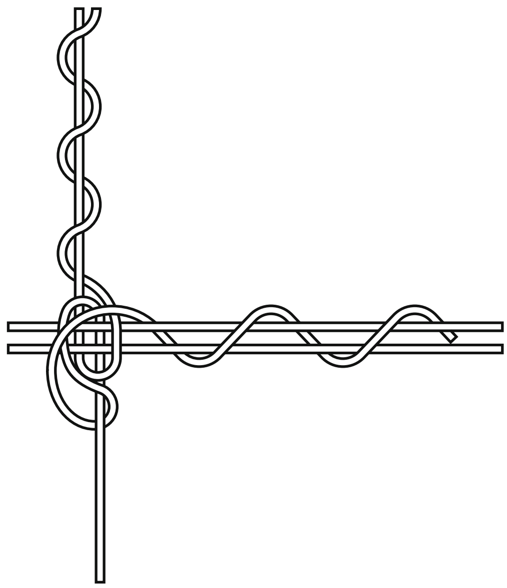

(a) Step 1



(b) Step 2



(c) Step 3



(d) Step 4

FIGURE 4.4.3(B) LONGITUDINAL WIRE JOINTING

### 4.4.4 Installation methods

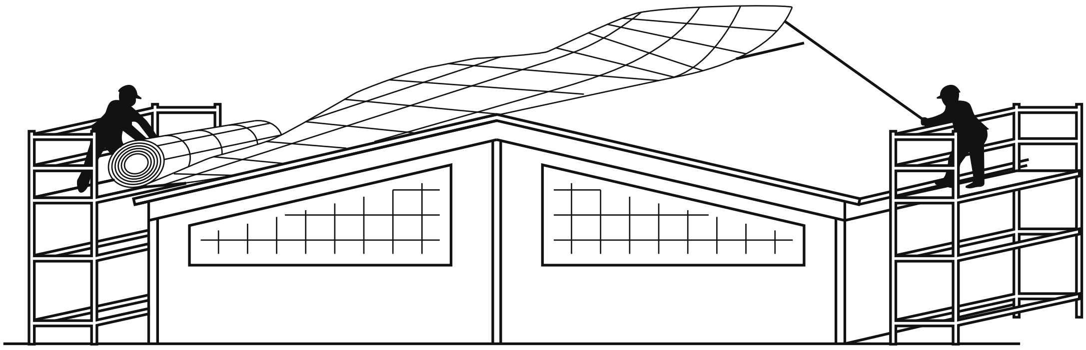
For best practice, when installing the mesh, the installation is to be as follows:

(a) The mesh is to be cut to length from the roll or ordered (from the supplier) to the correct length, and run out over the roof using a continuous rope system. Fall protection will be required by the person(s) running out the continuous rope. Figure 4.4.4 shows a method for installing the mesh.

(b) As much of the roof framing as is erected is to be meshed out prior to loading the roof with bundles of decking.

(c) Persons engaged in the installation of the mesh to be protected against injury by falling, taking into consideration the fall protection measures of Clause 4.2.

(d) Mesh to not be over-tightened, as difficulty in laying roof decking may occur.



Mesh can be installed safely from scaffolding positioned at each end of the roof

FIGURE 4.4.4 INSTALLATION OF MESH

### 4.4.5 Mesh requirements

The mesh and its installation is to comply with AS/NZS 4389, a summary of which is as follows:

(a) The mesh to comprise 2 mm diameter wire of not less than 450 MPa tensile strength, electrically welded into a mesh having longitudinal wire spacings of 150 mm and cross-wire spacings of 300 mm.

(b) A test certificate issued by a laboratory accredited by signatories to the International Laboratory Accreditation Corporation (ILAC) through their Mutual Recognition Agreement (MRA) showing that a sample of the mesh had successfully undergone the required tests is to be available.

NOTES:

1 In Australia, an ILAC MRA signatory is the National Association of Testing Authorities (NATA).

2 This evidence is not required if the safety mesh has been labelled by the manufacturer to indicate that it complies with AS/NZS 4389.

(c) Mesh to be used under translucent sheeting to be able to withstand a 1.1 kN concentrated load placed at any location.

Certified and approved securely fixed mesh, as detailed above, will offer long-term protection against falling for maintenance and repair workers in addition to providing fall protection for the original roof installation workers.

## 4.5 HARNESSES

Safety harnesses are an essential part of a roofer’s equipment and should be used when there are no other options available, such as when re-roofing (an information course in the correct choice and use of safety harnesses is recommended).

For best practice, the following is to be applied:

(a) Parachute type full body harness, connected to the lanyard or lifeline at the top dorsal position, to be worn by the roofer.

(b) An alternative attachment position to be provided; that is where a lifeline and rope-grab device to be used on steeply sloping roofs, whereby the user will need to manually operate the mechanism by having the device in front, and connect the device onto a D-ring on the side of the belt.

(c) Waist type belts not to be used for roof work.

(d) Inertia reels are not designed for continuous support but become effective by locking the system in the event of a fall. They are not to be used to support the user during normal work.

(e) Independent fall arrest systems and safety harnesses to be used only when the individual components are known to be compatible.

## 4.6 USE OF LADDERS

When using ladders, particular attention is to be paid to the following points, which have been summarised from AS/NZS 1892.5:

(a) Ladders are to comply with AS/NZS 1892.5 and be of industrial grade.

(b) Ladders are to be maintained in a sound condition; damaged or defective ladders are not to be used.

(c) No ladders other than a trestle ladder are to be used to support a plank from which a person is required to work.

(d) Ladders are not to be used as guys, braces, struts, beams or skids or for anything other than their proper purpose.

(e) Access to and from all working platforms and intermediate access platforms is to be provided by means of single ladders (not exceeding 9 m) unless permanent or temporary stairways, suitable ramps or runways, or other suitable means of access is provided.

## 4.7 ACCESS LADDERS AND ACCESS PLATFORMS

When using access ladders and access platforms, the following steps, summarised from AS 1657, are to be implemented:

(a) Aluminium ladders are not to be used in the vicinity of uninsulated electrical power lines.

(b) Ladders are to be provided at horizontal intervals not greater than 15 m.

(c) Ladders are to extend to the full height of the scaffolding.

(d) Ladders are to rise above each access platform to the minimum prescribed distance, in accordance with each State's and Territory's legislation.

(e) Ladders are to be secured against displacement at their head and base.

(f) Access platforms are to be provided at working platforms levels and at the head and base of every ladder in such a manner as not to encroach on the working platform.

(g) Access platforms are to be constructed as in a bay of the class of scaffolding being used.

(h) Where a ladder passes through an opening in the floor of an access platform, such opening is to be as small as is reasonably practicable and is to be protected by handrails where necessary.

(i) The vertical distance between any two successive platforms is not to exceed 7.3 m.

(j) Each ladder is to be offset from the ladder immediately below.

(k) Each ladder is to be fixed so that the slope is not less than one horizontally to four vertically (1 in 4) and not more than 1 in 6.

## 4.8 SINGLE AND EXTENSION LADDERS

For best practice, when using single and extension ladders, the following steps are to be observed:

(a) Single ladders greater than 9 m in length are not to be used.

(b) Extension ladders of maximum extended length greater than 15 m are not to be used.

(c) Single extension ladders are to be secured so that they cannot move from their top or bottom points of rest or alternatively a person is to be stationed at the base to prevent the ladder from slipping.

(d) Where work is performed from a single or extension ladder, the ladder is to rise to a height of at least 1 m above the highest rung to be reached by the feet of the person working from the ladder.

NOTE: Checks should be made with the relevant authority for additional requirements when working from single and extension ladders.

(e) Single ladders are not to be joined together to make a longer ladder.

(f) Single and extension ladders with a wire-reinforced stile are not to be used in the vicinity of uninsulated electrical power lines.

(g) Single and extension ladders are to be used at such an angle that the horizontal distance from the top support to the foot of the ladder is approximately equal to one quarter of the length of the ladder.

# SECTION 5 ROOF DRAINAGE

## 5.1 SCOPE OF SECTION

This Section sets out guidelines for the manufacture and fitting of internal and external metal gutters, downpipes, sumps and rainheads.

Although box gutters, valley gutters and soaker gutters are all forms of internal gutter, they are each discussed separately for the sake of clarity.

Siphonic drainage systems are site-specific roof stormwater drainage systems outside the scope of this document. They are designed by a manufacturer of the system for specific application. Siphonic drainage systems, associated gutters, overflow provisions and auxiliary components are to be designed and installed in accordance with the hydraulic engineers design specifications in line with the manufacturer's specifications.

## 5.2 MATERIALS

AS/NZS 2179.1 provides requirements for materials, particularly with respect to the compatibility of materials in contact with run-off from the roof (see also Section 2).

In addition, the following criteria, summarised from AS/NZS 2179.1, are to be applied for all roof drainage components:

(a) *Minimum thickness of sheet material*—to be as given in the appropriate Tables in Section 2.

(b) *Expansion components*—to be installed so that longitudinal expansion and contraction does not damage any components, joint or accessory.

(c) *Sizing of components*—as specified in AS/NZS 3500.3 or AS/NZS 3500.5.

The jointing of components is to be as stated in Clause 5.8.

## 5.3 BOX GUTTERS UP TO 600 mm

### 5.3.1 Overflow provision and size

To protect buildings from a total or partial blockage of outlets, downpipes or stormwater drains, it is essential that box gutters discharge all roof water clear of the building via overflows. To ensure that adequate overflow provisions are made and any surcharge is accommodated, the overflow weir of any rainhead is to be not less than 25 mm below the sole of the gutter discharging to the rainhead. Box gutter sumps are to be fitted with overflow ducts, overflow channels or high capacity overflow devices [see Figure 5.3.1(a)].

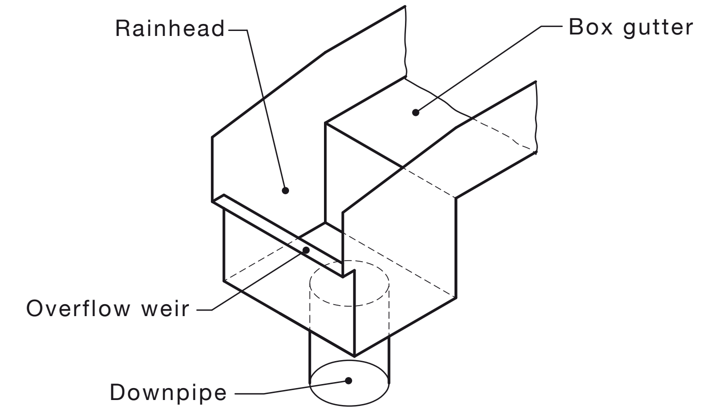
Particular attention is to be paid to the following (see Figure 5.3.1):

(a) The size of overflows are to be calculated in accordance with AS/NZS 3500.3.

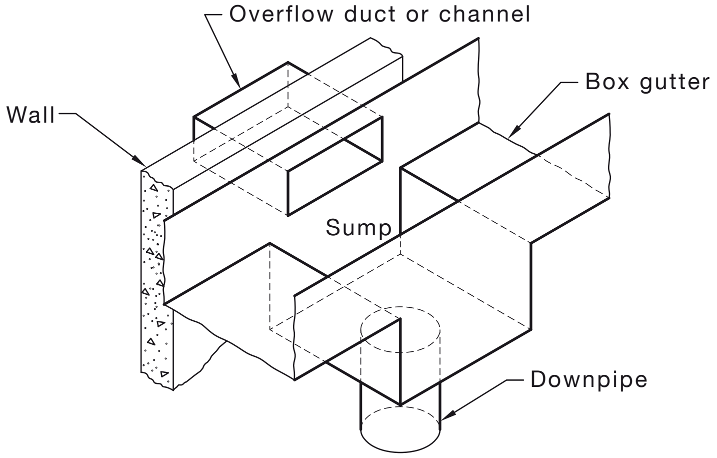
(b) Overflows are to be terminated in such a way as to prevent damage to buildings and property.

(c) The hydraulic capacity of overflow devices are to be not less than the design flow for the associated gutter outlets and discharge to atmosphere.

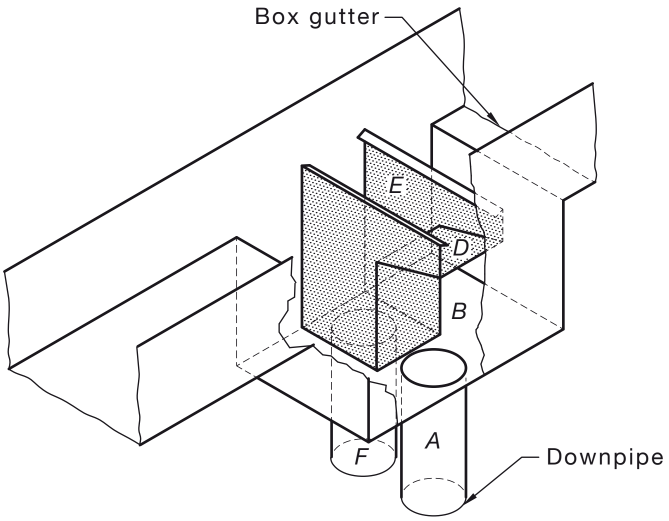
NOTE: For the design of appropriate overflow devices such as rainhead, sump/side overflow and sump/high capacity overflow devices, see AS/NZS 3500.3.




(a) Rainhead




(b) Sump/side overflow device




(c) Sump/high capacity overflow device (see AS/NZS 3500.3)

NOTES:

1 Layout of sump/side overflow device may have to be varied due to constraints [see Figure 5.3.1(b)].

2 Where desired, the sides of the sump/high-capacity overflow device may be perforated to flush the downpipe (*F*), as shown in illustration (c).

3 The normal outlet may be moved longitudinally to enable better inspection and maintenance access.

FIGURE 5.3.1 OVERFLOW PROVISION

### 5.3.2 Gutter installation

In the installation of box gutters, lear gutters and associated equipment and components, particular attention is to be paid to the following:

(a) *Lead gutters* Lead gutters are not to be used on any roof if the roof is part of a drinking water catchment area.

(b) *Maximum length* Box gutters are not to exceed the recommended maximum length given in Table 5.3.2 without the installation of appropriate expansion joints [see Figures 5.3.2(B) and 5.3.2(C)].

(c) *Expansion joint saddles* Double-ended expansion joints will need a clear space of not less than 50 mm between stop ends. The whole of the expansion joint is to be flashed with a saddle flashing as shown in Figure 5.3.2(B). The saddle flashing is not to be fastened to the gutter.

(d) *Synthetic rubber* Where utilised, synthetic rubber expansion joints are to be installed at the highest point of the gutter and fully supported by the gutter profile to prevent damage to the joint. Gutters are to be lapped under the synthetic joint by a minimum 50 mm.

(e) *Freeboard* Box gutters calculated in accordance with AS/NZS 3500.3 include a 25 mm allowance for freeboard to prevent wind driven spillages.

(f) *Overhang* Roof coverings discharging into a gutter to overhang the back of gutters by 50 mm and have the trays turned down 20 mm at 45°.

NOTE: Roof coverings discharging into the gutter should not overhang the gutter so as to make maintenance of the gutter impracticable.

(g) *Sizing* Box gutters to be sized in accordance with AS/NZS 3500.3 to effectively collect and discharge all roof water with an overflow risk of 1 in 100 y. As an example for commercial or industrial installations, gutters to have a minimum size of not less than 300 mm wide and 75 mm deep at the high end; for domestic installations a minimum size of not less than 200 mm wide and 75 mm deep at the high end commensurate to the roof catchment area serviced by that gutter.

NOTE: Box gutters 200 mm wide are more prone to blockages and should be subject to more frequent inspections and maintenance.

(h) *Tapered sole* The sole width of a box gutter is not to be reduced towards the outlet without a proportional increase in depth.

(i) *Fall or slope* Box gutters to be installed with a minimum uniform fall of 1 in 200 towards the outlet.

(j) *Gutter upstand* Box gutters to be installed without gaps between the underside of the roof coverings and the top of the gutter upstand or alternatively be provided with suitable hanging flashings with weathering folds.

(k) *Lap joints* Lap joints of box gutters to have 25 mm laps sealed and fastened in the direction of fall.

(l) *Gutter design* Box gutters to be straight (without change of direction).

(m) *Gutter profile* In a cross-section, box gutters to have a horizontal constant width base (sole) with vertical sides. [For gutter profiles see Figure 5.3.2(A).]

NOTE: Where appropriate, internal gutters may also be constructed with a horizontal constant width base (sole), single vertical side with lear, double vertical side with lear or single or double lear extending under the roof coverings, provided the effective area of the gutter is appropriately sized for the roof catchment area. V-shaped gutters are not to be used.

(n) *Gutters seal* Box gutters to be sealed to the rainheads and sumps.

(o) *Sarking* To enter the box gutter 25 mm.

(p) *Structural strength* The sides of any box gutters to have adequate structural strength so that water pressure will not cause deformation that can affect water surface levels and hence the hydraulic capacity of the gutter.

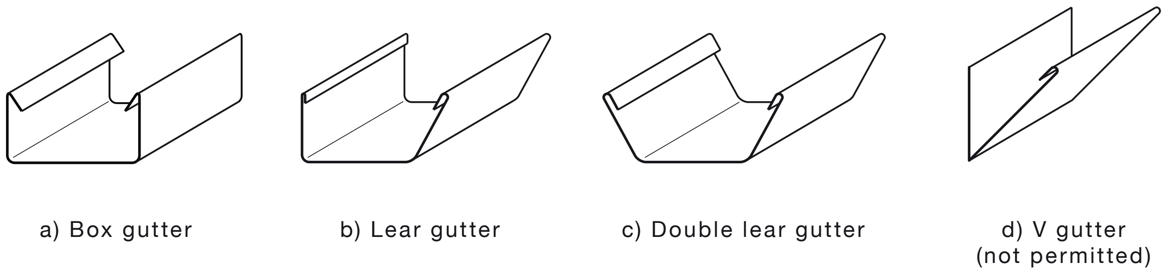



FIGURE 5.3.2(A) INTERNAL GUTTER PROFILES

**TABLE 5.3.2**

**BOX GUTTERS—RECOMMENDED MAXIMUM LENGTH BETWEEN EXPANSION JOINTS AND MINIMUM EXPANSION SPACE**

| Material | Base metal thickness<br>mm | Maximum length between expansion joints<br>m | Minimum expansion space<br>mm |
|---|---|---|---|
| Aluminium | 0.90<br>1.00 | 12<br>12 | 50 |
| Copper | 0.60<br>0.80<br>1.00 | 9<br>15<br>26 | 50 |
| Steel | 0.42*<br>0.55*<br>0.75* | 15<br>20<br>25 | 50 |
| Stainless steel | 0.55 | 20 | 50 |
| PVC | — | 10 | 30 |
| Zinc | 0.80 | 10 | 50 |

* These are base metal thicknesses.

NOTE: The temperature variation experienced by products will depend upon geographical location, extent of shading and absorptivity and surface colour. During summer, in most parts of Australia, the temperature of products exposed to direct sunlight may exceed 80°C.

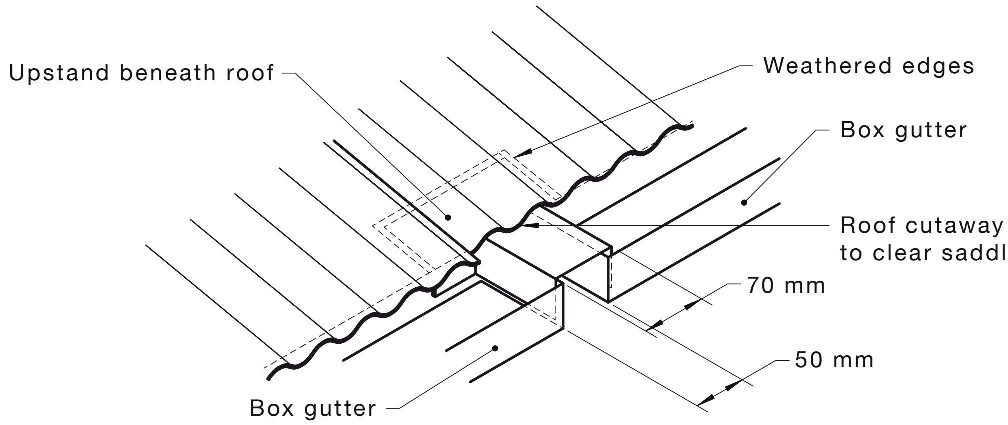



FIGURE 5.3.2(B) EXPANSION JOINTS AND SADDLE FLASHING

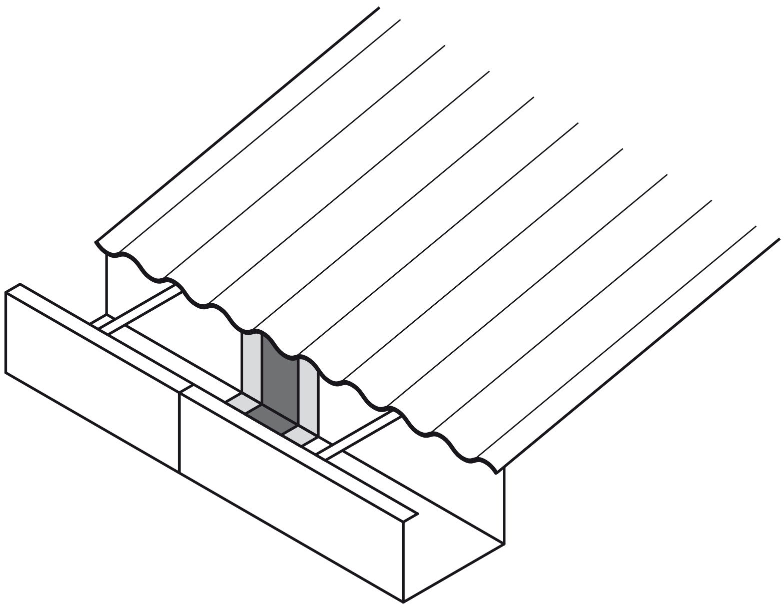


FIGURE 5.3.2(C) SYNTHETIC RUBBER EXPANSION JOINT

### 5.3.3 Outlets

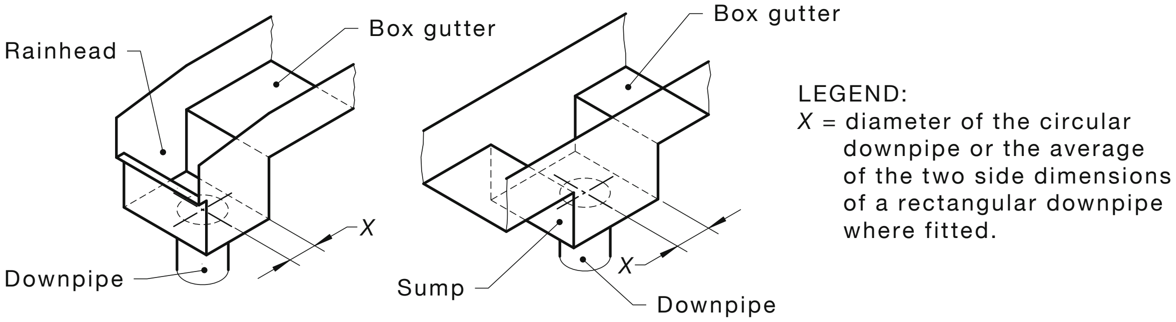
In addition to the recommendations of Clause 5.2, particular attention is to be paid to the following:

(a) *Sumps and rainheads* All box gutters to discharge through either a sump or a rainhead as shown in Figure 5.3.3.

(b) *Gutter discharge* Box gutters to discharge at the downstream end without change of direction (i.e. not to the side).

(c) *Outlet* Outlets from sumps and rainheads to be tapered to fit into the top section of downpipes.

(d) *Installation* To permit total drainage of the rainhead or sump, outlets to be installed so that, on completion of the installation, no part of the outlet is above the sole of the sump or rainhead.




NOTE: Sump overflow provision has been left out for clarity.

FIGURE 5.3.3 OUTLETS

### 5.3.4 Gutter support systems

#### 5.3.4.1 *General*

Gutter support systems are to be designed and manufactured so as to be able to support the entire weight of the gutter and sumps when full of water as well as a trafficable load at any point in the gutter and sumps and installed as follows:

(a) Supports are to be manufactured from materials compatible with the gutter.

(b) Supports are to be installed with a uniform fall of not less than 1 in 200 towards the outlets.

(c) Support systems are to be securely fastened to structures to resist all appropriate live and dead loads.

(d) Supports are to be spaced to prevent ponding or permanent deflection of the sole of the gutter.

#### 5.3.4.2 *Position and spacing*

Support brackets are to be spaced, positioned and fastened—

(a) at stop-ends;

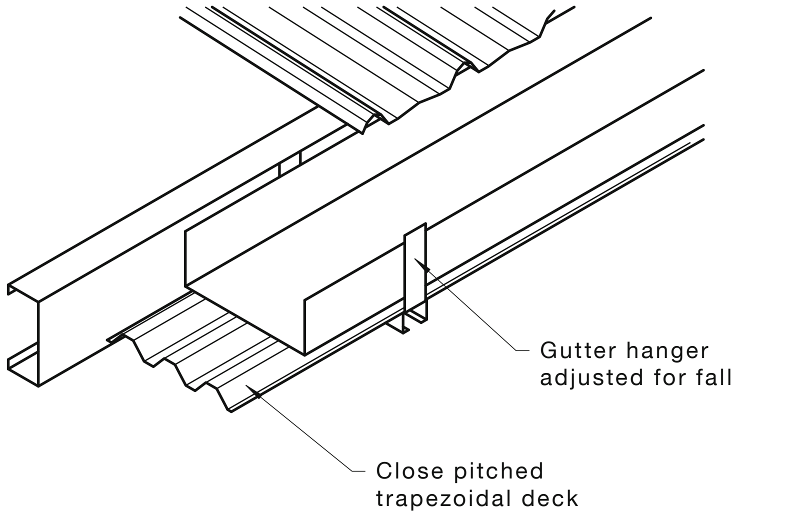
(b) at both ends of sumps;

(c) at rainheads;

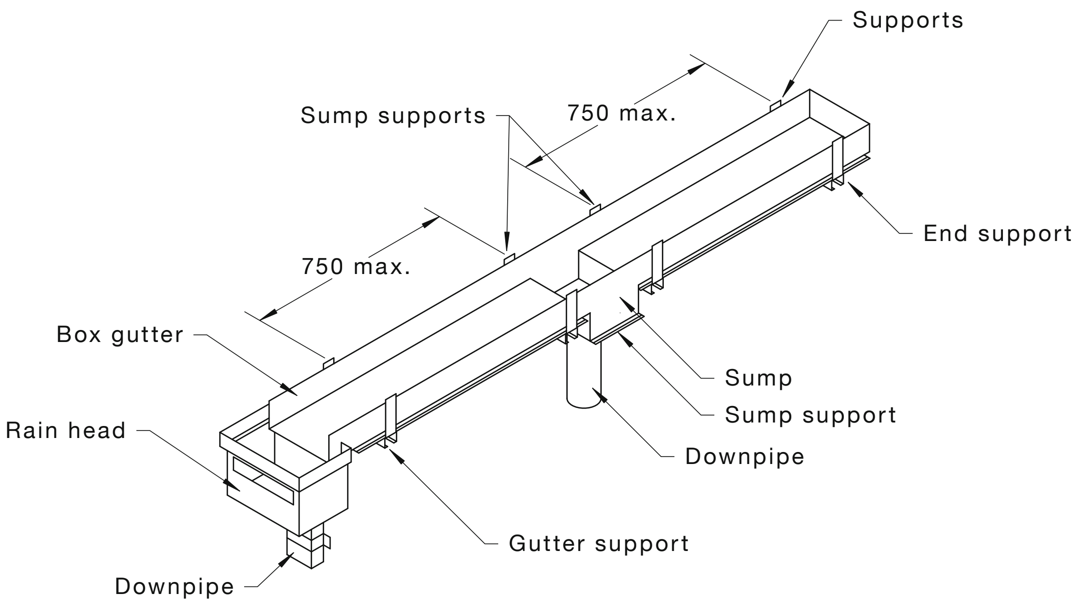
(d) between stop-ends at intervals not exceeding 750 mm for 300 mm and wider gutters, unless fully engineered; and

(e) so that the full sole of the gutter is supported with multi-ribbed metal roof sheeting or materials compatible with the gutter as shown in Figure 5.3.4(a).

The above will ensure the criteria of Clause 5.3.4.1 is accomplished.




(a) Support bracket detail




(b) Support spacing

NOTE: Sump overflow provision has been left out for clarity.

DIMENSIONS IN MILLIMETRES

FIGURE 5.3.4 SUPPORT BRACKETS

### 5.3.5 Workmanship

Gutters, sumps and rainheads are to be kept free of debris during installation. On completion of the installation, all debris is to be removed to prevent it entering the downpipes and stormwater drains.

## 5.4 VALLEY GUTTERS

In addition to the recommendations of Clause 5.2, particular attention is to be given to the following:

(a) *Shape* Valley gutters to be sized and manufactured in accordance with Figures 5.4(A) and 5.4(B).

(b) *Valley head* Valley head to be securely fastened at the ridge and turned up against the weather.

(c) *Discharge at eaves gutters* Valley gutters to enter eaves gutters 50 mm and have the ends turned down into the eaves gutter.

(d) *Fastening* Valley gutters to be fixed with galvanised nails bent over or cleated at the outer edges of the gutter at intervals not exceeding 500 mm.

(e) *Lap joints* Lap joints to have a lap of 25 mm in the direction of flow sealed and fastened.

(f) *Discharge width* The effective discharge width of the valley gutter on completion is to be a minimum 150 mm as shown in Figure 5.4(B).

(g) *Pierced roofs* Pierce-fastened roof covers to overlap the valley gutter by not less than 100 mm. Roof fastening screws to be clear of valley gutters.

(h) *Supports* Valley boards, if used to support the whole of the valley gutter, to have a thickness not less than 19 mm. Alternatively, valley supports may be suitably rigidised galvanised, aluminium/zinc alloy-coated or aluminium/zinc/magnesium-coated steel.

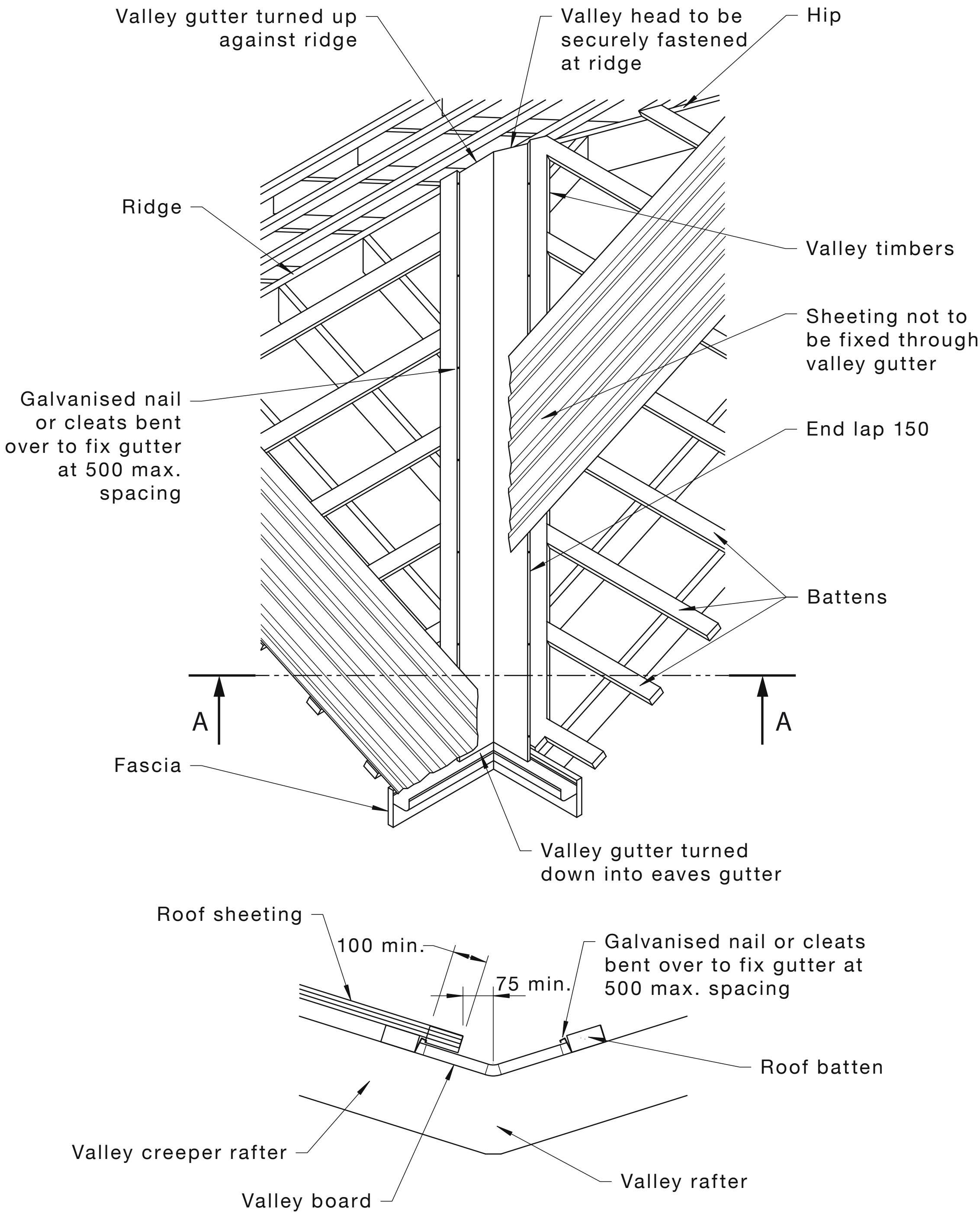
(i) *Grade* Valley gutters not to be installed on roof slopes less than 1:4.5 (12.5°).

(j) *Angle* Valley gutters to have a nominal side angle of 1:3.4 (16.5°).

(k) *Catchment* Valley gutters to be installed so as to service a catchment area not exceeding 20 $m^2$.

NOTE: Valley gutters discharging into eaves gutters should not overhang the gutter so as to make maintenance of the gutter impracticable.

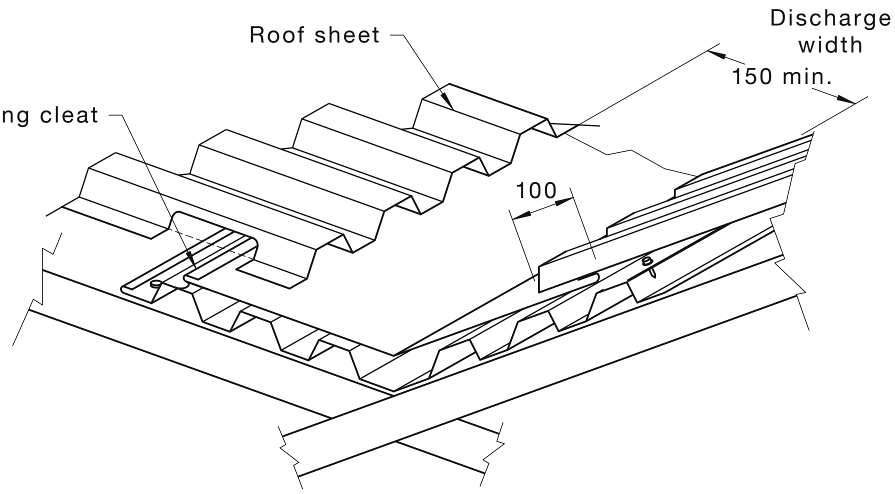
 




SECTION A-A

DIMENSIONS IN MILLIMETRES

FIGURE 5.4(A) VALLEY GUTTERS




DIMENSIONS IN MILLIMETRES

FIGURE 5.4(B) DISCHARGE WIDTH

## 5.5 SOAKER GUTTERS

In addition to the recommendations of Clause 5.2, particular attention is to be given to the following:

(a) *Support* The whole of the area around a penetration, including a soaker gutter or apron flashing to be supported so that the roof surface retains its former strength.

(b) *Drainage* Soaker gutters of adequate strength appropriately sized for the roof catchment area above the penetration to be installed so that all roof water is collected and drained into two or more full trays at the sides of the penetration. The waterway upstream of any penetration to be not less than 100 mm [see Figure 5.5(A)].

(c) *Stop-ending* The roof covering sheets on the lower side of the penetration to be stop-ended to the full height of the corrugations or ribs.

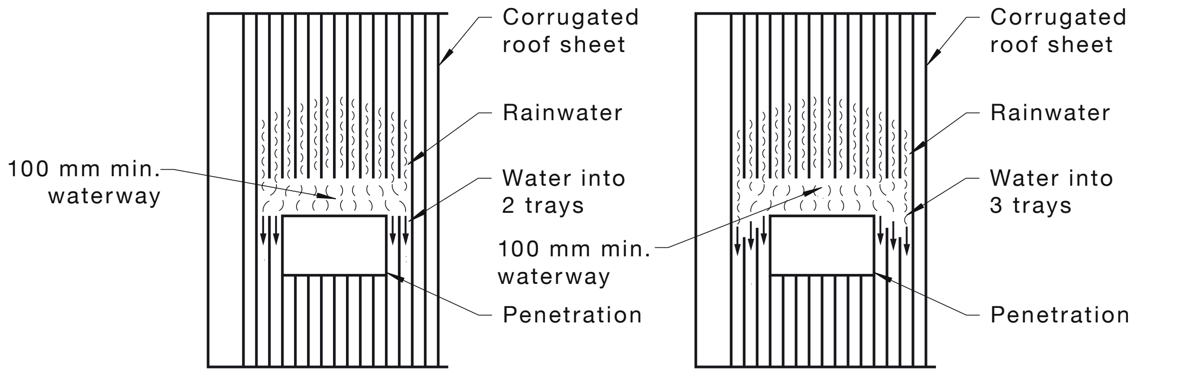
(d) *Rib sealing* Where ribs have been cut above the soaker gutter, the rib openings to be sealed and fastened as shown in Figure 5.5(B) or proprietary rib end closers used.

(e) *Clearance* The opening in the roof coverings and upstand to be sized so as to leave an annular gap of not less than 20 mm between the service and the penetration [see Figure 5.5(C)]. No service to be directly fastened to, or suspended from, the roof surface, soaker gutter or its apron flashing.

(f) *Apron* The penetration to be flashed with an appropriately sized soaker and apron flashing around its perimeter as specified in Clause 8.6.1(f).

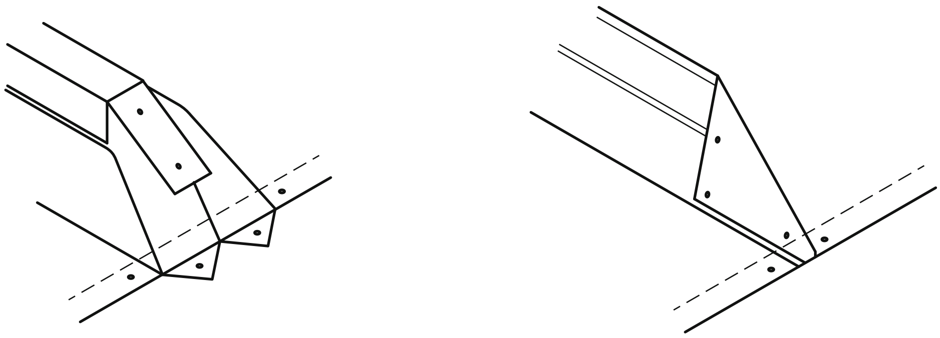
(g) *Upstand height* The minimum upstand height for a soaker gutter and apron flashing to be 100 mm. This is to allow for a minimum hanging flashing overlap of not less than 50 mm and a minimum clearance of not less than 25 mm from the hanging flashing and soaker gutter and or apron flashing.

(h) *Workmanship* On completion of the installation of the soaker gutter and apron flashing all debris to be removed and the penetration is to be weathertight.



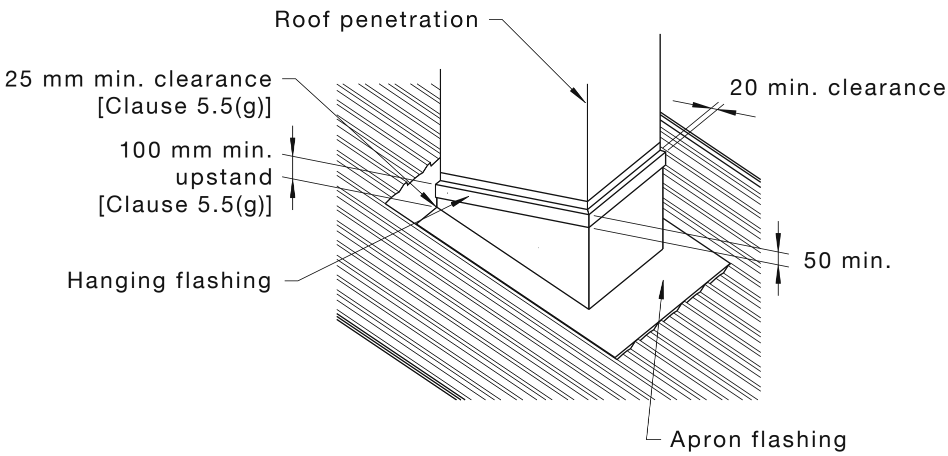

(a) Catchment up to 40 m$^2$ (b) Catchment 40 to 80 m$^2$

FIGURE 5.5(A) DIVERSION OF FLOW AROUND LARGE PENETRATIONS



(a) Wide rib (b) Narrow rib

FIGURE 5.5(B) RIB SEALING




DIMENSIONS IN MILLIMETRES

FIGURE 5.5(C) CLEARANCE GAP

## 5.6 EAVES GUTTERS

In addition to the recommendations of Clause 5.2, the following apply to eaves gutters, including any fascia-mounted eaves gutters, deck-mounted eaves gutter and concealed eaves gutters [see Figure 5.6(A)]:

(a) *Overflow provision and size* In the event of a total blockage of its downpipes or stormwater drains, eaves gutters to discharge all excess roof water clear of the building via overflows.

Eaves gutters with higher fronts than backs and eaves gutters that could potentially discharge overflow into any building to be provided with fixed overflows calculated in accordance with AS/NZS 3500.3.

(b) *Mitres* On completion, mitres fabricated from eaves gutter materials, to be true in size, shape and angle and have all beads reinforced with purpose-made gussets or custom-designed cast or pressed metal angles.

(c) *Maximum length* Eaves gutters to not exceed 20 m without provision of an appropriate expansion joint [see Figures 5.3.2(B) and 5.3.2(C)], and eaves gutters to be installed to permit expansion and contraction so as to protect cast-type bolted clamp angles where installed.

(d) *Cleating* Eaves gutters to be fastened using purpose-made fixings to permit longitudinal expansion and contraction [see Figure 5.6(B)].

The back of eaves gutters to be cleated by means of galvanised nails bent over to allow expansion and contraction of the gutter or snap/spring clips at 1200 mm centres.

(e) *Bracket spacing* Eaves gutter brackets to be securely fastened to the building structure or fascias at stop-ends and between stop-ends at intervals not exceeding 1200 mm.

(f) *Fall* Eaves gutter brackets to be installed with a uniform fall towards the outlets to achieve an effective gradient with no permanent ponding. A minimum fall of 1:500 will ensure complete drainage.

(g) *Roof cover discharge over eaves gutters* Roof covers discharging into an eaves gutter to enter the eaves gutter 50 mm. Flat pans or valleys to have the lower edge turned down 20 mm at 45° into the eaves gutter [see Figure 5.6(C)].

(h) *Lap joints* Lap joints to have a lap of 25 mm in the direction of flow and to be sealed and fastened.

(i) *Baffles* Where the low edge of the roof cover is subject to excessive direct wind forces, baffle flashings to be installed to prevent the entry of roof water under the roof cover.

(j) *Sarking* To enter the eaves gutter 25 mm [see Figure 5.6(C)].

NOTE: Care should be taken when installing insulation blanket to ensure that the blanket does not protrude into the gutter.

(k) *Outlets* Gutter outlets to be as follows:

(i) Gutters to have outlets of an appropriate size for the roof catchment area, spaced at intervals that permit the efficient, effective discharge of all roof water.

(ii) The minimum size of the outlet to be in accordance with Table 5.6.

(iii) Eaves gutter outlets to be tapered to fit inside the upper end of the downpipe.

(iv) Outlets to be fastened and sealed in accordance with Clause 5.8.

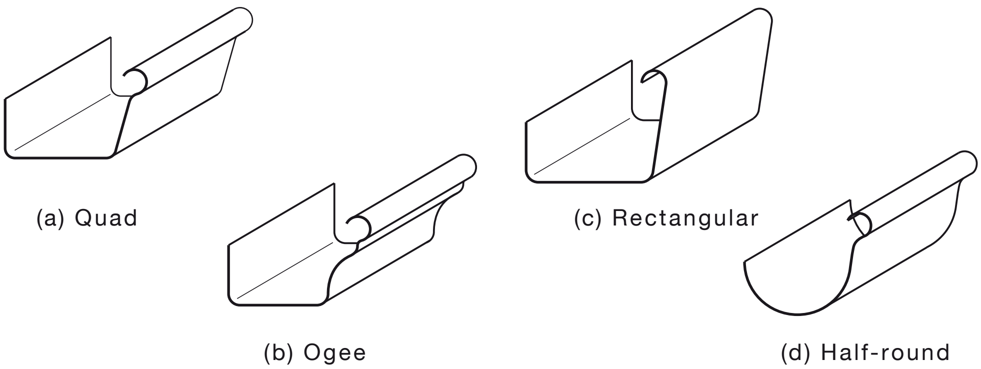



FIGURE 5.6(A) TYPICAL EXTERNAL EAVES GUTTERS

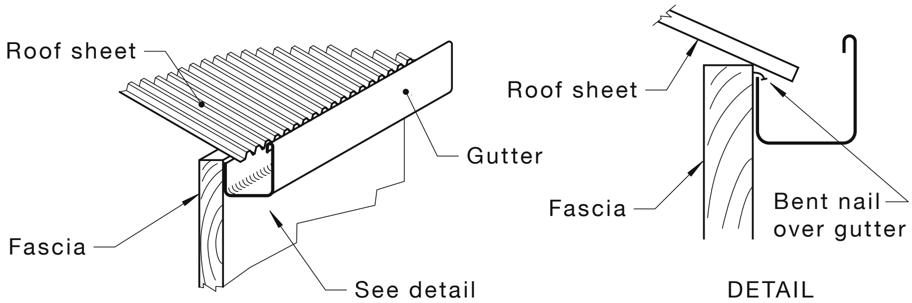
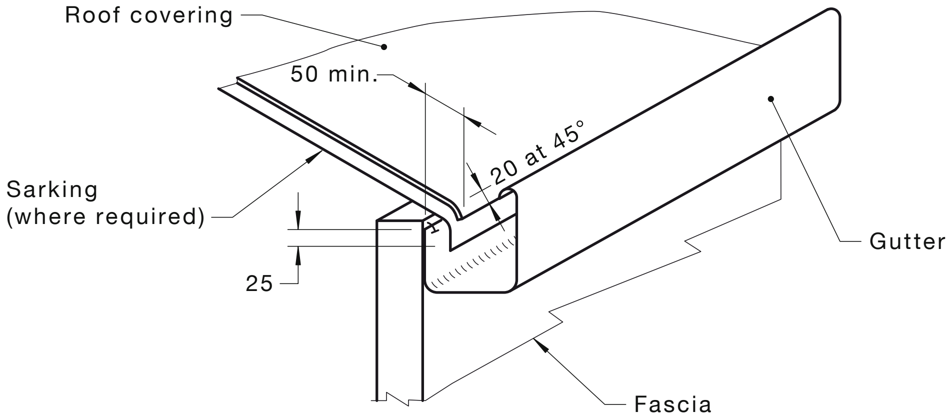



FIGURE 5.6(B) CLEATING




DIMENSIONS IN MILLIMETRES

FIGURE 5.6(C) EAVES GUTTER DETAIL

## TABLE 5.6

## EAVES GUTTER—SIZE OF VERTICAL DOWNPIPE

| Maximum effective cross-sectional area of an eaves gutter ($A_e$)* Nearest 100 mm² | | Internal size of vertical downpipe mm | |
|---|---|---|---|
| Gradient | | Cross-section | |
| 1:500 and steeper | Flatter than 1:500 | Circular | Rectangular or square |
| 3500 | 4700 | 65 | 65 × 50 |
| 4200 | 5600 | 75 | 65 × 50 |
| 4600 | 6200 | 75 | 75 × 50 |
| 4800 | 6400 | 80 | 75 × 50 |
| 5200 | 7000 | 80 | 100 × 50 |
| 5900 | 7900 | 85 | 100 × 50 |
| 6400 | 8600 | 90 | 100 × 50 |
| 6600 | 8900 | 90 | 75 × 70 |
| 6700 | 9000 | 100 | 75 × 70 |
| 8200 | 11 000 | 100 | 100 × 75 |
| 9600 | 12 900 | 125 | 100 × 75 |
| 12 800 | 17 100 | 125 | 100 × 100 |
| 12 800 | 17 200 | 150 | 100 × 100 |
| 16 000 | 21 500 | 150 | 125 × 100 |
| 18 400 | 24 700 | 150 | 150 × 100 |
| 19 200 | 25 800 | — | 150 × 100 |
| 20 000 | 26 800 | — | 125 × 125 |

* See AS/NZS 2179.1 and AS/NZS 3500.3.

NOTES:

1 This table assumes—

(a) an effective width to depth is a ratio of about 2:1;

(b) a gradient in the direction of flow, 1:500 or steeper;

(c) the least favourable positioning of the downpipe and bends within the gutter length;

(d) a cross-section or half-round, quad, ogee or square; and

(e) the outlet to a vertical downpipe is located centrally in the sole of the eaves gutter.

2 The required eaves gutter discharge areas do not allow for loss of waterway due to internal brackets.

## 5.7 DOWNPIPES, SUMPS AND RAINHEADS

### 5.7.1 General

AS/NZS 3500.3 sets out requirements for the sizing of downpipes, sumps and rainheads. For best practice, the recommendations of Clause 5.2 should be taken into consideration.

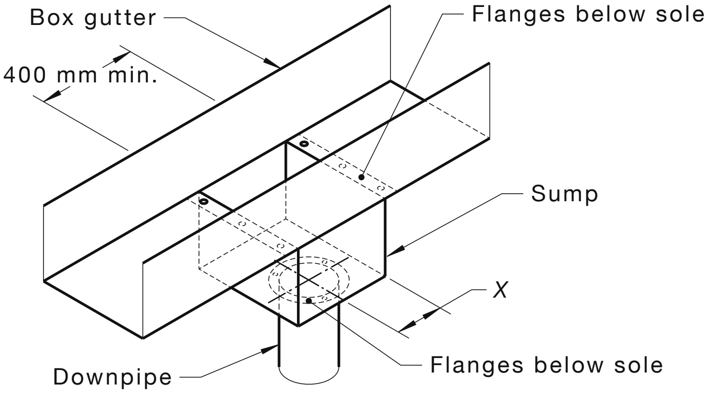
### 5.7.2 Sumps

The sump is to be sealed to the box gutter on all sides (see Figure 5.7.2). The centre-line of the downpipe is to be no further from the nearest vertical side of the sump than either—

(a) the diameter of a circular downpipe; or

(b) the average of the two side dimensions of a rectangular downpipe (see Figure 5.7.2).

Sumps are to be appropriately sized in accordance with AS/NZS 3500.3 with a minimum length of 400 mm and when fitted with a high capacity overflow device with a depth of not less than 150 mm when either one or two box gutters enter the sump.




NOTES:

1 Dimension *X* not greater than downpipe size (see Clause 5.3.3).

2 Overflow, sump and gutter support have been left out for clarity.

FIGURE 5.7.2 SUMP

### 5.7.3 Rainheads

The purpose of a rainhead is to ensure that, in the event of a blockage or extreme flow conditions, all excess flow is discharged to the outside of the building. The centre-line of the downpipe is to be not further from the nearest vertical side of the rainhead than either—

(a) the diameter of a circular downpipe; or

(b) the average of the two side dimensions of a rectangular downpipe (see Figure 5.7.3).

The width of the rainhead is to be at least equal to the width of the box gutter.

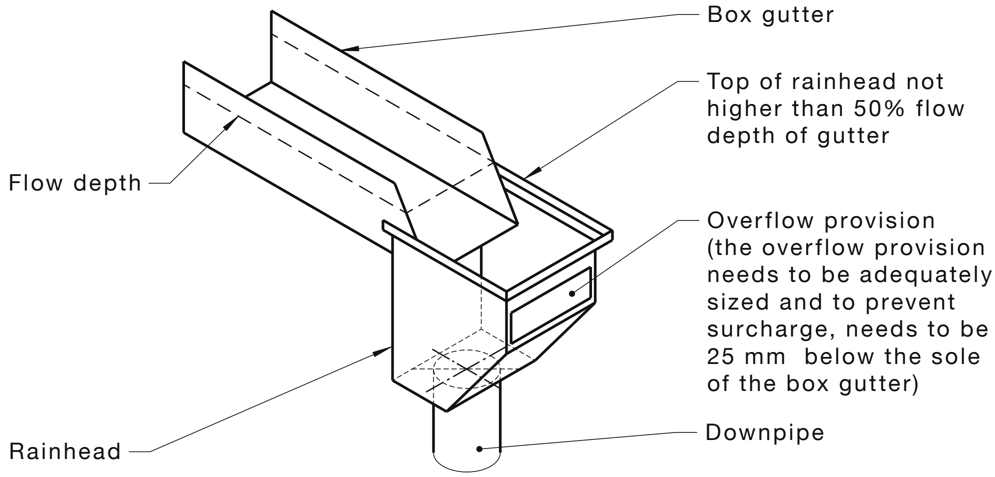



FIGURE 5.7.3 RAINHEAD

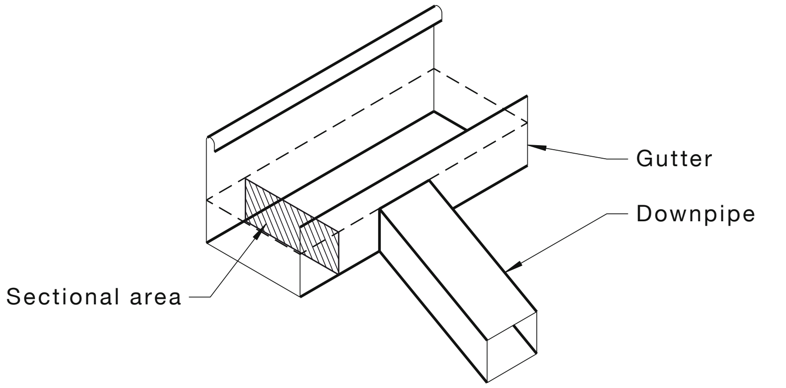
### 5.7.4 Eaves gutter outlets

All parts of eaves gutter outlets, including flanges, sealants and fasteners installed to the sole of the eaves gutter are to be level with or below the sole of the gutter.

### 5.7.5 Back outlets

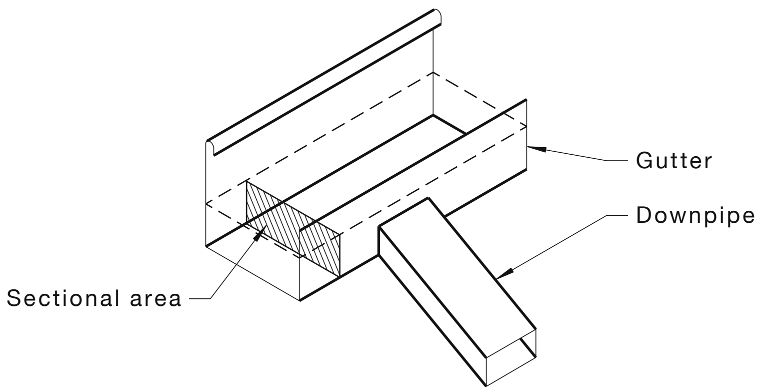
An outlet at the back of an eaves gutters with a fall of less than 1 in 25 is to be of a size equal to the effective cross-sectional area of the eaves gutter it drains (see Figure 5.7.5).

An outlet at the back of an eaves gutter with a fall of equal to or greater than 1:25 is to be not less than 50% of the effective cross-sectional area of the eaves gutter it drains, and where vertical, sized in accordance with Table 5.6.




Area of downpipe = sectional area of gutter

(a) Fall of downpipe <1:25




Area of downpipe = 50% sectional area of gutter

(b) Fall of downpipe ≥1:25

FIGURE 5.7.5 BACK OUTLETS

### 5.7.6 Downpipes

For the installation of downpipes, see AS/NZS 3500.3. In addition, for best practice, the following are to be observed:

(a) *Connection* Downpipes to be connected to the base of rainheads or sumps.

(b) *Internal gutters* Downpipes from internal gutters to be connected to sumps with appropriate overflow provision, or discharged into external rainheads with appropriate overflow provision.

(c) *Fastening* Downpipes to be securely fastened to walls and structures so as to withstand movement due to thermal expansion or weight due to a partial or total blockage.

(d) *Spacing* Downpipe brackets to be spaced at intervals not exceeding 2 m vertically and 1 m on a slope (see Figure 5.7.6).

NOTE: Support systems for metal downpipes should comply with AS/NZS 2179.1.

(e) *Thickness of brackets* Brackets to be manufactured of materials compatible with downpipe materials and of thicknesses not less than that of downpipes. Downpipe brackets to be sized to permit thermal expansion and contraction.

(f) *Fixed points on PVC downpipes* PVC downpipes to have provision for expansion or expansion joints at spaces not exceeding 6 m. The expansion joints or couplings to be fixed to prevent lateral movement.

(g) *Slip joints on external downpipes* External downpipes manufactured of sheet metal materials to incorporate unsealed slip joints of 75 mm within 2 m of the point of entry to the stormwater drain.

(h) *Graded downpipes for eaves gutters* Downpipes other than vertical downpipes to have a minimum gradient of 1 in 25. Downpipes with gradients less than 1 in 25 to have a minimum cross-sectional area equal to the effective eaves gutter cross-sectional area it serves.

(i) *Seams* Downpipes manufactured of sheet metal materials with seams to have the seams positioned uppermost on graded downpipes and concealed from front view on all other downpipes.

(j) *Concealed downpipes* Concealed downpipes or downpipes with limited access made of sheet materials with seams, to have such seams sealed. Concealed downpipes to be made of copper, copper alloy, stainless steel or PVC. Incompatibility with gutter material to be avoided and a downpipe pop constructed from a neutral material may be necessary.

(k) *Joints in PVC* Joints in downpipes made of PVC to be socket and spigot joints sealed as given in Clause 5.8.9.

(l) *Joints in sheet metal* Straight joints in downpipes made of sheet metal to be lapped not less than 50 mm in the direction of flow.

(m) *Jointing* Joints in downpipes to be as stated in Clause 5.8 and the following:

  (i) Laps in offsets and branches to be manufactured so as to avoid obstruction to rainwater flow and lapped not less than 6 mm.

  (ii) Circular downpipes to have their offset joints lapped not less than 6 mm.

  (iii) Straight, on-grade downpipes to be sealed.

(n) *Overflow protection* Downpipes directly connected to stormwater drains to be installed so as to ensure the stormwater drain has overflow protection.

NOTE: The stormwater overflow relief should not cause flooding to habitable or storage areas and vehicle garages.

(o) *Heavy duty applications* Downpipes exposed to heavy vehicular traffic to be appropriately protected or manufactured from appropriate materials such cast iron or galvanised wrought iron.

(p) *Shoes* Downpipes connecting to earthenware stormwater drains, or where stormwater drains are installed so as to be mounted off the wall, to have shoes installed at an angle not exceeding 30° to the vertical at the base of the downpipe.

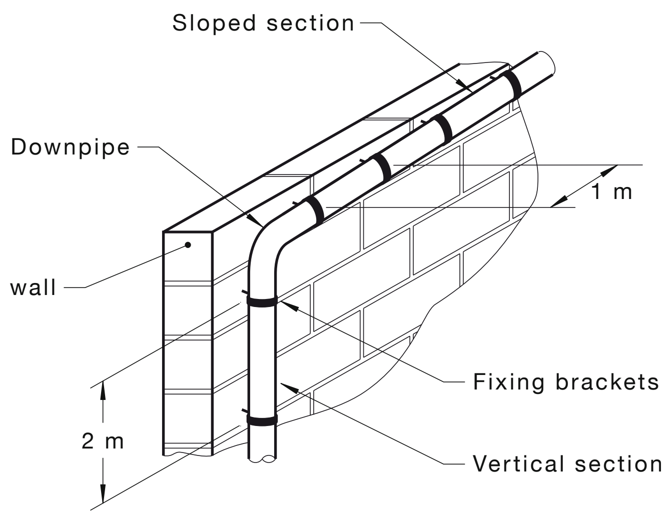



FIGURE 5.7.6 DOWNPIPE FASTENINGS

### 5.7.7 Spreaders

Spreaders may be used to drain rainwater from a higher roof surface with a catchment area not exceeding 15 m$^2$ provided the following conditions are satisfied (see Figure 5.7.7):

(a) When discharging onto a tiled roof, the lower section is sarked a minimum width of 1800 mm, either side of the point of discharge extending down to the eaves gutter.

(b) When discharging onto a corrugated roof, a minimum width of 1800 mm on either side of the point of discharge is sealed for the full length of the side laps.

(c) The increased roof water volume from the upper roof is not to enter any seam of the roof coverings of the lower roof.

(d) No spreader is to discharge roof water onto or over ridge tiles, mortar jointed tiles, flashings, timber fascia or a roof sheet side lap.

(e) No spreader is to have its discharge entering any part of any building.

(f) Spreaders are to discharge all roof water onto roof coverings in the direction of flow, avoiding discharging onto laps on lower roof sheets and tiles.

When discharging an upper roof catchment onto a lower roof, the total roof area including the additional upper roof catchment area is to be considered for inclusion when sizing the lower roofing, gutters and downpipes.

(g) Spreaders do not discharge on sheets or tiles discharging to valleys.

NOTE: The spreader catchment area indicated above is based on a standard corrugated roofing profile and may only exceed 15 m$^2$ provided the additional upper roof discharge does not exceed the lower roof profile manufacturer’s design-carrying capacity.

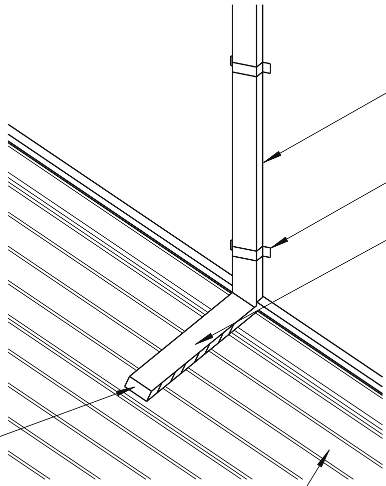



FIGURE 5.7.7 SPREADERS

## 5.8 JOINTING METHODS FOR OTHER THAN ALUMINIUM

### 5.8.1 General

Soldered or silicone-sealed joints are to be flush and lapped 25 mm in the direction of the fall, with one row of rivets or other approved type fastenings at spacings not exceeding 40 mm. This applies to all joints in flashings, cappings, eaves, gutters, box gutters, soakers, penetrations and all general sheet metal joints in roof plumbing.

### 5.8.2 Soldered joints

#### 5.8.2.1 *Preparation of joint surfaces*

The surface area to be jointed is to be clean and free from oils, grease and rust.

#### 5.8.2.2 *Fluxing for soldering*

Immediately after being cleaned, the surfaces to be jointed are to be prepared with the appropriate flux as set out in Table 5.8.

NOTE: For health and safety reasons, care should be taken during any soldering operation.

**TABLE 5.8**

**SUITABILITY OF FLUXES**

| Material to be joined | Type of flux |
|---|---|
| Zinc-coated steel | Zinc chloride or diluted hydrochloric acid —3 parts water (muriatic acid) |
| Copper | Zinc chloride (killed spirits) |
| Stainless steel | Phosphoric acid |
| Zinc | Zinc chloride (killed spirits) |
| Lead | Candle wax |

NOTE: Aluminium/zinc and aluminium/zinc/magnesium alloy-coated steel cannot be soldered.

**5.8.2.3** *Soft soldering*

Joints in zinc-coated steel, copper, stainless steel and sheet lead flashings and cappings are to be fastened at intervals not exceeding 40 mm and sweat-soldered with 50/50 lead/tin solders.

NOTE: For improved appearance, joints in stainless steel may be soldered with 60/40 or 80/20 solders.

**5.8.2.4** *Cleaning joints after soldering*

Flux-affected areas are to be thoroughly cleaned with a clean wet cloth to remove all traces of flux. Damaged zinc coating is to be coated with heavy-duty zinc paint.

### 5.8.3 Fastening and sealing with sealants

Joints in aluminium/zinc, aluminium/zinc/magnesium alloy-coated steel, prepainted steel flashings and cappings are to be fastened at intervals not exceeding 40 mm and silicone sealed with neutral cure silicone sealants.

Sealants are to be sandwiched between the laps of the joint to provide a positive seal and to protect the silicone sealant from exposure to ultraviolet radiation (see Figure 2.9).

### 5.8.4 Fastening and sealing with rivets for steel sheets

Rivets are to be of plugged or waterproof aluminium alloy or zinc-coated steel sealed over with solder or neutral curing silicone sealant. While steel rivets need to be soldered over, this is not necessary if the waterproof aluminium rivets are located so they pass through wet silicone rubber sealant.

### 5.8.5 Fastening and sealing with rivets for stainless steel

Rivets are to be of stainless steel or Monel sealed with solder or neutral curing silicone sealants.

### 5.8.6 Fastening with spot welds

Rivets may be substituted with spot welds.

### 5.8.7 Fastening and sealing with welded joints

Joints in aluminium, stainless steel and copper may be welded using the tungsten-inert gas welding process, provided the thickness of the metals permits such welding. After welding, the joint and gutter profile is to not be deformed.

### 5.8.8 Fastening with aluminium and zinc-alloy diecasts

Rainwater goods, including internal and external angle joints, stop-ends, joiners and outlets may be jointed with aluminium castings or zinc alloy pressure diecast accessories provided such fittings are installed to comply with manufacturer's instructions.

### 5.8.9 Sealing PVC downpipes

For best practice, the PVC downpipes are to be sealed as follows:

(a) Cut pipe ends square, and clean both spigot and sockets with cleaning fluid.

(b) Coat both spigot and socket with solvent cement.

(c) After joining, keep the joint immobile until solvent cement sets.

(d) Remove excess solvent cement.

### 5.8.10 Performance of joints

On completion of the installation, all joints are to be watertight and not prohibit water flow.

## 5.9 JOINTING METHODS FOR ALUMINIUM RAINWATER GOODS AND ACCESSORIES

### 5.9.1 Mechanically fastened lap joints

Silicone sealed joints are to be flush and lapped 25 mm in the direction of the fall, with one row of rivets at spacings not exceeding 40 mm. Fasteners are to be rivets or bolts in aluminium alloys or 300 series stainless steel. All joints are to be made watertight with neutral cure silicone sealants.

### 5.9.2 Brazed joints

Brazed joints are to have a minimum lap and be brazed with aluminium/silicon alloy filler containing 10% to 13% silicon. After brazing, all traces of flux are to be removed by washing with water to prevent subsequent corrosion of the joint by residual flux.

NOTE: Lower melting point silicon aluminium brazing alloys containing copper cannot be used for brazing aluminium.

### 5.9.3 Soldered joints

The use of soldered joints for joining aluminium rainwater goods in the presence of moisture is to be avoided, as both soft and hard solders will suffer corrosion at the joint.

### 5.9.4 Welded joints

Welded joints are to be either GMAW (MIG) or GTAW (TIG) type in accordance with AS/NZS 1665 and all assemblies shop fabricated. The use of field welding is to be avoided, unless the joints are fully protected from air movement and moisture. After welding, the joint and gutter profile is to not be deformed.

### 5.9.5 Jointing with aluminium and zinc-alloy castings

Rainwater goods, including internal and external angle joints, stop-ends, joiners and outlets may be joined with aluminium castings or zinc alloy pressure diecast accessories provided such fittings are installed to comply with manufacturer’s instructions.

## 5.10 FIXING AND JOINING ROOF FLASHINGS AND CAPPINGS

As aluminium expands twice as much as steel for the same temperature change, greater attention is to be given to the correct procedure for the installation and provision for thermal movement of aluminium roof flashings and cappings. For this reason, the relevant manufacturer’s technical literature is to be referred to for details as to method of fixing roof flashings and cappings, length of lap and how to provide for thermal movement associated with the particular roofing profile to be installed. Standard practices for installing steel roof flashings are not to be used for aluminium roof flashings.

## 5.11 RAINFALL INTENSITIES

### 5.11.1 General

Consideration of rainfall intensities is very important to ensure that the installed roof is designed to prevent the ingress of rainwater into the building.

The greater the intensity of rainfall in a given period of time, the more it is essential that the roof system (including rainwater goods) has a greater capacity to convey the rain into the stormwater system (which in some cases means to ground).

Precautions are to be taken to prevent the accumulation of debris and, if hail is common, from hailstones impeding the flow of rainwater.

### 5.11.2 Average recurrence interval (ARI)

The appropriate recurrence intervals are as follows:

(a) For external eaves gutters ........................................................................ 20 ari.

(b) For external eaves gutter overflow provision.................................................. 100 ari.

(c) For internal gutters .......................................................................... 100 ari.

(d) For internal eaves gutter overflow provision .................................................. 100 ari.

# SECTION 6 THERMAL AND CONDENSATION CONTROL

## 6.1 SCOPE OF SECTION

This Section provides guidance on the various methods of insulating metal roofs, including assessing the amount of insulation required and controlling condensation.

## 6.2 INSULATION OF STEEL ROOFING

Steel roofing systems are used for a wide range of building types, as follows:

(a) Houses.

(b) Supermarkets.

(c) Factories and warehouses.

(d) Offices.

(e) Farm buildings including buildings for animals and birds.

In each case, details of the roof systems should be determined to provide adequate performance in respect of energy conservation and condensation of atmospheric moisture.

Insulation is essential to reduce fuel bills, control condensation, retard heat flow and reduce unwanted noise created by rain and expansion and contraction of the roof.

Rain noise is particularly noticeable with metal roofing and it is strongly recommended that blanket insulation be pressed up tightly against the underside of the sheeting to minimise noise caused by rain and hail. This then serves the dual requirements of thermal insulation and sound reduction.

Every new metal roof construction should have an insulation blanket with an appropriate R-value and a suitable vapour barrier incorporated in the original design. (It is difficult and expensive to insulate a flat or raked roof area after construction.)

## 6.3 CONDENSATION

The roof is the most exposed element of the building fabric. Overnight its temperature often drops below the outdoor air temperature as it loses heat via radiation to the cold night sky. The cold roof combined with the vast amount of moisture generated at times within homes makes condensation within roofs likely. While it is good practice to minimise the risk of condensation, significant problems will usually only eventuate if condensation is persistent or becomes trapped. The risk of persistent condensation in roofs is greatest in the coolest climate zones (NCC climate zones 7 and 8) although persistent condensation can also eventuate in all climate zones when high internal moisture levels are not controlled. [Reference: ABCB handbook, *Condensation in Buildings Handbook*].

The underside of metal deck roofs provide conditions under which condensation of water vapour can occur. Insulation, such as sarking, reflective foil or blanket and foil placed beneath the sheeting are of value in reducing the likelihood of condensation forming and potentially dripping from under roof sheeting. When blanket insulation is placed beneath a metal deck roof it should incorporate foil or similar to provide a vapour barrier underneath the blanket insulation to prevent moisture from inside the building penetrating and wetting the blanket insulation.

It is commonly thought that condensation under metal roofing can be eliminated by liberal ventilation of the roof space. Whilst this can help any condensation to dry quickly avoiding persistent condensation, it will not eliminate condensation that may form from time to time on clear still nights. In line with the ABCB handbook, *Condensation in Buildings Handbook*, a minimum of sarking or reflective foil laminate is recommended in climates with cold winters (NCC climate zones 4, 7 and 8), noting that blanket and foil will provide greater protection.

Uninsulated steel roof systems may be unsuitable for farm buildings used for intensive husbandry of animals or birds. These buildings, because of their occupation and purpose, may contain air of a high relative humidity together with high concentrations of ammonia and organic acids. When these occur in the water condensed on the underside of the roof, rapid corrosion will occur.

NOTE: Aluminium roofing for farm buildings may be used without an insulation/moisture barrier, although the buildings benefit from its inclusion both thermally and protection against condensation.

In buildings without ceilings, such as factories, condensed water may fall on the machinery or stock in the space below and consideration should be given to the inclusion of insulation beneath the metal deck roof to reduce this risk.

NOTES:

1 For further details on condensation in buildings and metal roofs, see ABCB handbook, *Condensation in Buildings Handbook*.

2 The information above may not be applicable to the tropics or homes that are built to meet bushfire requirements. Please refer to the ABCB handbook, *Condensation in Buildings Handbook* for further information.

3 Specialist advice should be sought for buildings that are not constructed in typical environments and or for special purpose buildings, such as where high humidity may exist, e.g. swimming pools or where high levels of environmental control are required, e.g. museum.

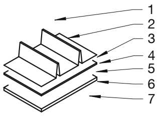
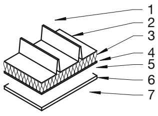
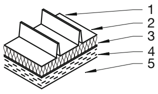
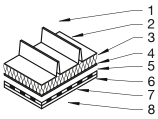
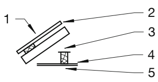
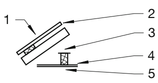
## 6.4 RESISTANCE TO HEAT TRANSFER (R-VALUES)

In selecting the thickness of the insulation blanket to give the necessary resistance to heat flow (R-value) it should be noted that where the blanket will be compressed by the roof sheeting for rain noise reduction, the R-value will be reduced and other devices such as reflective air spaces may need to be added to achieve the necessary R-value for the roof system.

Table 6.4(A) gives the R-value for various roof systems as a guide only. Note that blanket compression has not been taken into account in these examples. The various types of insulation available and their applicability are given in Table 6.4(B). To design a roof system to achieve a specific R-value, refer to the NCC.

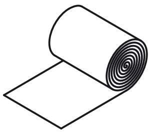
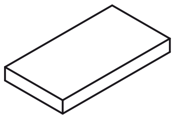
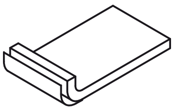
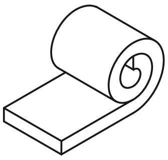
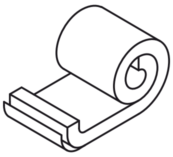
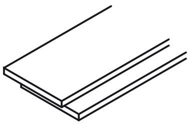
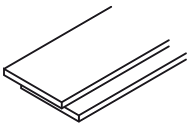
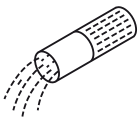
## TABLE 6.4(A)

## THERMAL RESISTANCE (R) OF FLAT AND SLOPING ROOFS FOR DIRECTION OF HEAT FLOW

| Type | Construction | Component | | Thermal resistance (R) $m^2.K/W$ (up winter) | Thermal resistance (R) $m^2.K/W$ (down summer) |
|---|---|---|---|---|---|
| | | Key | Material | | |
| Metal deck, reflective foil laminate, gypsum plasterboard |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal deck | 0 | 0 |
| | | 3 | Reflective foil laminate covered with dust | 0 | 0 |
| | | 4 | Airspace, 25 mm | 0.30 | 0.37 |
| | | 5 | Reflective airspace, 100 mm | 0.36 | 1.42 |
| | | 6 | Gypsum plasterboard, 13 mm | 0.08 | 0.08 |
| | | 7 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 0.88 | 2.06 |
| Metal deck, bulk insulation, reflective foil laminate, gypsum plasterboard |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal deck | 0 | 0 |
| | | 3 | Bulk insulation | 1.50 | 1.50 |
| | | 4 | Reflective foil laminate | 0 | 0 |
| | | 5 | Airspace, 25 mm | 0.36 | 1.42 |
| | | 6 | Reflective airspace, 100 mm | 0.08 | 0.08 |
| | | | Gypsum plasterboard, 13 mm | 0 | 0 |
| | | 7 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 2.20 | 3.03 |
| Metal deck, bulk insulation, compressed straw (faced) |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal deck | 0 | 0 |
| | | 3 | Bulk insulation | 1.50 | 1.50 |
| | | 4 | Compressed straw | 0.62 | 0.62 |
| | | 5 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 2.26 | 2.31 |
| Metal deck, bulk insulation, reflective foil laminate, bulk insulation, gypsum plasterboard |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal deck | 0 | 0 |
| | | 3 | Bulk insulation | 1.50 | 1.50 |
| | | 4 | Reflective foil laminate | 0 | 0 |
| | | 5 | Airspace, 50 mm | 0.42 | 1.00 |
| | | 6 | Glass fibre | 1.30 | 1.30 |
| | | 7 | Gypsum plasterboard | 0.08 | 0.08 |
| | | 8 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 3.44 | 4.07 |
| Galvanised iron, softboard |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal roof | 0.00 | 0.00 |
| | | 3 | Roof space | 0.18 | 0.28 |
| | | 4 | Softboard, 12.7 mm | 0.25 | 0.25 |
| | | 5 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 0.57 | 0.72 |
| Aluminium roofing softboard |  | 1 | Outdoor air film | 0.03 | 0.03 |
| | | 2 | Metal roof | 0.00 | 0.00 |
| | | 3 | Roof space | 0.29 | 0.37 |
| | | 4 | Softboard, 12.7 mm | 0.25 | 0.25 |
| | | 5 | Indoor air film | 0.11 | 0.16 |
| | | *Ro* | Total thermal resistance | 0.68 | 0.81 |

## TABLE 6.4(B)

## INSULATION TYPES AND APPLICABILITY

| Types of insulation suitable for application | | Application | |
|---|---|---|---|
| | | Ceilings pitched roofs | Cathedral ceilings and metal deck roofs |
|  | Reflective foil laminate (RFL)—Rolls | ✓ | ✓ |
|  | Fibreglass batts | ✓ | ✓ |
|  | Fibreglass batts foil facing | ✓ | ✓ |
|  | Fibreglass blankets | | ✓ |
|  | Fibreglass blankets with RFL facing | | ✓ |
|  | Expanded polystyrene (EPS) | | ✓ |
|  | Extruded polystyrene boards | | ✓ |
|  | Loose fill material | ✓ | ✓ |

# SECTION 7 ROOF SHEETING

## 7.1 TYPICAL PROFILES

### 7.1.1 General

This Section is not restricted to existing profiles. There are many variations with different fixing methods, and new decks are appearing regularly. Therefore, only the common and typical profiles are considered (see Figure 7.1).

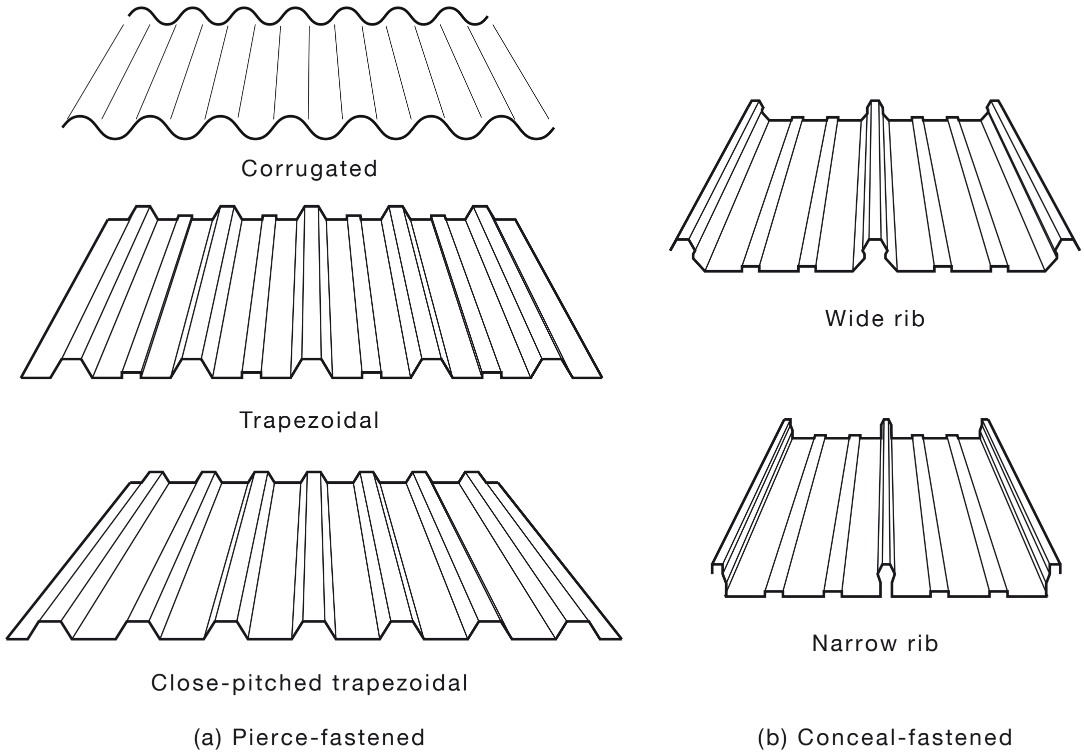



FIGURE 7.1 TYPICAL PROFILES

### 7.1.2 Insulated roof panels

An insulated panel is any laminated panel that is manufactured from different materials, permanently bonded together so that they act as a single element (see Appendix B for further information).

A typical description of an insulated panel (sandwich panel or composite panel) refers to building cladding panels having metal facings to both surfaces, with an insulation core that completely fills the space between the two facings and is continuously and permanently bonded to both.

Laying sequences and fixing methods and patterns are sometimes different to traditional built-up roofing systems and the panel manufacturer should be consulted prior to installation.

Practitioners/installers are to refer to manufacturer's specifications when selecting, installing and working safely with these products.

## 7.2 ROOF SHEET SPAN INFORMATION

The roof sheet support spacing for the various profiles are listed in the typical span information of the manufacturer's product literature. This information varies depending on whether it is an end span, internal span or overhang (see Figure 7.2).

The typical span charts are based on the ability of sheeting to withstand the wind loads specified in AS/NZS 1562.1 and the NCC. All sheeting is to be fixed to each and every roof support unless otherwise approved by a structural engineer.

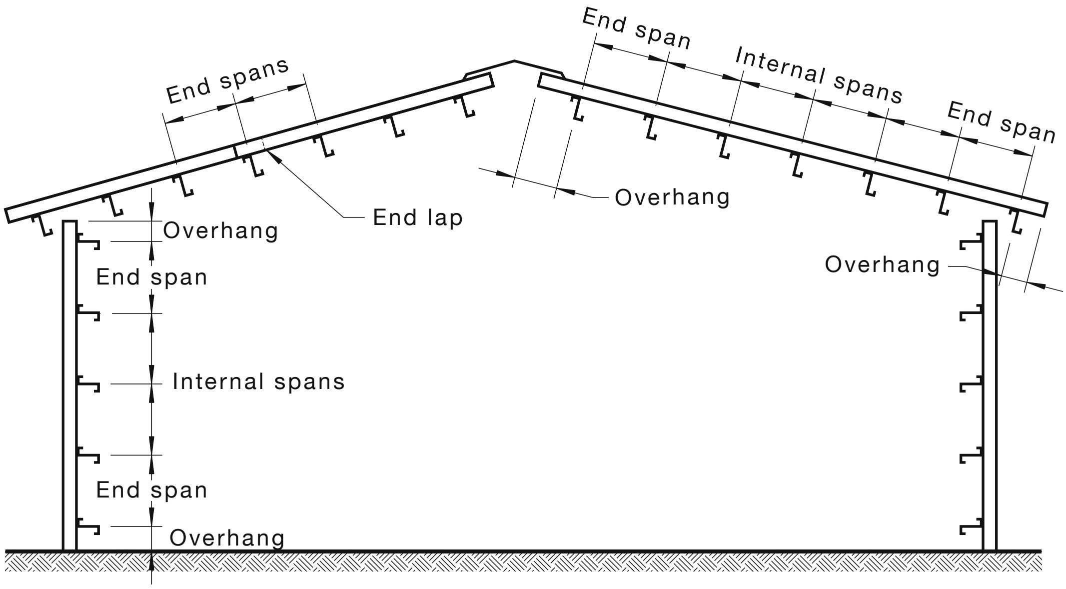



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FIGURE 7.2 EFFECTIVE SPANS

## 7.3 WIND FORCES ON ROOFS

Winds create considerable forces on both the upper surface and the underside of a roof cladding system. These forces may take the form of positive pressure or suction (negative pressure) and are to be considered in the overall design and fixing of a roof. Generally, the greatest wind forces imposed on roofs are due to suction, tending to lift the roof cladding from its framing and the entire roof structure from its supports. As the dead weight of roofing materials is relatively small, the suction forces are to be resisted by the roof fastenings.

Before replacing the roof sheeting on an existing structure, it is important to check the condition of the roof framing and fixings. Framing members that are structurally unsound are to be replaced and in, some cases, the existing members or fixings may need to be upgraded to meet local building regulations. The local building regulatory authority will need to be consulted to determine their requirements for this type of work before commencing any work.

## 7.4 CYCLONE AREAS

### 7.4.1 Testing requirements

For applications in cyclone areas, the loading or performance capacity of the respective products are to be determined from testing to AS 1562.1.

### 7.4.2 Installation

In general, the same products and installation procedures previously detailed for normal conditions are also applicable for cyclonic conditions.

Cyclone washers or caps for pierce-fastened decks are recommended to improve the performance of the sheeting and also give greater allowance for thermal expansion (see Figure 7.4.2).

For aluminium pierce-fastened roofing, the fixing screws are to be fitted with a washer system similar to that illustrated in illustration (c) of Figure 7.4.2 for both normal and cyclonic wind regions. For support spacings, the manufacturer's technical data is to be consulted.

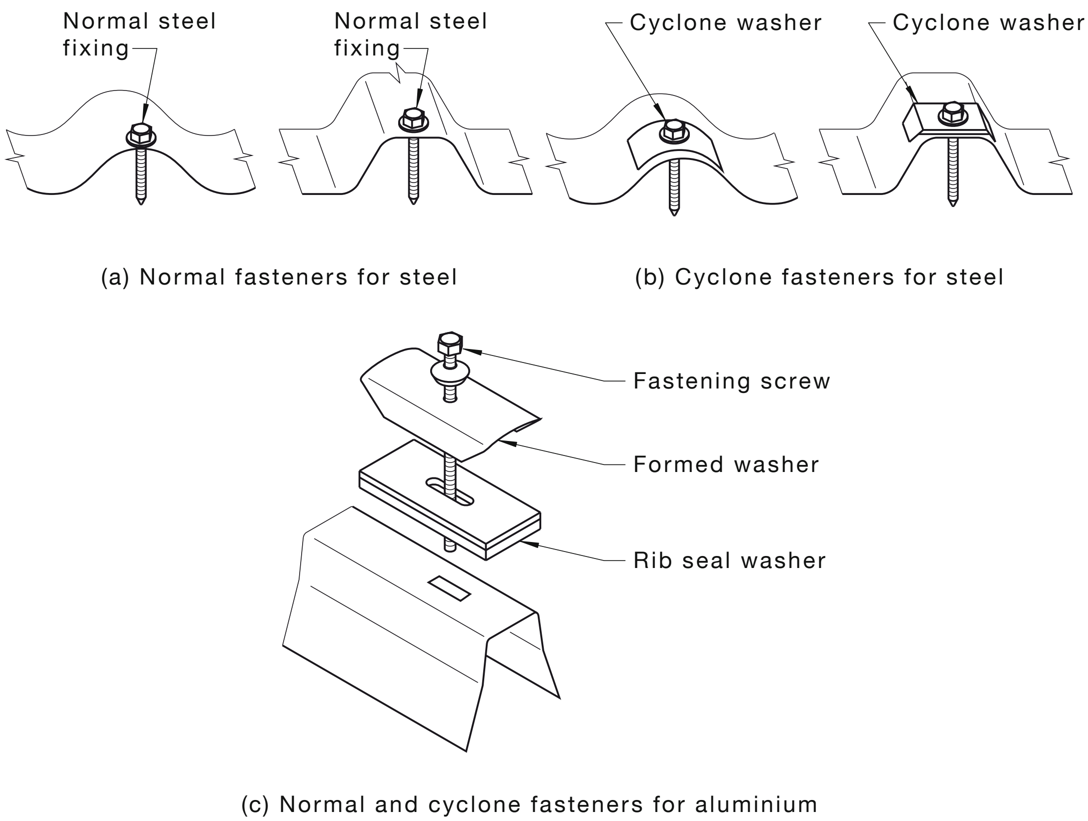



FIGURE 7.4.2 FIXING DETAILS

## 7.5 STANDARDS AND PERFORMANCE TESTS

AS 1562.1 and the NCC specify the performance requirements for a metal roofing system.

These requirements include the ability of the roof to resist wind uplift forces and concentrated loads with an appropriate factor of safety. AS 1562.1 requires that compliance of steel roofing products be checked by stringent proof tests, carried out in accordance with a specified procedure. All specified products are to comply with AS 1562.1 in all respects.

## 7.6 ROOF SLOPE AND PITCH

The drainage or run-off capacity of roof sheeting is a roof design and construction criteria to be considered in determining the roof slope and total length of a sheet run.

The minimum roof pitch is to be limited to ensure that the depth of water in the roof slopes does not exceed the height of the overlaps.

The roof slope or pitch of a roof profile is dependent on a number of factors, as follows:

(a) Profile configuration, dimensions and stiffness or flexibility.

(b) Number and type of ribs, rib height, tray width.

(c) Rainfall intensity.

(d) Maximum length of the roof sheet and roof slope for the various sheet profiles, which are typically based on CSIRO research, proprietary calculations and experience of the behaviour of specific profiles under peak rainfall conditions. The following points are to be noted:

  (i) The roof sheeting to be designed for a minimum of 1 in 100 y average recurrence interval rainfall intensity for the particular location.

  (ii) Total roofing run length is the sum of all roof components and roof lengths that form a common drainage path.

  (iii) Where an upper roof or downpipe outlet discharges directly onto a lower roof, damming, splashing or standing waves may occur on the lower roof. Special design and construction attention to prevent water intrusion may be necessary in these circumstances. Care is also to be taken for roofs with skylights, vents, etc., where the water flow from several ‘pans’ may be concentrated into one or two pans and standing waves may again occur.

  (iv) To minimise the risk of ponding and surcharge of side laps, care is to be taken to ensure that purlins and battens are in plane.

## 7.7 THERMAL EXPANSION/CONTRACTION

Because the movement in a length of fastened sheeting would normally take place from the centre towards each end of the sheet, the actual expansion or contraction movement between the end of a sheet and the last support would only be a fraction of the figures shown in Table 7.7. The movement at each end is then only half the expansion or contraction. The level of thermal expansion will also be related to the thermal response of the roof sheet; steels with higher solar reflectance will experience fewer issues with thermal expansion. Refer to manufacturer’s data for information on solar reflectance. Also, there would be some temperature change in the building structure or roof frame, although not as much as in the roof sheeting, which would reduce the movement between the sheeting and its supports. Finally, flexing of the purlins, particularly at their mid-span, would further reduce the differential movement (see Figure 7.7).

Transverse thermal expansion poses no problem in ribbed roofing profiles because each rib absorbs some transverse movement; however, transverse flashings and the structure supporting the roof may require an expansion joint.

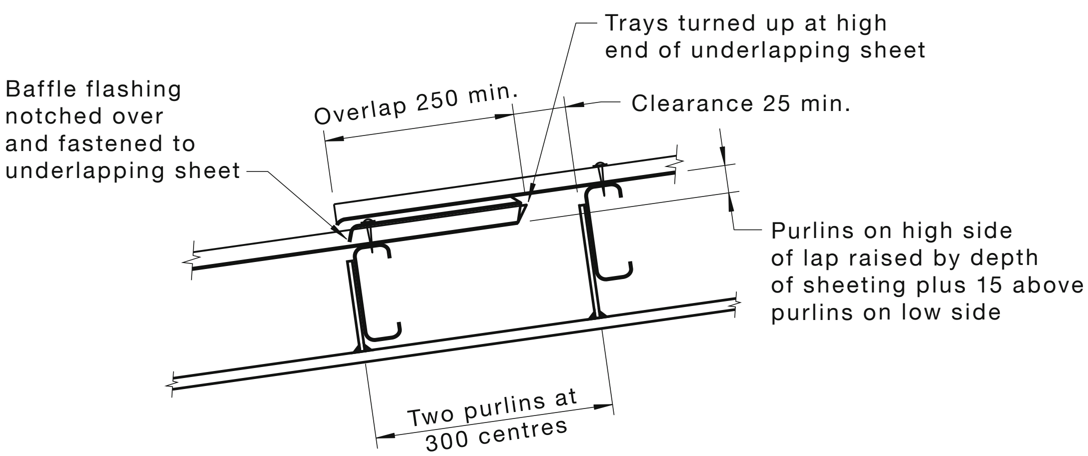
 

**TABLE 7.7**

**EXPANSION AND CONTRACTION OF METAL SHEETING**

| Sheet length m | 50°C change | | 75°C change | |
|---|---|---|---|---|
| | Steel, mm | Aluminium, mm | Steel, mm | Aluminium, mm |
| 5 | 2.9 | 5.8 | 4.3 | 8.6 |
| 10 | 5.7 | 11.4 | 8.6 | 17.2 |
| 15 | 8.6 | 17.2 | 12.8 | 25.6 |
| 20 | 11.4 | 22.8 | 17.1 | 34.2 |
| 25 | 14.3 | 28.6 | 21.4 | 42.8 |
| 30 | 17.1 | 34.2 | 25.7 | 51.4 |

NOTE: Where provision for thermal expansion is required, appropriate manufacturers' product specifications should be referred to.

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DIMENSIONS IN MILLIMETRES

FIGURE 7.7 EXPANSION DETAIL

## 7.8 ROOF LOADING AND SETTING OUT

### 7.8.1 Roof loading

When unloading packs of sheets and transferring to the roof, the following steps are to be taken:

(a) Ensure all operations of cranage and lifting are carried out in accordance with relevant Commonwealth/State and Territory health and safety legislation.

(b) Consider the weight of the pack before placing it on the roof.

(c) Securely fasten each pack with metal bands or ropes as they are placed on the roof to prevent their becoming airborne due to sudden gusts of wind.

(d) On steep roofs, make solid fixings at the end of each pack to stop it from slipping down the roof.

(e) To avoid possible damage to the frame, note the position of the portal frame or roof purlins/rafters when lowering the pack on to the roof. Check that all frame ties and bracing are complete before the roof is loaded.

(f) Centre the packs adjacent to the portal frames or roof purlins/rafters. Do not place packs on purlins without direct portal support.

(g) When loading gable roofs, place packs equally on each side of the gable as loading progresses. This is particularly necessary where timber trusses are used.

(h) Where required, lay any mesh before roofing packs are placed on the roof and lay insulation just prior to deck fixing.

(i) To avoid having to turn sheets around on the roof, load the packs onto the roof with the underlap facing toward the direction of laying. The overlap is to be near the starting point.

(j) Place the first pack to be laid at least the width of the number of sheets in the pack from the starting point of the roof. This will avoid the need to shift the pack unnecessarily as laying progresses.

### 7.8.2 Setting out

When setting out the roof, the following steps are to be taken (see Figure 7.8.2):

(a) Lay the sheets toward the prevailing weather with the side laps facing away from the prevailing weather.

(b) If the length of the roof is such that two or more sheets are required, place the packs end to end on the roof. Lay bottom roof sheets first and then lap top sheets over lower sheets. Fix the full length of the roof as laying progresses across the roof.

(c) Ensure that the underlap sheet does not protrude beyond the overlap sheet or water ‘draw back’ may occur. If necessary, snip away the corner of the underlap sheet.

(d) Ensure that the first sheet is set square to the cover area with the ridgeline square, and that the roof gutter overhang is 50 mm.

(e) Ensure that packs placed on the roofs are not too large.

(f) Make regular checks by measuring as covering progresses, to ensure sheets are running parallel to the frame. This allows for correction of creep of the sheets.

(g) Use a string line along the roof to line up roof fixings. This ensures that roof fixings are in the centre of the purlins, avoids unnecessary holes in sheets and adds to the aesthetics of the finished roof. Do not use a black pencil to mark the fastener line (see Clause 2.8).

(h) On small roofs where sheets are pulled up on to the roof and crane lifting is not used, protect the gutters from collapse by means of blocks of wood, or similar, placed within them. Also use this practice for ladders.

To ensure the roofs rigidity is maintained, all sheets, when fixed, are to have full bearing on all purlins or battens which they cross.

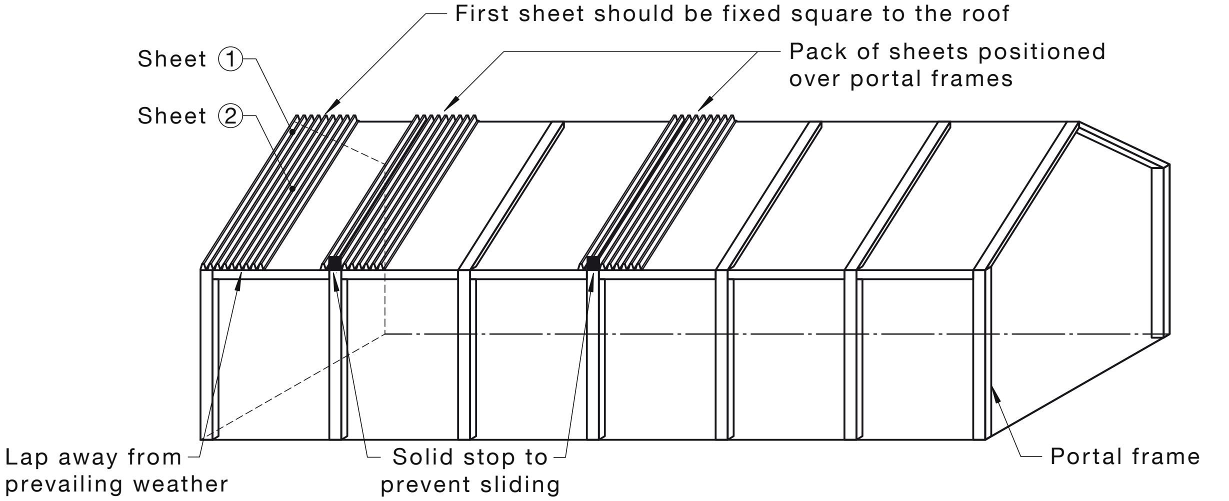


FIGURE 7.8.2 SETTING OUT

### 7.8.3 Checklists

Checklists for loading and setting out are as follows:

(a) *Roof loading*:

(i) Plan the loading.

(ii) Load the sheets to suit the direction of laying.

(iii) Spread the load evenly.

(iv) Be aware of stress loads on the frame.

(v) Check for creep when laying.

(vi) Balance load on the roof on both sides of the ridgeline if possible.

(vii) Secure the sheets as they are loaded onto the roof.

(b) *Setting out*:

(i) Ensure the first sheet is set square to the cover area.

(ii) Ensure purlins and battens are fixed securely and in plane, and the maximum span is not exceeded.

(iii) Check that sheets at ridge and gutter-line are square and that the roof gutter overhang is 50 mm.

(iv) Use string-lines for straight fastener placement.

## 7.9 PIERCE-FASTENED DECKS

Corrugated, trapezoidal and close pitch trapezoidal profiles of pierce-fastened decks may have some form of anti-capillary roof feature formed in the side laps (see proprietary product literature). This is to provide an air break in the side laps to prevent capillary action drawing moisture into the lap. The break also drains water from the laps.

For this feature to operate effectively it is important that these profiles be lapped in the right direction with regard to ‘over’ versus ‘under’ lap, i.e. the side lap is to face away from prevailing weather. (For respective details, see Figure 7.9 and proprietary literature.)

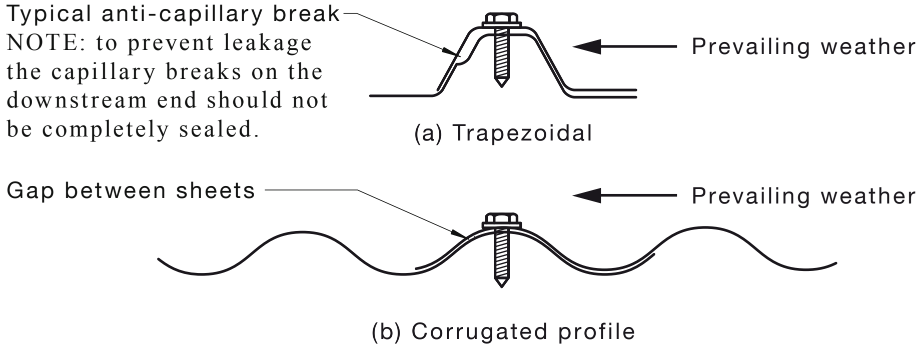


FIGURE 7.9 SIDE LAPPING

## 7.10 CONCEALED-FASTENED DECKS

The concealed-fastened deck profiles follow the same general installation procedures as described in Clauses 7.8 and 7.9 except that a row of fixing clips is positioned and fastened to the support before the first sheet is located over them and locked in position (see Figure 7.10).

As there are differences in the clipping, assembly and locking between most concealed-fastened decks, the installation of each is to be in accordance with the manufacturer's specifications.

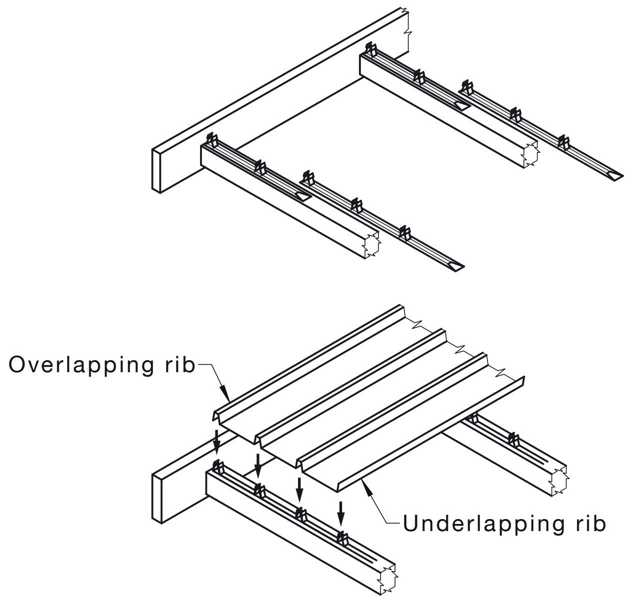


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FIGURE 7.10 CONCEALED-FASTENED DECK—WIDE RIB

## 7.11 MEASURING UP (ESTIMATING QUANTITIES)

The required length of sheets is obtained by measuring the distance from the ridge to the fascia and adding 50 mm for overhang at the gutter or low end. The number of sheets required is obtained by measuring the width of the roof area in metres and multiplying by the factor recommended by the sheet manufacturer. When calculating the number of sheets, allowance is to be made for the fact that the whole of the centre rib is to be left in place by cutting only in the tray area.

## 7.12 FIXING CLIPS AND FASTENERS

### 7.12.1 Fixings

The deck is to be secured by concealed fixing clips, which eliminate the necessity for penetrations through the decking thereby ensuring maximum water protection (see Figure 7.12.1).

The following points are to be noted:

(a) One clip per sheet per support is to be used (see span table for support spacings from individual proprietary product literature).

(b) There are variations in the clips for the different proprietary products (see individual proprietary product literature). The correct clip is to be used for each decking product.

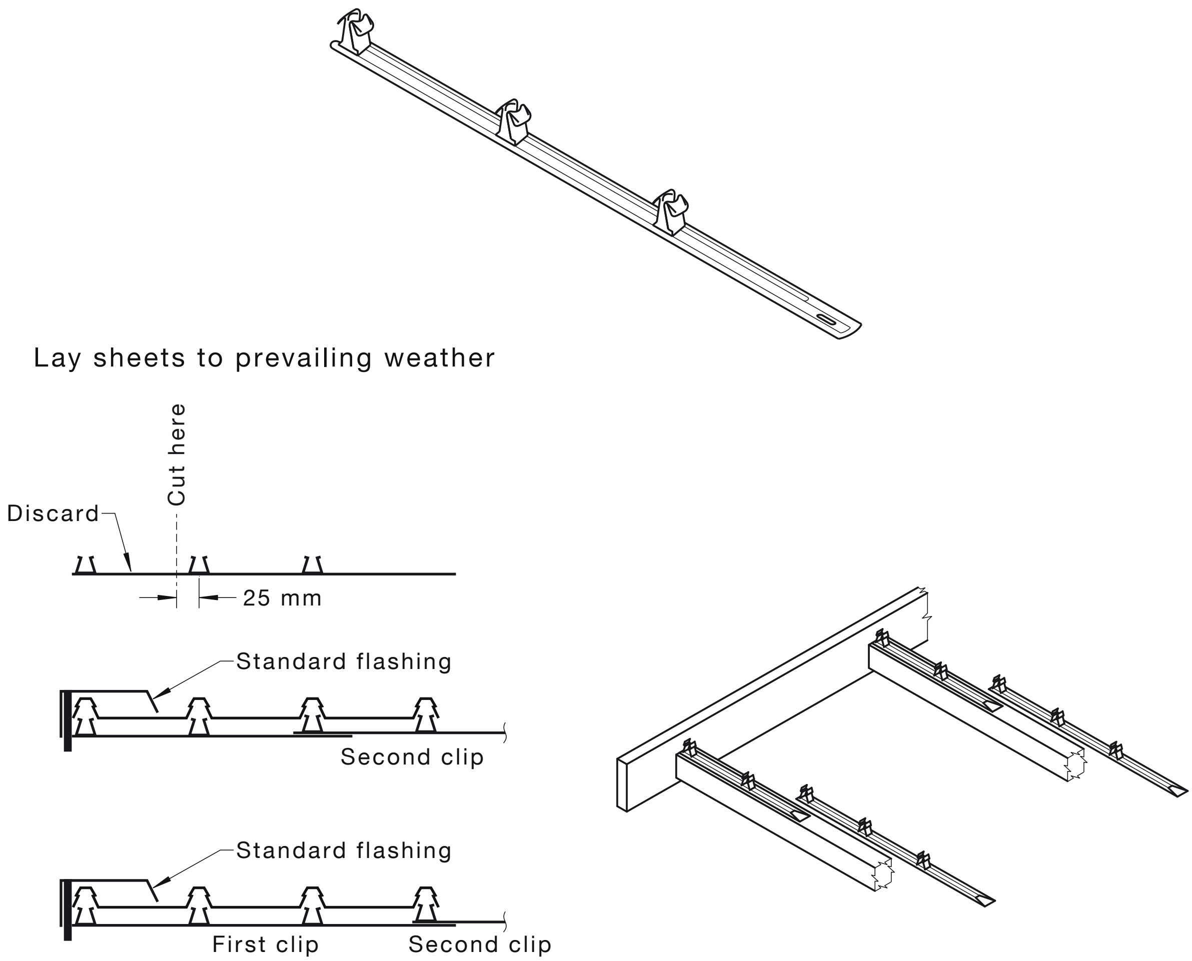


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FIGURE 7.12.1 FIXING CLIPS

### 7.12.2 Installation

For the installation of fixing clips and fasteners, the following is to be observed:

(a) The standard clip is recommended for most applications as it anchors the male rib and the centre rib of each deck. Clips are to be fixed at each purlin and at every rib position for each deck (see Figure 7.12.2).

(b) If clips are accidentally damaged or deformed during application, they are to be replaced before placing the deck in position.

(c) If other types of fasteners such as rivets, bolts or screws are used to anchor clips, they are to have a guaranteed holding force equal to the manufacturer's specification or approval. Such guarantees are to be obtained from the manufacturer or supplier.

Recommended fasteners for concealed-fastened deck (wide-rib) clips in non-cyclone areas are given in Table 7.12.2.

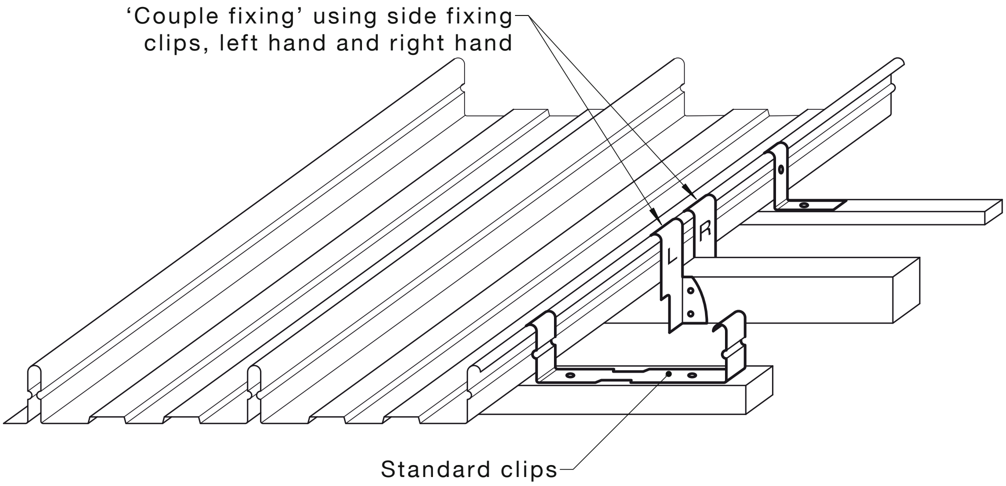



FIGURE 7.12.2 COUPLE FIXING

**TABLE 7.12.2**

**RECOMMENDED FASTENERS FOR WIDE RIB CLIPS IN NON-CYCLONE AREAS**

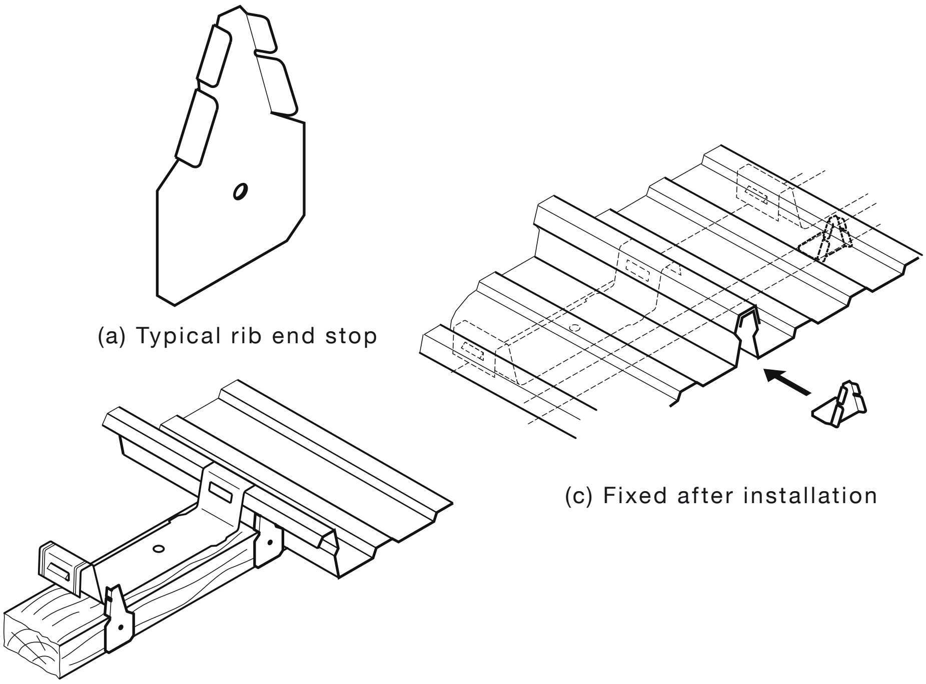
| Support | Fastener |
|---|---|
| Steel supports ≤2.5 mm thick | No. 10–16 × 16 mm wafer head self-drilling thread-forming screws (use 22 mm long screws over insulation blanket if thread of 16 mm screws does not fully penetrate steel supports) |
| Steel supports >2.5 ≤5 mm thick | No. 10–24 × 16 mm wafer head self-drilling thread-forming screws over insulation blanket if thread of 16 mm screws does not fully penetrate steel support |
| Steel supports >5 mm thick | No. 10–24 × 16 mm wafer head thread-cutting screws. Drill 4.5 mm diameter tapping hole through support |
| Timber supports—hardwood | • 3.75 mm diameter × 50 mm long flat head galvanised spiral threaded or annular grooved roofing nails with washers; *or* • No. 10–12 × 25 mm wafer head type 17 self-drilling wood screws (use 60 mm long nails and 45 mm long screws over insulation blanket). |
| Timber supports—softwood | • No. 10–12 × 45 mm wafer head type 17 self-drilling wood screws; *or* • 3.75 × 60 mm long flat head galvanised spiral threaded or annular grooved nails with washers. |

NOTE: Reference is to be made to the manufacturer's specifications and installation instructions.

## 7.13 RIB END STOPS

Rib end stops [see Figure 7.13(a)] may be used to close the rib cavities on the underside of wide rib concealed-fastened deck. They may be fitted during installation [see Figure 7.13(b)] by fastening two end stops to the side of the eaves support after the fastening clip has been secured in position. One end stop is aligned with the centre rib upstand of the clip and the other is positioned under the male rib of the previous sheet.

The rib end stops may be fitted after the deck is installed [see Figure 7.13(c)] by bending the flat of the end stops to 90° and forcing one into the end of each rib until the flat is wedged firmly between the underside of the wide rib concealed-fastened deck and support.



(a) Typical rib end stop

(b) Fixed during installation

(c) Fixed after installation

FIGURE 7.13 RIB END STOPS

## 7.14 FILLER STRIPS

Profiled foam plastic filler or sealing strips are used to match the top and bottom profile of much of the range of roof sheeting. When sandwiched between the eaves support and underside of the roof sheeting (as an eaves closure), a bottom profile strip seals against dust, snow, insects, birds, rodents and wind-driven rain entering the cavities (see Figure 7.14).

To provide a similar seal along the underside of flashing and capping, a top profile strip may be sandwiched between the underside of the flashing and the topside of the roof sheeting.

Filler strips that can absorb water are to be avoided as retained moisture may lead to deterioration of the sheet coating. Closed cell types are recommended. Filler strips made from easily ignited materials are also to be avoided, particularly in bushfire prone areas (see AS 3959).

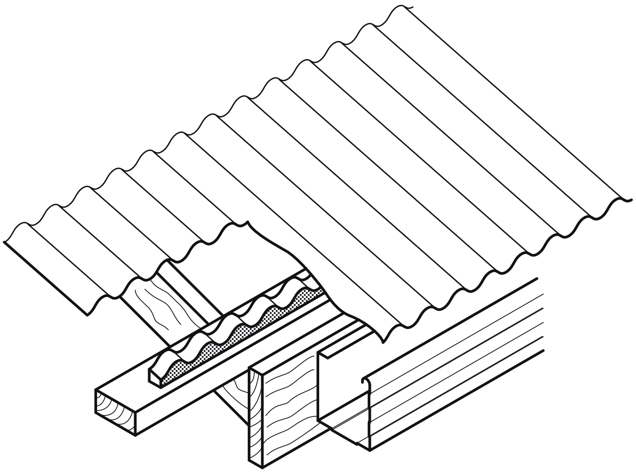


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FIGURE 7.14 PROFILED FOAM PLASTIC FILLER STRIP

## 7.15 END LAPPING

As metal roofing and walling profiles are manufactured by continuous processes, sheet lengths long enough to cover the roof without lapping may be supplied (subject to the limits of transport regulations). Particular attention is to be paid to the following:

(a) Should it be necessary, due to handling or transport considerations, to use shorter end-lapped sheets to provide full length coverage, it is preferred that each run of sheets be laid in turn from bottom to top before moving on to the next run [see Figure 7.15(A)].

(b) The minimum recommended end lap for roof sheeting is 200 mm for roof slopes between 1 in 12 (5°) and 1 in 4 (15°) and 150 mm for slopes above 1 in 4. For walls, the minimum is 100 mm.

(c) The spacing between supports on either side of an end lap in roof sheeting is to be as recommended for end spans.

(d) For pierce-fastened roof, the end lap is to be positioned over a support and the fastener frequency to be as for an end support [see Figure 7.15(B)].

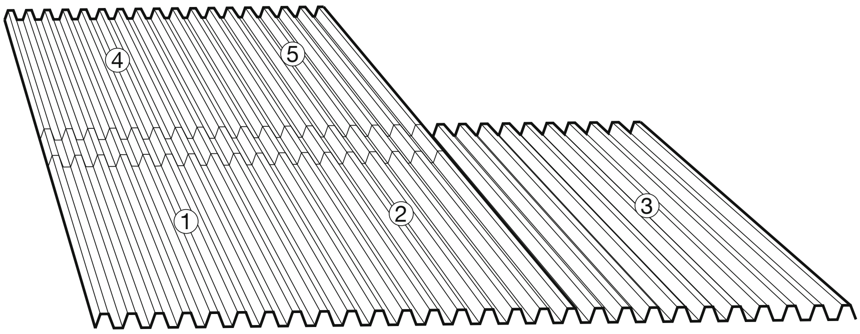
(e) Concealed-fastened sheeting is not to be lapped. Use the stepped expansion detail in Clause 7.7, with the overlap positioned just clear, and on the high side of the support. This will allow normal concealed fastening at the support.

(f) As concealed-fastened sheeting requires pierce fasteners through the trays to secure an end lap; on steel supports the lap could alternatively be positioned over the support with valley fasteners both securing the lap and fastening the sheeting. In this case, the sheets would be anchored at the lap and thermal movement would be towards the ends of the sheet as it would be for a single sheet length [see Figure 7.15(C)].

(g) Condensation forming on the underside of the roof can cause corrosion at the point where it reaches the cut edge of the underlapping sheet. Over time, this can perforate the top sheet as the coating is consumed by reason of it trying to sacrificially protect the cut edge. Roofs with relatively low pitches are more prone to this corrosion.

A continuous strip of plastic, such as polythene dampcourse inserted into the lap so that it projects from the top of the lap by 50 mm, will deflect the condensation from the cut edge [see Figure 7.15(D)].

Alternatively, a generous bead of neutral-cure silicone rubber sealant extruded on the top edge of the bottom sheet prior to being overlapped will both seal the edge and deflect the condensate.




NOTE: Numbers show sheet laying sequence sheet 3 goes under sheet 2 and sheet 5 goes under sheet 4.

FIGURE 7.15(A) PREFERRED SHEET LAYING SEQUENCE

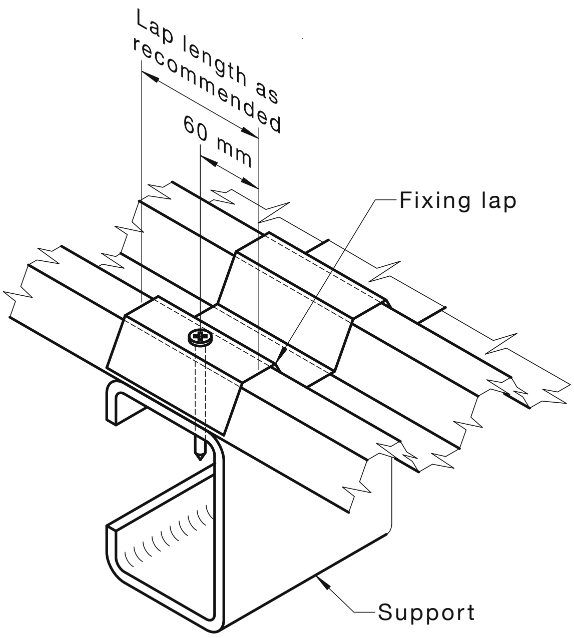



FIGURE 7.15(B) PIERCE-FASTENED SHEETING AND END LAP

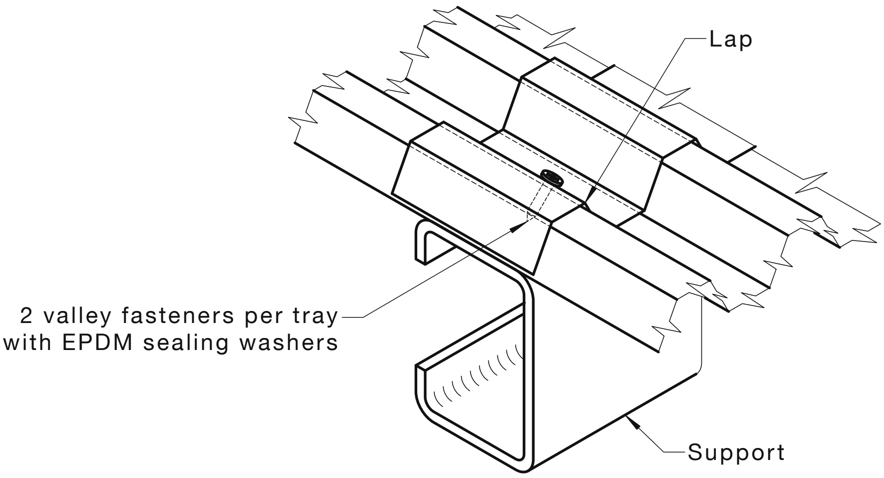



FIGURE 7.15(C) CONCEALED-FASTENED OR TRAY DECKING

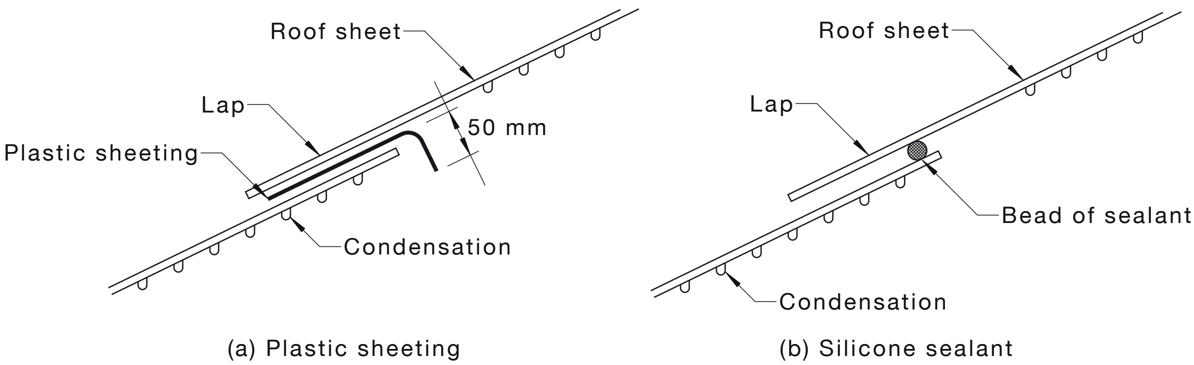



FIGURE 7.15(D) CAPILLARY ACTION PROTECTION

## 7.16 BULL NOSING AND CURVING

### 7.16.1 General

The process of curving roofing profiles, whether by roll or crimp curving, requires skill and specialised equipment. The following points are to be noted:

(a) Roofing materials that are to be curved are to be curved in accordance with the manufacturer's specifications in a manner that will not have a detrimental effect on the material.

NOTE: Most curving of corrugated steel is performed by specialists in their field such as tank makers.

(b) Steel roofing may be curved to produce a remarkable variety of interesting effects both modern and colonial [see Figure 7.16(A)].

(c) The most commonly curved roofing profile is corrugated. In this process, special ductile corrugated steel sheets (ordered as curving quality or brand name) are passed through profile matching curving rolls. These can be progressively adjusted to form curves in a wide range of configurations and radii.

(d) Ductile-grade corrugated steel (curving-quality corrugated) has a lower strength than the standard high tensile product and is thicker (commonly 0.60 mm base thickness) to compensate. The thicker material is easier to curve smoothly. Non-curving quality is usually 0.42 mm base thickness and made from high-tensile steel.

(e) In all cases where large quantities, small radii (less than 500 mm) or unusual curving effects are involved, designers and specifiers are advised to consult local curving specialists to ensure the required results can be achieved [see Figure 7.16(B)].

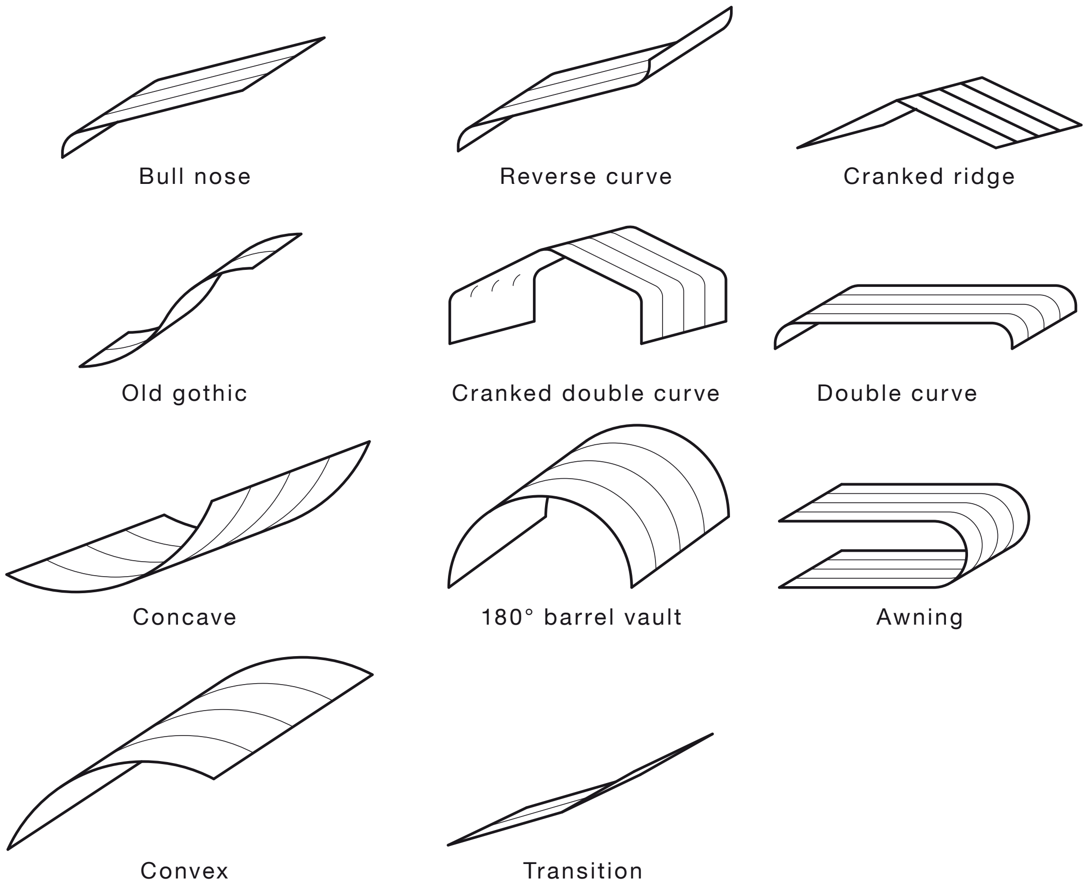



FIGURE 7.16(A) CURVING STYLES

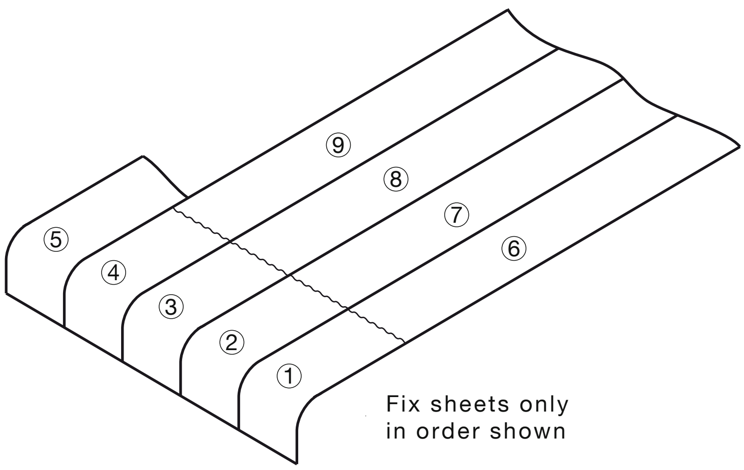



FIGURE 7.16(B) SHEET CURVING

### 7.16.2 Crimp curving of steel and aluminium

Trapezoidal and concealed-fastened deck profiles are normally too deep to be roll-curved and therefore are crimped. In most States, crimp curvers can produce sheets curved to the same contours. By pressing small crimps in either the tops of the ribs or the pans of the sheeting, the profile is progressively shortened at those points causing it to deflect in that direction. The radius is adjusted by varying the depth and the frequency of the crimps.

## 7.17 SPRING-CURVED SHEETING

### 7.17.1 Spring-curved ridges

A suitable method of sheeting low slope gable roofs is to run continuous lengths of roof sheeting from eaves to eaves, across the full width of the roof, allowing them to spring curve between ridge purlins that are spaced widely apart. This method provides a particularly neat, attractive roof and eliminates the ridge capping, so avoiding any possibility of leakage along this fitting (see Figure 7.17.1).

For this type of roof construction, the following are to be observed:

(a) The roof slope for each steel sheeting profile is to be within the limits shown in Table 7.17.1(A).

(b) Minimum spacing between the ridge purlins is to be as shown in Table 7.17.1(A).

(c) If the roof construction has a ridgeboard, its top face is to be no higher than the line of curvature that the roof sheeting will assume between the ridge purlins.

Side laps are to be sealed for the entire length of the curvature (i.e. between the two centre purlins) with a recommended sealant. Each sheet is to be first fastened to one side of the roof and then pulled down and fastened to the slope on the other side of the ridge curve. Alternate sheets are to be laid from the opposite sides of the roof.

NOTE: It should be noted that over the ridge purlins or battens, very slight crease marks may appear in the tray or valleys of the curved sheeting when subjected to foot traffic.

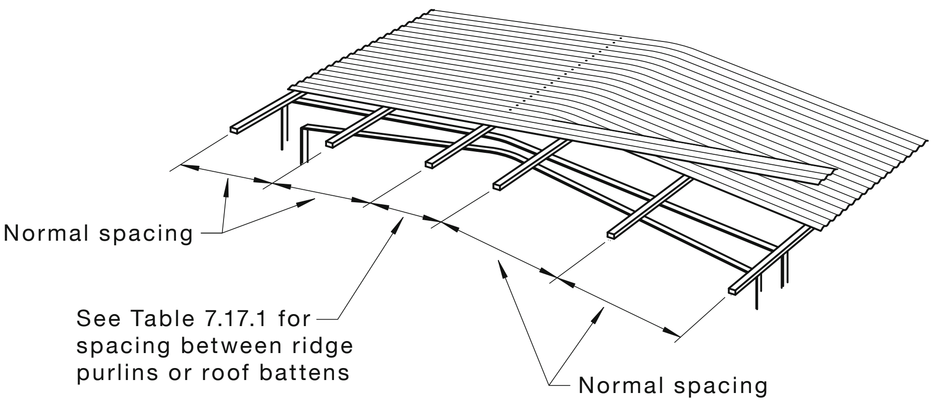



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FIGURE 7.17.1 SPRING-CURVED RIDGE

The trays of wide panned decks tend to wave or oil-can when spring-curved so only the sheet profiles shown in Table 7.17.1(B) are recommended for spring-curving. If some oil-canning in the trays is considered to be acceptable, concealed-fastened decks may be sprung over a ridge up to a maximum slope of 1 in 30 (2°) with the spacing between ridge purlins at slightly less than the recommended internal span for the sheet profile. However, it should be noted that an oil-canned tray may retain water, which could lead to discolouration or deterioration of the sheet coating and also contribute to thermally induced roof noise.

**TABLE 7.17.1(A)**

**MINIMUM RIDGE PURLIN SPACING FOR SPRING-CURVED RIDGE**

millimetres

| Roof slope | Close pitched trapezoidal | | Corrugated | | Curving corrugated |
|---|---|---|---|---|---|
| | 0.42 | 0.48 | 0.42 | 0.48 | 0.60 |
| 1 in 20 (3°) | 1400 | 1500 | | | |
| 1 in 15 (4°) | 1500 | 1600 | | | |
| 1 in 12 (5°) | | 1700 | 1200 | 1300 | 1200 |
| 1 in 10 (6°) | | | | 1400 | 1300 |
| 1 in 8 (7°) | | | | | 1400 |

NOTE: Steel thicknesses are expressed as base metal thickness (BMT) not total coated thickness.

**TABLE 7.17.1(B)**

**RECOMMENDED RADIUS FOR CONVEX SPRING-CURVING**

millimetres

| Sheet profile | Minimum radius | Purlin spacing at minimum radius* | Maximum radius† |
|---|---|---|---|
| 0.42 Close pitch trapezoidal | 20 | 1200 | 60 |
| 0.48 Close pitch trapezoidal | 20 | 1400 | 60 |
| 0.42 Corrugated | 12 | 800 | 35 |
| 0.48 Corrugated | 10 | 1000 | 35 |
| 0.60 Curving corrugated | 9 | 900 | 35 |

* For radii curvature greater than the recommended minimum, the purlin spacing may be increased provided it does not exceed the specified maximum for straight sheeting given on individual proprietary product literature.

† Maximum recommended radius to provide sufficient drainage near crest of curvature.

### 7.17.2 Sprung concave roofs

Sheeting may be spring-curved to the minimum radii shown in Table 7.17.3 for concave applications (see Figure 7.17.2).

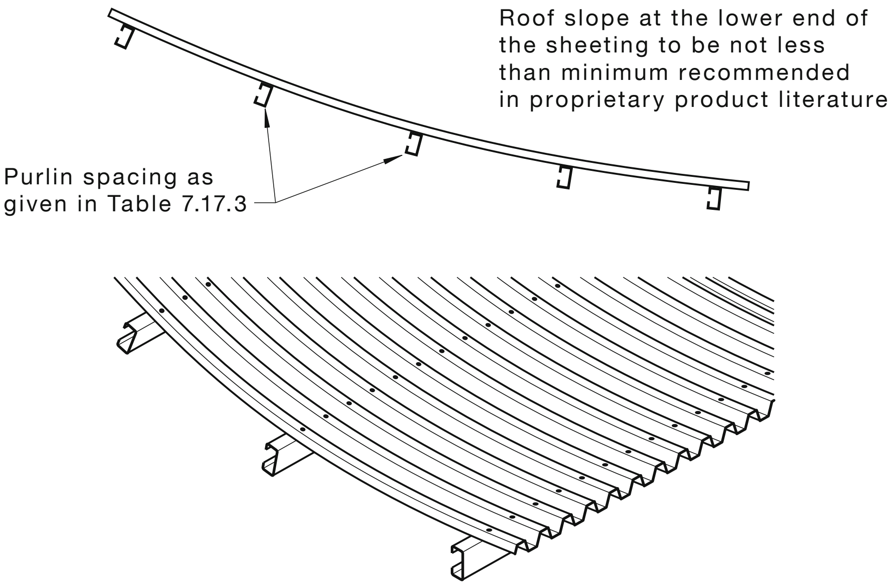


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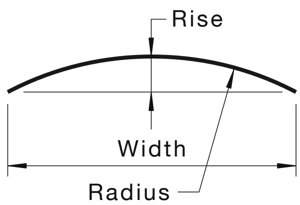
FIGURE 7.17.2 SPRUNG-CONCAVE ROOF

### 7.17.3 Arched roofing

Some of the roofing profiles may also be spring-curved over an arched roof, provided the radius of the arch is not less than the minimum listed for each profile in Table 7.17.3.

From the overall width and required rise of an arched roof, the radius of curvature may be calculated from the following equation:

$$\text{Radius} = \frac{\text{Width}^2 + 4(\text{Rise})^2}{8(\text{Rise})}$$

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Side laps are to be sealed with recommended sealants over the crest of the arch where the slope is less than the recommended minimum for that sheet profile. If end laps are necessary, they are not to be located at or near the crest of the arch and each sheet length is to span at least three purlin spacings. The top faces of all purlins are to accurately follow and be tangential to the arch curvature. Each alternate sheet is to be laid from opposite sides of the roof.

NOTE: Very slight crease marks may appear in the trays or valleys over the supports when curved sheeting is subjected to foot traffic.

**TABLE 7.17.3**

**RECOMMENDED RADII FOR CONCAVE SPRING-CURVING**

| Sheet profile<br>mm | Minimum radius<br>m | Purlin spacing at minimum radius*<br>mm |
|---|---|---|
| 0.55 narrow rib concealed | 25 | 1000 |
| 0.70 fix-deck | 26 | 1500 |
| 0.42 wide rib concealed-fastened deck | 24 | 1000 |
| 0.48 wide rib concealed-fastened deck | 26 | 1400 |
| 0.60 wide rib concealed-fastened deck | 28 | 1800 |
| 0.42 close pitch trapezoidal | 18 | 1200 |
| 0.48 close pitch trapezoidal | 20 | 1400 |
| 0.42 trapezoidal | 20 | 1000 |
| 0.48 trapezoidal | 22 | 1200 |
| 0.42 corrugated | 10 | 800 |
| 0.48 corrugated | 10 | 1000 |
| 0.60 curving corrugated | 8 | 800 |

* For radii of curvature greater than the recommended minimum, the purlin spacing may be increased provided it does not exceed the specified maximum for straight sheeting given in individual proprietary product literature.

## 7.18 CAPPED, BENT CONTINUOUS SHEETING

### 7.18.1 Gable roofs

On gable roofs, tray sheeting may be used in continuous lengths from eaves to eaves by cutting the ribs and bending the trays at the ridgeline. Caps are then fitted and sealed over the cut ribs, which would otherwise be opened up when the trays are bent.

This method provides a roof of very neat appearance with continuous trays and unbroken rib lines over the ridge and down each side of the roof slope; however, fitting the rib caps is time consuming and care is to be taken with sealing to avoid any possibility of leakage. Also, care is to be taken to ensure that the rib of each sheet is cut at 90° to the longitudinal axis of the sheet. The ribs are easily cut with a metal cutting steel blade, ‘not an abrasive disc’ in a power saw, set to the rib depth, minus 2 mm.

In some States and Territories, pressed steel caps, cut and bent to a limited angle range can be obtained to suit concealed-fastened deck ribs [see Figure 7.18.1(a)]. Alternatively, caps may be cut and folded from flat sheet, of the same finish as the roofing, to suit any angle [see Figure 7.18.1(b)].

This method may also be used for trapezoidal or even close pitch trapezoidal sheets in continuous lengths over a ridgeline. For these profiles, the rib caps may be made from pieces of rib profile cut from a short length of sheeting.

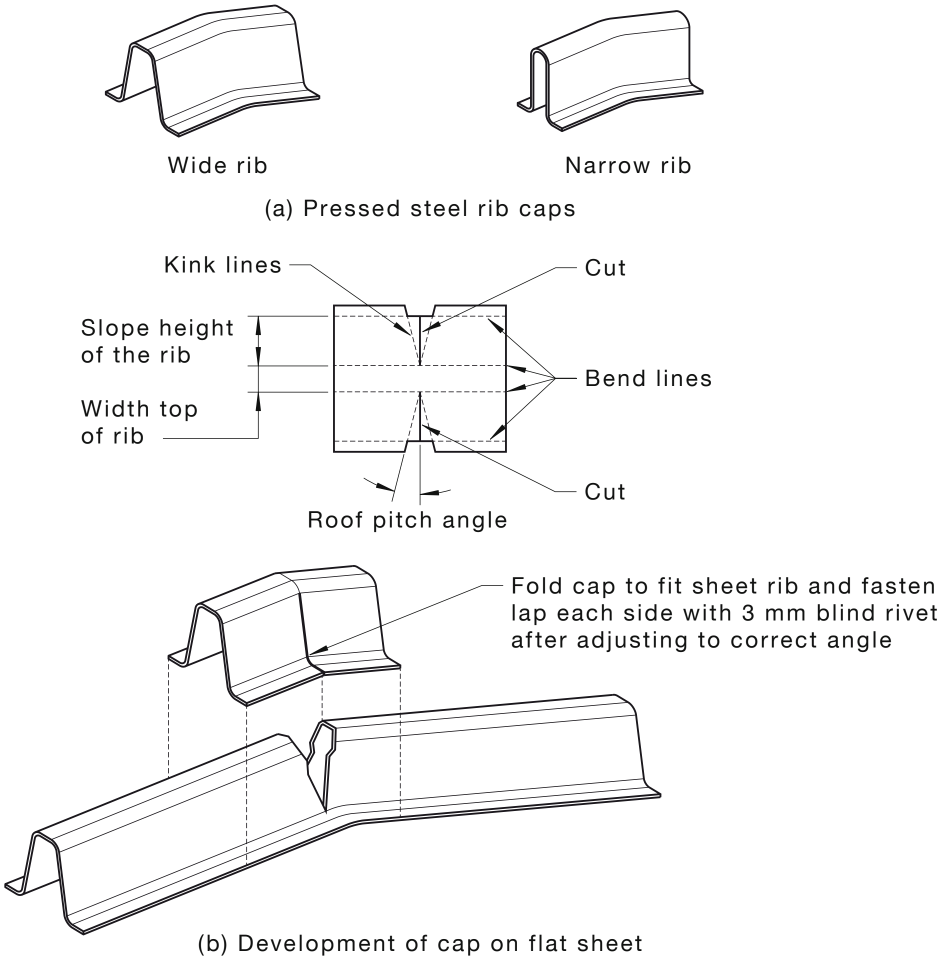


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FIGURE 7.18.1 RIB CAPS

### 7.18.2 A-frame

Where the pitch is steep, some method of preventing the decks sliding downward over the clips due to thermal movement is desirable. A rivet or screw under the ridge capping securing the upper ends of the decks to the framework is satisfactory (see Figure 7.18.2).

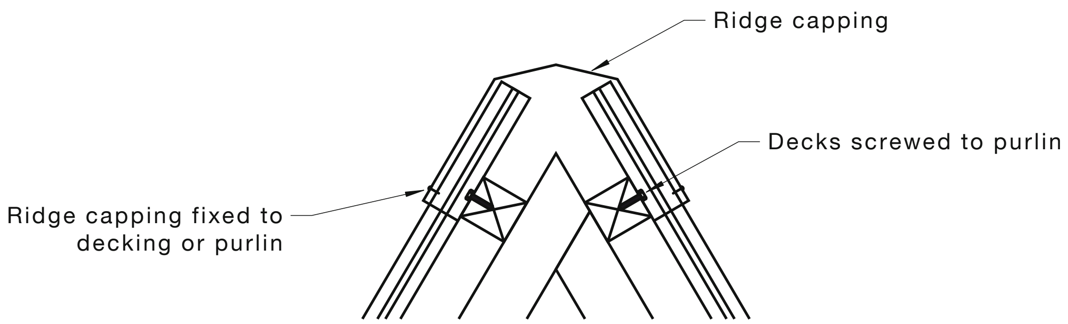


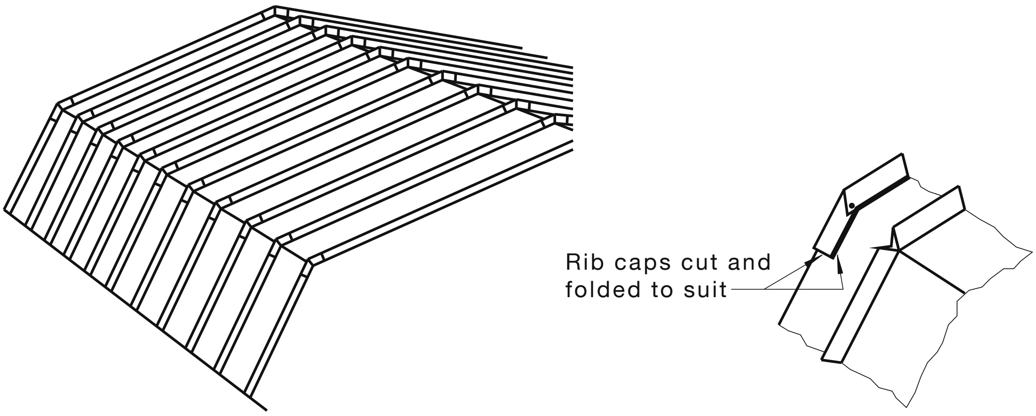

FIGURE 7.18.2 A-FRAME ROOF

## 7.19 MANSARD ROOFS

The same method of bending tray sheeting allows continuous lengths to be used on mansard roofs. For mansard applications, the rib caps have to be cut from flat sheet and folded to suit the angle required for a mansard bend (see Figure 7.19).

A recommended sealant is to be applied inside the ends and along the bottom of cap. The cap is then placed in position over the opened rib and fastened with appropriate fasteners. An anti-capillary cut is required to ensure rib drainage [see Figure 8.6.1(D)].

To form the junction between upper and lower slopes for aluminium trapezoidal profiles it is conventional to cut and TIG weld short lengths of the roof sheet together on a jig set at the correct angle. These sub-assemblies are then fastened properly to the upper and lower sheets to form the mansard shape.




NOTE: Aluminium would normally be shop-fabricated and welded.

FIGURE 7.19 MANSARD ROOF

## 7.20 WALL CLADDING

The following general considerations apply to wall cladding:

(a) *Material specification* Wall cladding materials are the same as those used for roofing applications; however, the material thickness may be reduced.

(b) *Maximum span charts* The maximum support spacings for the various profiles are listed in the maximum span charts of the individual proprietary product literature.

(c) *General care and handling* The same general principles that apply for roofing also apply for wall cladding.

(d) *Installation procedures* The same general principles that apply for roofing also apply for wall cladding.

(e) *Fastening locations* Fastening locations for pierce-fastened sheeting is normally in the pan or valley adjacent to a rib, rather than through the crest or rib; however, when fastening 10½ corrugated sheet, the fastener is to be located in the second valley, rather than the first, otherwise the underlap will be pierced.

NOTE: For aluminium wall cladding in cyclonic wind regions, valley fastening is generally not adequate to restrain sheeting at normal girt spacings and the same method used for crest fastening roofing is used to secure wall sheeting to girts.

# SECTION 8 ROOF FLASHINGS AND CAPPINGS

## 8.1 GENERAL

### 8.1.1 Material compatibility

Roof flashings and cappings other than zinc, sheet lead or pre-coated sheet lead are to be purpose-made, machine-folded sheet metal sections of materials compatible with all upstream and downstream roof coverings materials, including guttering and downpipes [see Tables 2.3(A) and 2.3(B)].

### 8.1.2 Minimum thickness

The sheet metal thickness for roof flashings and cappings is given in Table 8.1.2.

**TABLE 8.1.2**

**MINIMUM THICKNESS FOR FLASHINGS AND CAPPINGS**

| Materials | Coating class | Usage | Thickness min, mm |
|---|---|---|---|
| Aluminium | | Flashing and cappings | 0.7 |
| Copper and alloys | | Flashing and cappings | 0.55 |
| Fibre cement | | Flashing and cappings | 6 |
| Galvanised steel | Z275 | Flashing and cappings | 0.55* |
| Prepainted steel | AS150 | Flashing and cappings | 0.55* |
| PVC | | Flashing and cappings | 3 |
| Sheet lead | | Flashing | 1.7 |
| Stainless steel | | Flashing and cappings | 0.45 |
| Zinc | | Flashing | 0.65 |
| Aluminium/zinc coated steel | AS150 | Flashing and cappings | 0.55*/0.42 |
| Aluminium/zinc/magnesium coated steel | AM100/AM125 | Flashing and cappings | 0.55*/0.42 |

* Base metal thickness.

### 8.1.3 Expansion provision

Roof flashings and cappings are to be installed to permit longitudinal expansion and contraction without detrimental effect to the flashing themselves and other roof components, structures and covers.

Roof flashings and cappings fastened to metallic roof covering materials are not to be fastened to other building materials with other than sliding cleats or fixings.

### 8.1.4 Sizes and covers

The minimum cover over other materials and structures for roof flashings and cappings is given in Table 8.1.4. For jointing, see Clause 5.8.

## TABLE 8.1.4
## MINIMUM COVERS FOR FLASHINGS AND CAPPINGS FOR STEEL ROOFING

| Material to be covered | Type of cover | Special feature | Minimum cover mm |
|---|---|---|---|
| Pierce-fastened roof sheet | Transverse flashing | 30° × 20 mm brake pressed down | 150 |
| Pierce-fastened roof sheet | Sloping apron | 30° × 20 mm brake pressed down | 150 |
| Concealed-fastened roof sheet | Transverse capping | 90° × 40 mm down-fold | 150 |
| Concealed-fastened roof sheet | Sloping apron | 90° × 40 mm down-fold | Into full tray |
| Masonry wall | Parapet capping | Front face 135° × 10 mm plus 90° downturn | To suit wall |
| Masonry wall | Parapet capping | Back 90° anti-capillary break cover over apron flashing | 50 |
| Timber fasciae and bargeboard | Cappings | Front face 135° × 10 mm plus 90° downturn | 50 |
| Gutters and spoutings | Apron or wall flashing | Overlapping gutter and anti-capillary break | 50 |
| Apron flashings | Cappings and aprons | Overlapping gutter and anti-capillary break | 50 |
| Tops of steel sheet cladding | Cappings | 90° × 40 mm downturn | 150 |
| Tops of glass walls | Cappings | 135° × 10 mm fold | 150 |
| Over fibre cement | All covers | To manufacturer's instructions | See manuals |
| Flashing and cappings | Laps | Laps of all types and sealing method | 25 |
| Roof tiles | Aprons lead | Dressed to suit contours and cleats | 150 |
| Flat surfaces | Any flashing and capping | 30° × 10 mm plus anti-capillary break | 50 |

NOTE: For aluminium, refer to manufacturers' technical literature.

### 8.2 FASTENING

Roof flashings and cappings are to be fastened as given in Table 8.2(A) and the following:

(a) Roof flashings and cappings to be fastened to withstand wind pressures and thermal movements.

(b) Roof flashings and cappings to be tightly notched and fastened at the intervals given in Table 8.2(B).

(c) Intersections, laps and mitres are to be fastened and sealed as per Clause 5.8.

(d) Joints in flashings and cappings other than straight joints to be accurately mitred, fastened and sealed.

(e) Joints in flashings and cappings to be 25 mm and fastened at intervals not exceeding 40 mm. Joints are to lap in the direction of fall.

(f) All joints are to be watertight on completion of the installations.

TABLE 8.2(A)

FASTENER TYPES

| Atmospheric exposure (AS/NZS 2312) | Atmospheric exposure (ISO 9223) | Roof sheeting: Aluminium/zinc (AZ150) or aluminium/zinc/magnesium (AM125) alloy-coated steel | Roof sheeting: Prepainted aluminium/zinc (AZ150) or prepainted aluminium/zinc/magnesium (AM100) alloy-coated steel | Roof sheeting: Prepainted aluminium/zinc (AZ200) or prepainted aluminium/zinc/magnesium (AM150) alloy-coated steel | Roof sheeting: Prepainted stainless steel |
|---|---|---|---|---|---|
| Very severe marine | C5 | N/R | N/R | MR | C |
| Severe industrial | C5 | N/R | N/R | MR | C |
| Severe | C4 | B | B | B | C |
| Marine | C3 | B | B | B | C* |
| Moderate | C2 | A,B | A,B | A,B* | C* |
| Benign | C1 | A,B | A,B | A,B* | C* |

LEGEND:
A AS 3566.1 Class 3 zinc coated
B AS 3566.1 Class 4 zinc coated
C Class 4 stainless steel fastener
N/R Product not recommended for atmospheric exposure
MR Seek fastener manufacturers recommendation
* Typically this product is not used in this environment

NOTE: Special consideration should be given to areas not washed by natural rainfall. General guide if in doubt seek manufacturer's recommendations.

TABLE 8.2(B)

FASTENER FREQUENCY FOR TRANSVERSE FLASHINGS AND CAPPINGS

| Roof type | Frequency | Fastener type |
|---|---|---|
| On concealed-fastened roofs | At no more than 300 mm centres | Rivets |
| On pierce-fastened roofs | At no more than 300 mm centres | SD screws or rivets |
| On corrugated | Every fourth rib | SD screws or rivets |

LEGEND:
SD = self-drive tapping screws

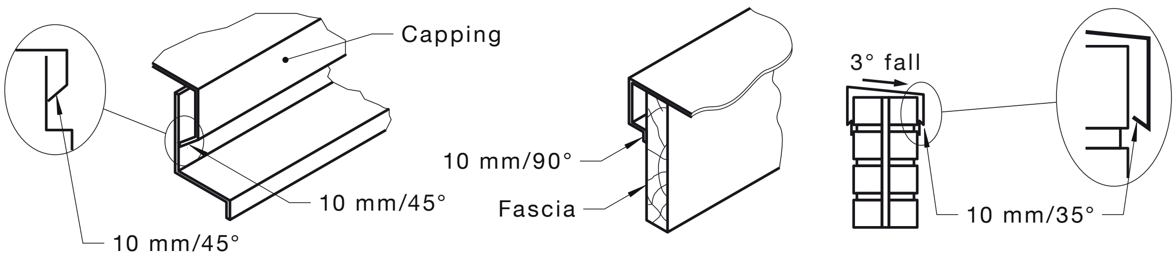
## 8.3 SPECIAL FOLDS

Particular attention is to be paid to the following:

(a) Roof flashings and cappings to be provided with anti-capillary breaks of 10 mm/30° against flat surfaces and 10 mm 90°/135° breaks against all other surfaces (see Figure 8.3).

(b) Roof cappings are to overlap underlying vertical flashings and gutters by at least 50 mm.

(c) When flashings and cappings exceed 150 mm width, continuous supports are to be provided at intervals not exceeding 500 mm centres.




(a) Roof capping (b) Fascia flashing (c) Wall flashing (See Note)

NOTE: Parapet cappings to fall back to roof coverings or internal gutters.

FIGURE 8.3 ANTI-CAPILLARY BREAKS

## 8.4 WALL AND STEP FLASHINGS

For material, thickness, expansion, sizes and covers, and wall and step flashings, see Clause 8.1. Additionally, the following points are to be observed [see Figure 8.4(A)]:

(a) Fixings to align with roof covering ribs. Wall and step flashings are to incorporate a weathering fold, enter the masonry walls by at least 22 mm and the capping sealed as stated in Clause 8.3(b) [see Figure 8.4(A)(a)].

(b) All wall and step flashings are to be fastened into masonry walls with galvanised, aluminium/zinc alloy-coated or aluminium/zinc/magnesium-coated sheet metal wedges at each end of each length and other than pressure flashings at intermittent intervals not exceeding 500 mm [see Figure 8.4(B)]. Alternatively, sloped wall flashings are to incorporate a weathering fold fixed into a 25 mm deep saw cut [see Figure 8.4(A)(b)].

(c) Pressure flashings may be used in lieu of cutting grooves into walls, provided they are used only with smooth surface finished walls, e.g. smooth finished concrete or smooth finished brickwork with flush pointed mortar courses, provided [see Figure 8.4(C)]—

   (i) the pressure flashings are purpose-made machine folded with a safety/stiffening fold at the upper edge or alternatively constructed with a safety/stiffening fold at 45° from vertical to allow for the placement of a silicone filler;

   (ii) the sealant is applied in a sandwiched seal of approximately 20 mm wide;

   (iii) the fixing of the flashing will ensure a durable seal is maintained;

   (iv) the seal is protected from excessive movement due to expansion and contraction;

   (v) the fixing centres are at no more than 100 mm spacings; and

   (vi) the fixing devices are fit for purpose and compatible with the flashing material.

(d) Where high wind velocities are frequently encountered, extra cover and fasteners may be required to prevent wind-driven rain from entering the building.

   The joint containing the weathering fold is to be made watertight using mortar or appropriate sealants.

(e) Walls and step flashings to overlap by 50 mm in the direction of flow and be cleated by means of holding cuts [see Figure 8.4(D)].

(f) Step flashings to be of uniform size and incorporate the angle of the roof slope.

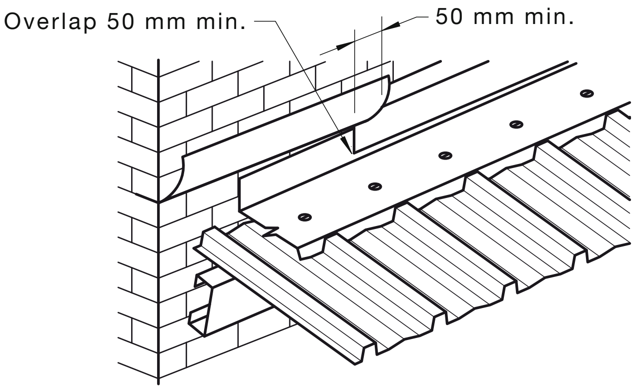
(g) No wall or step flashing to be fastened to other roof components or flashings.

(h) On completion of the installation of all flashings and cappings—

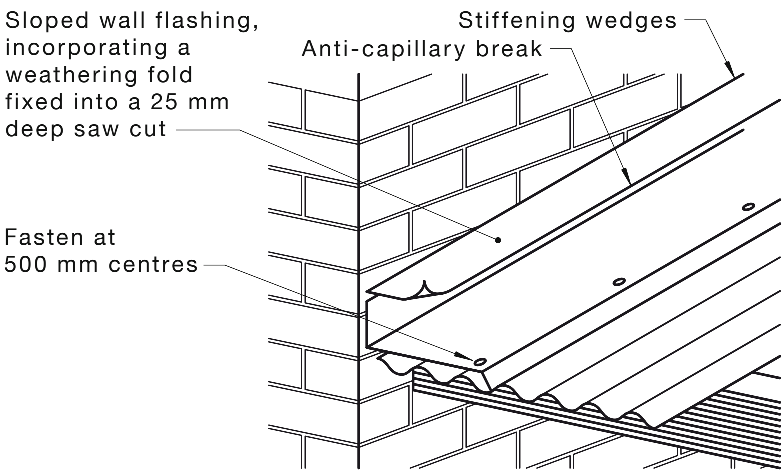
(i) all debris to be removed from the roof;

(ii) all traces of cement and mortar to be washed off; and

(iii) excess sealant protruding from any roof component or roof covering, including flashings and cappings, to be removed.




(a) Parapet flashing




(b) Sloped wall flashing

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FIGURE 8.4(A) WALL AND STEP FLASHINGS

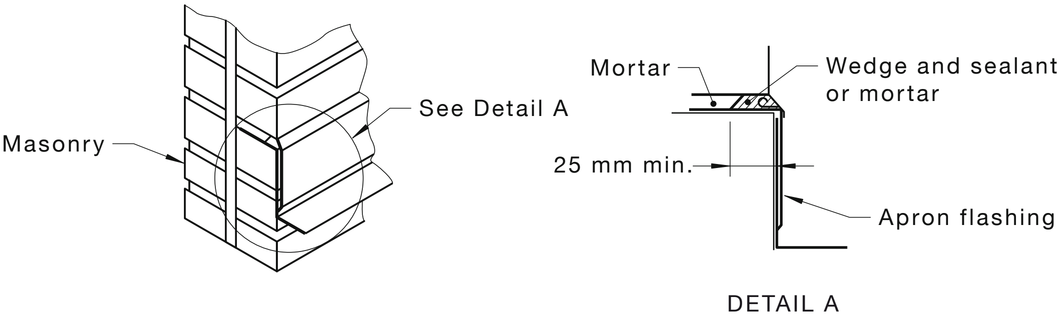
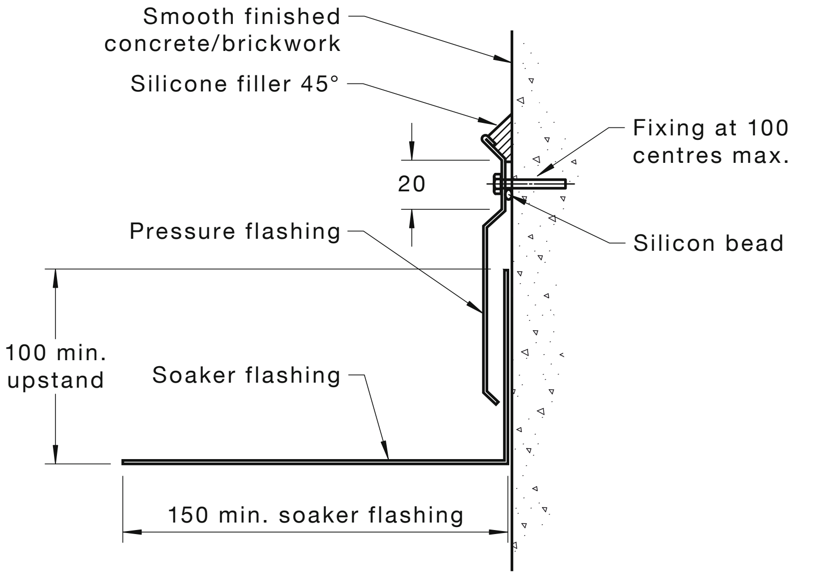



FIGURE 8.4(B) MASONRY FIXING WEDGES




DIMENSIONS IN MILLIMETRES

FIGURE 8.4(C) PRESSURE FLASHING

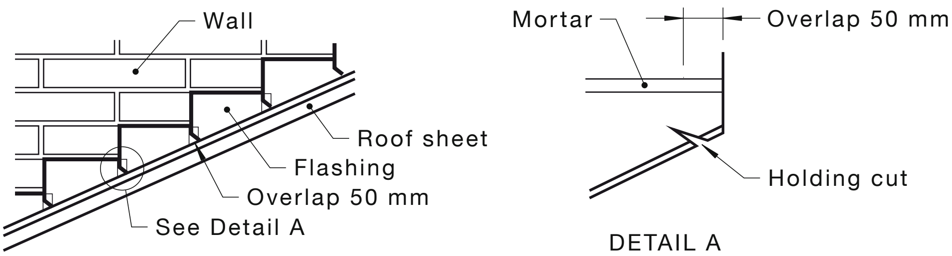



FIGURE 8.4(D) FLASHING OVERLAP

## 8.5 LEAD FLASHINGS

Lead flashings are to have a permanent protective coating when used with prepainted steel, aluminium/zinc alloy-coated or aluminium/zinc/magnesium alloy-coated steel or aluminium. Uncoated lead flashings are not to be used on any roof if the roof is part of a drinking water catchment.

## 8.6 PENETRATIONS

### 8.6.1 Collar flashings

Where any part of the roof surface is penetrated by any pipe, pole, duct, flue, shaft, cable or tank support, the penetration is to be flashed to prevent the entry of rainwater, and to permit the roof surface and penetrating object to expand and contract without detrimental effect to any part of the roof [see Figure 8.6.1(A)]. In addition, particular attention is to be paid to the following:

(a) *Pondage* Collar flashings to permit the total drainage of the area above the penetration.

(b) *Material compatibility* All collar flashing materials to be as given in Tables 2.3(A) and 2.3(B).

(c) *Inert catchments* All collar flashings to be manufactured from materials that—

(i) are not adversely affected by the run-off from inert catchments; and

(ii) have no detrimental effect on the roof surface, including gutters, soakers and downpipes.

Galvanised steel collar flashings to be installed only on a galvanised steel roof surface.

(d) *Expansion and contraction* Penetrations other than synthetic rubber flashings, to leave annular clearances of not less than 20 mm around the perimeter of the penetrating object.

(e) *Baseplate upstand* Collar flashings to be either metal spun of one piece or consist of an upstand of not less than 100 mm with a baseplate fastened and sealed to the roof surface.

(f) *Apron* A weathering apron to be fastened and sealed around the penetrating object and to overlap the upstand by not less than 50 mm. Sheet metal aprons to be purpose-made and compatible with other roof-covering materials.

(g) *Synthetic rubber collar* Flashings of the synthetic rubber type to be installed to manufacturer's specifications [see Figure 8.6.1(B)]. Upstands other than the synthetic rubber type are to be fastened and sealed to the roof surface only.

(h) *Soaker support* On completion of the installation, the roof surface to be restored to its initial strength [see Figure 8.6.1(C)].

(i) *Pre-coated lead collars* Pre-coated lead collars may be used with aluminium, aluminium/zinc alloy-coated, aluminium/zinc/magnesium alloy-coated or prepainted steel.

(j) *Rib sealing* Where the installation of collars requires ribs to be cut, the ribs to be fastened and sealed.

(k) *Hot and vibrating pipes* Where the penetrating object is subject to excessive heat or vibration, the apron to be physically separated from the upstand.

(l) *Gutter penetrations* Penetrations other than overflow provisions and downpipes are not permitted in any part of a roof gutter.

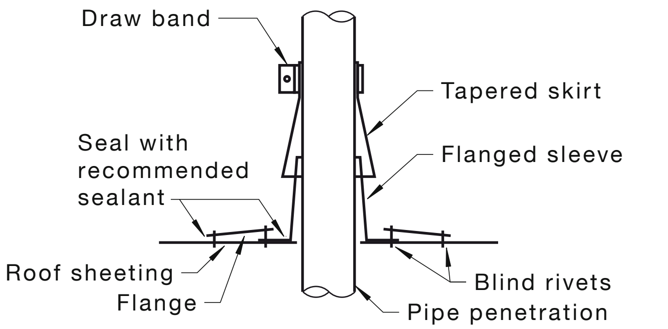



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FIGURE 8.6.1(A) COLLAR FLASHING

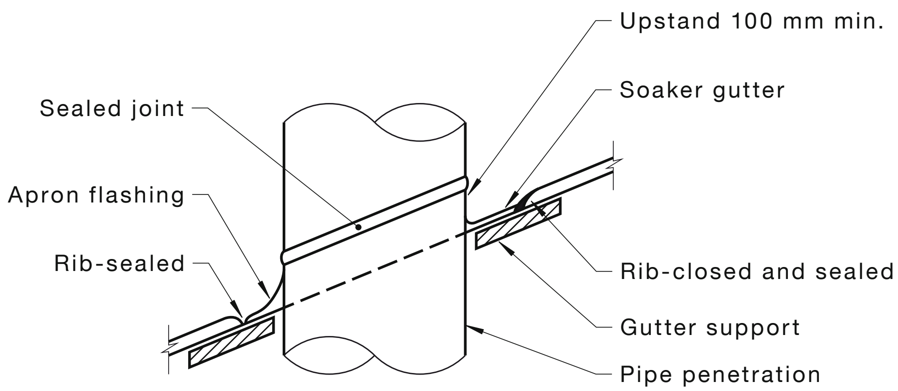


FIGURE 8.6.1(B) RUBBER/SYNTHETIC RUBBER FLASHING

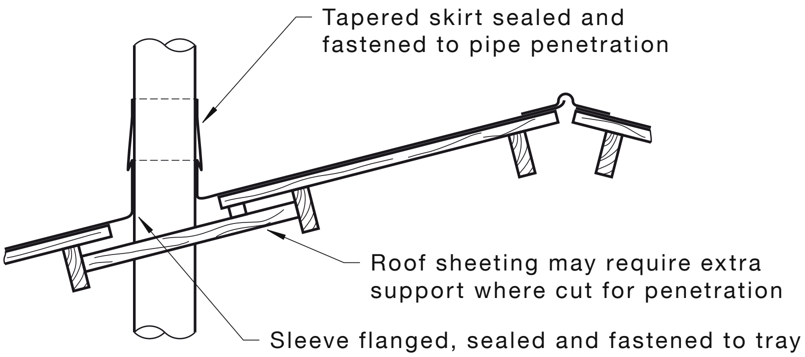


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FIGURE 8.6.1(C) SOAKER SUPPORT

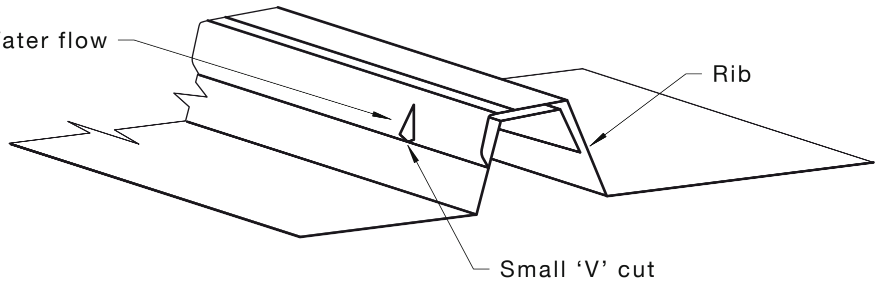



FIGURE 8.6.1(D) ANTI-CAPILLARY CUTS ASSOCIATED WITH OVERFLASHING ON SHEET JOIN

### 8.6.2 Large penetrations

Where large ducts/pipes penetrate the surface, adequate drainage of the roof surface above the penetration is to be provided (see Clause 5.5). In addition, particular attention is to be paid to the following:

(a) *Penetration support* Where a section of a roof has been removed for a penetration, the perimeter of the opening to be restored to the original strength of the roof.

(b) *Cutting penetration* Where power cutters are used to cut out the roof section, all occupational health and safety regulations to be observed. On completion of the cut, the whole of the area to be thoroughly cleaned to remove all traces of metal cuttings and swarf.

(c) *Rib sealing* Ribs on the upstream end of the penetrations to be folded, fastened and sealed as shown in Figure 5.5(B) or proprietary rib sealers to be fitted, fastened and sealed to render the rib end watertight (see Figure 7.13).

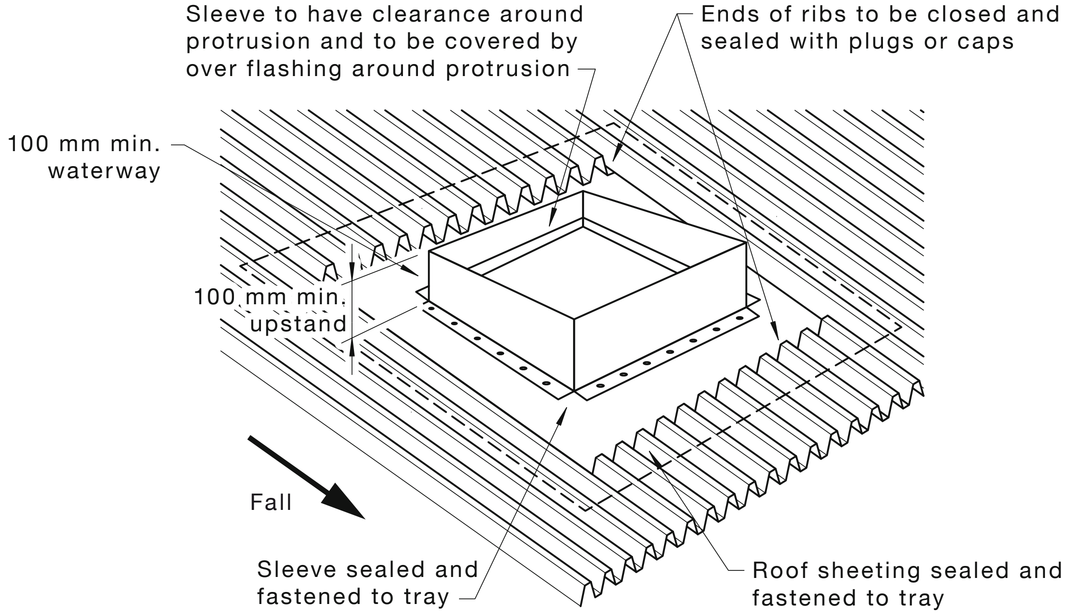
(d) *Anti-capillary cuts* All female ribs in crest-fixed sheeting to have an anti-capillary cut above all soakers to allow drainage to the roof thus preventing seepage of water into the rib-lap [see Figure 8.6.1(D)].

(e) *Stop-ending* Roof coverings at the lower end of penetrations to have their trays or valleys stop-ended to the full height of the ribs.

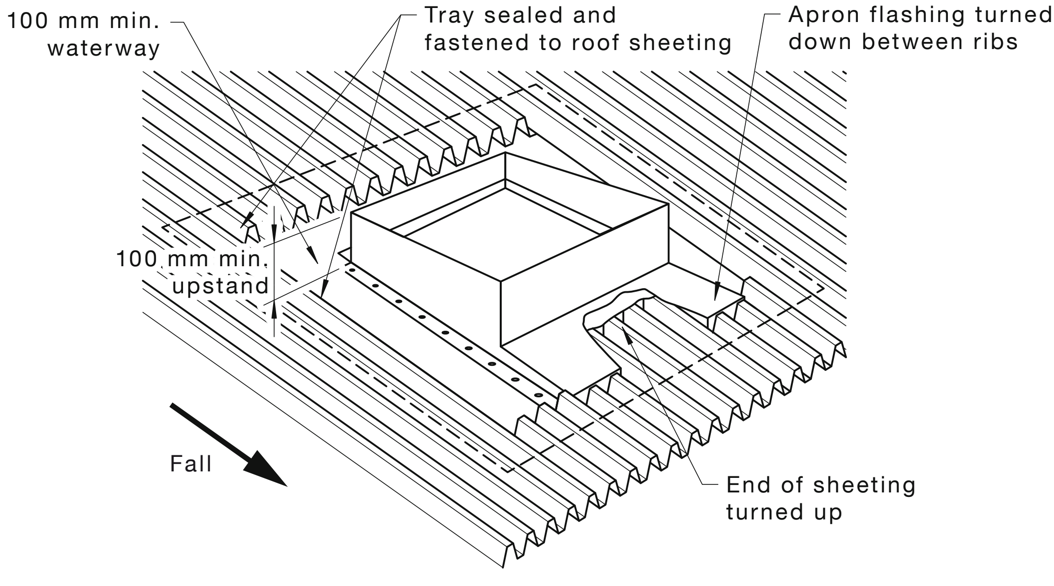
(f) *Stop-ended sheets* The high end of all roof covering sheets to be turned up to the full height of the ribs under transverse cappings to provide a watertight tray [see Figure 8.6.2(A)].

(g) *Cut-end sheets* Where longitudinal ribs have been removed on any sloping side, the pan is to be turned up to the height of the rib to form a new narrower tray.

(h) *Synthetic rubber* Synthetic rubber flashings to be installed to the manufacturer’s specification and are not to block the flow path of the water on the roof.




(a) Ribs stop-ended and sealed




(b) Using apron

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FIGURE 8.6.2(A) LARGE PENETRATIONS

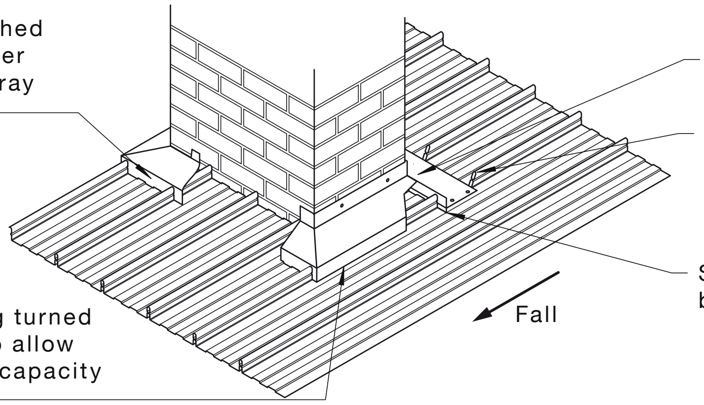


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FIGURE 8.6.2(B) STOP-ENDS ON TRANSVERSE CAPPING

## 8.7 ALL OTHER FLASHINGS AND CAPPINGS

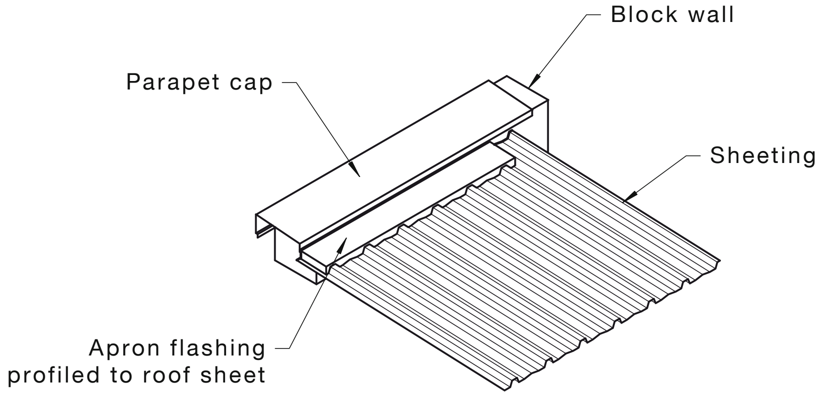
All other flashings and cappings to be fastened to the metal roof cover at intervals not exceeding 500 mm with self-drilling roof screws into the roof supports or rivets into the roof cover. All self-drilling self-tapping roof screws are to be fastened on crests of roof covers. For particular situations, the following is to be taken into consideration:

(a) *Thermal movement* Where thermal movement is likely to be a problem, cappings fixed to metal roof covers to be fastened with cleats or sliding supports to all other surfaces.

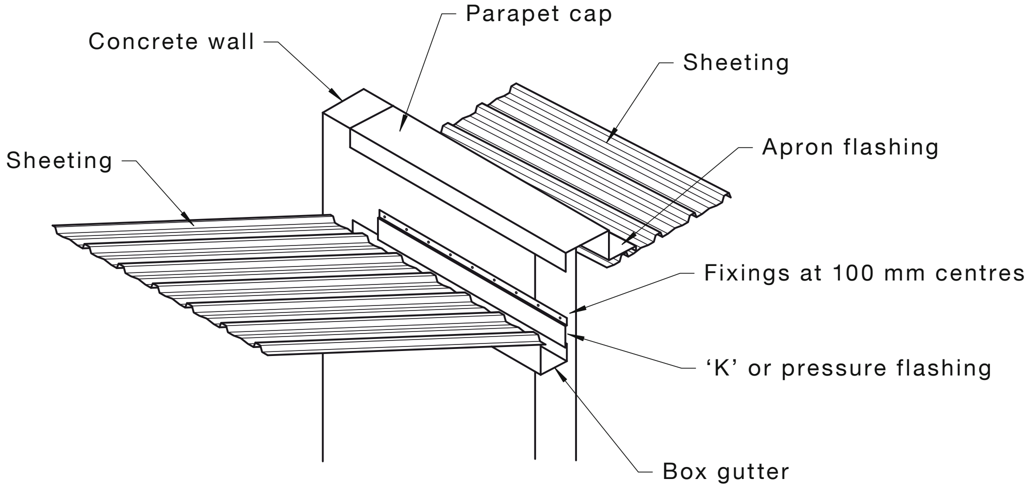
(b) *Parapet cappings* Parapet cappings to be fixed to parapet walls at intervals not exceeding 500 mm with masonry anchors and cleats that permit longitudinal expansion and contraction. A minimum fall of 3° to be provided across the width of the flashing, to divert water back onto the roof coverings so as to prevent the water from dripping down the fascia causing unsightly staining [see Figure 8.7(A)].

(c) *Barge capping* Barge capping to be installed as shown in Figure 8.7(B).

(d) *Synthetic rubber strip flashing* Synthetic rubber strip flashing installed as a transverse flashing or capping to be sealed with flexible sealant and fixed in a manner that will ensure a snug fit to the profile of the sheet. Where it is used as a sloping apron flashing it is to be fixed at 50 mm centres [see Figure 8.7(C)].

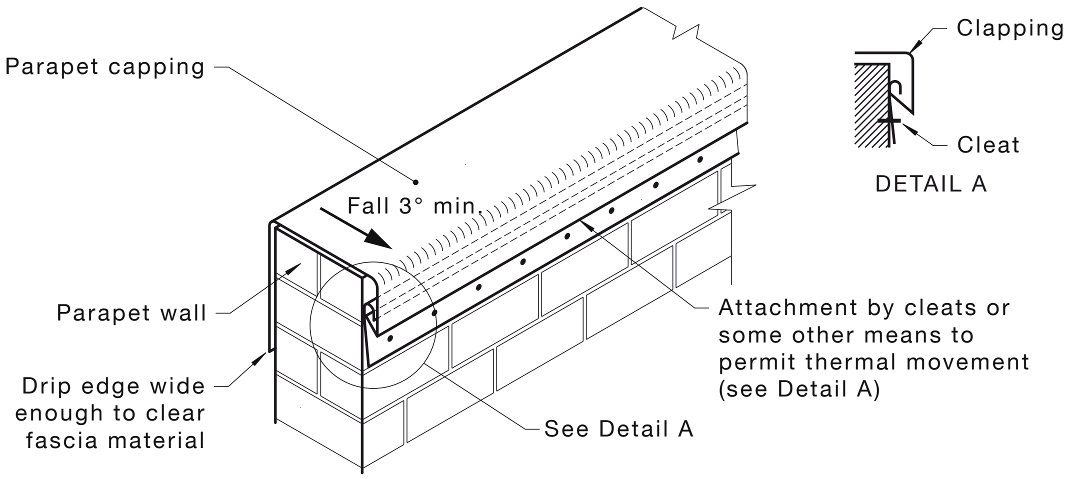



(a) Parapet caps




Fixings at 100 mm centres

(b) Concrete parapet wall




(c) Parapet capping

FIGURE 8.7(A) PARAPET CAPPING

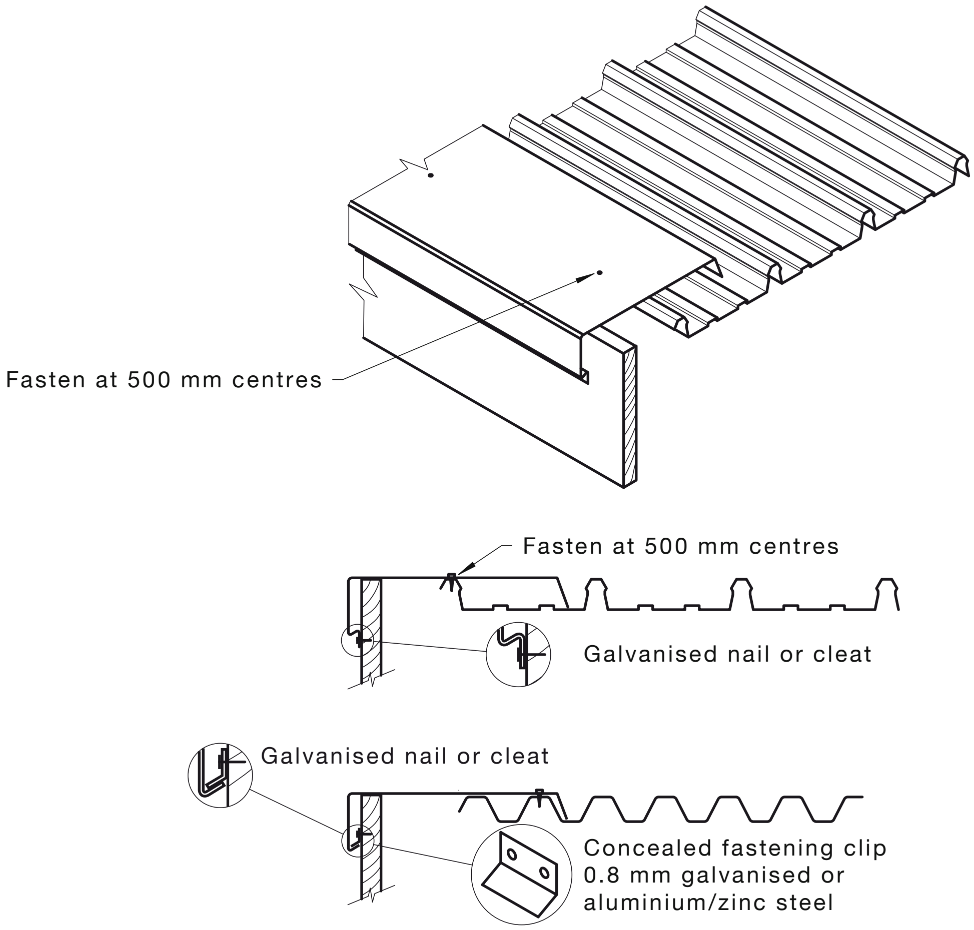



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FIGURE 8.7(B) BARGE CAPPING

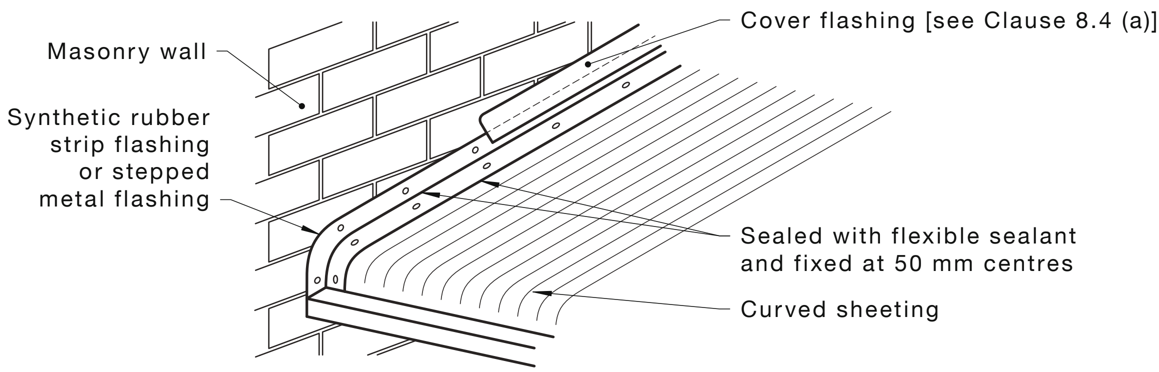


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FIGURE 8.7(C) SYNTHETIC RUBBER STRIP FLASHING

# SECTION 9 ROOF LIGHTS

## 9.1 TRANSLUCENT ROOF MATERIALS

Roof lights are generally manufactured from fibreglass reinforced polyester (FRP) and profiled polycarbonate as well as multi-walled polycarbonate to match the profiles of the roofing materials being installed and are to comply with AS/NZS 4256.3, AS 4256.2 and AS 4256.5. When selecting the roof light, attention is to be paid to the following:

(a) *Weights* Roof lights are available in weights ranging from 1800 g/$m^2$ to 4880 g/$m^2$ but 1800 g/$m^2$ material is not to be used on roof areas since it will not support the required dead load of 112 kg as required in AS/NZS 1170.1 over a purlin spacing of only 1200 mm.

(b) *Finishes* Roof lights are available in a range of types and finishes including fire-retardant sheeting and heat-reflecting sheeting to reduce solar heat load in buildings, as well as special formulations to provide chemical resistance.

(c) *Curving* Only the corrugated profile can be successfully curved over a barrel vault and the determining factor is the measurement of the opening and the rise above the horizontal [see Figure 9.1(A)].

Preferably, the fixing members are to be at no more than 900 mm spacings to promote a smooth arc when the sheeting is fastened. The sheet weight should be 2400 g/$m^2$ and the manufacturers consulted to determine if the required arc is achievable. Some arcs could be too tight for the product.

(d) *Spanning* It is important that the correct weight of sheeting be used on roofs to accommodate the purlin spacings. The sheets are to be of sufficient weight (and hence strength) to accommodate the following [see Figure 9.1(B)]:

(i) The purlin spacing.

(ii) The expansion and waving that can occur in the sheets as a result of the purlins and fasteners being too widely spaced.

(iii) The pressure and consequent surface crazing that can occur on the sheet as a result of safety mesh impinging on the underside of the sheet due to—

(A) purlins canting and forcing one edge upward into the sheeting; and

(B) purlins being fastened out of plane requiring the sheet to be cranked over the purlin.

(iv) The high rate of movement that occurs in a roof structure as a result of wind pressure and thermal expansion.

NOTE: Some translucent sheet manufacturers publish a recommended weight for span chart over most popular profiles.

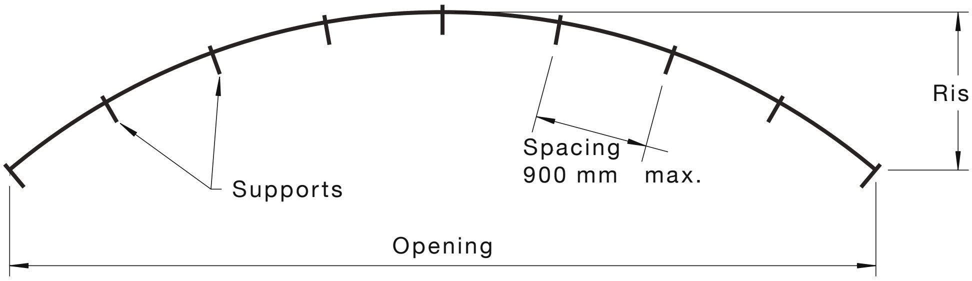
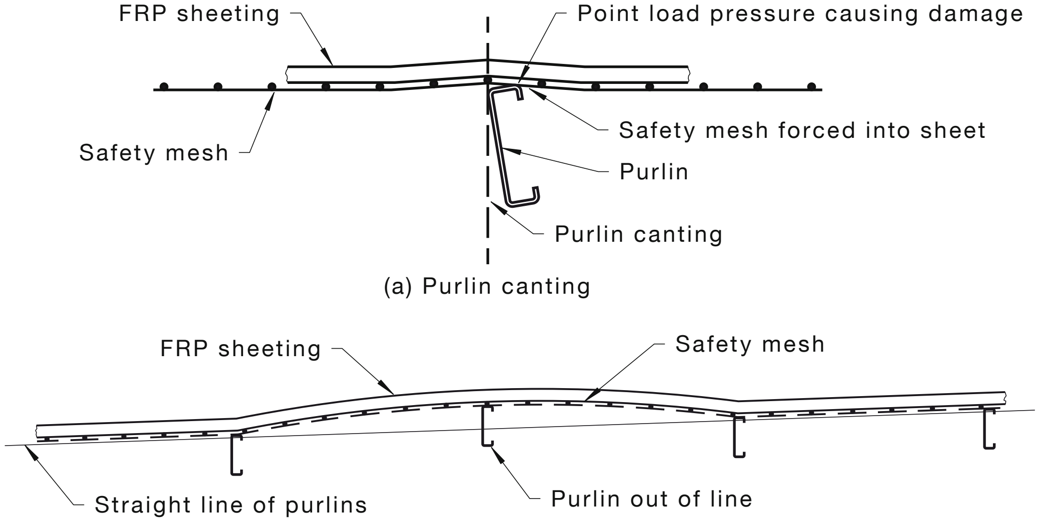



FIGURE 9.1(A) CURVING




NOTE: The out of line purlin pushes the FRP sheet out of line causing adjacent spans to deflect down, resulting in a wave effect along the length of sheeting.

(b) Purlin out of plane

FIGURE 9.1(B) SPANNING

## 9.2 INSTALLATION PROCEDURE

### 9.2.1 Inspection

Prior to fixing, the installer is to ensure that purlins and girts are in plane and the weight of sheeting to be used is in accordance with the manufacturer's specification or, in the absence of a specification, in accordance with load span charts relating to the actual purlin or girt spacings and wind load characteristics of the site and building.

### 9.2.2 Laying on roofs

When laying translucent sheets on roofs, particular attention is to be paid to the following:

(a) For the best internal light distribution, the roof lights is to be laid from ridge to gutter or second last purlin.

(b) For safety reasons, the bulk of the metal roofing is to be laid in position and the sheets pinned at the positions where the roof lights are required.

(c) Where required in AS 1562.3, safety mesh is to be fitted in roof construction under plastic roof cladding materials for Class 1 and Class 10, and Class 2 and Class 9 buildings, as defined in the NCC.

(d) A compressive foam strip of minimum 6 mm thickness or solid strip of minimum 1 mm thickness for the full width of the face of the supporting member and extended at least 25 mm past both edges of the plastic roof sheeting is to be provided between the safety mesh and the roof light at the purlins [see Figure 9.2.3.6(A)] to avoid rubbing of the sheet on the safety mesh.

(e) The roof light to be then laid in position, supported by the metal roofing at both side laps, and fastened.

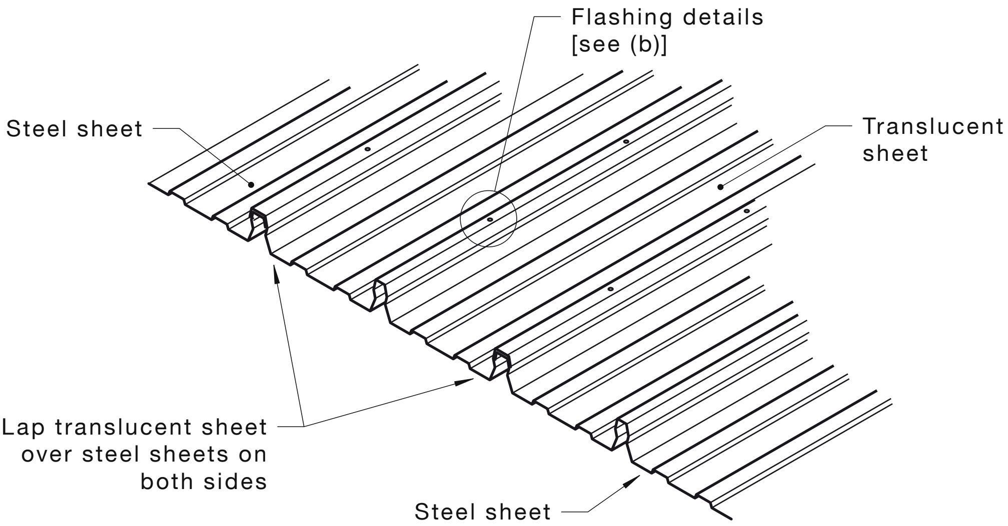
(f) Roof light profiles, including the corrugated profile, to be manufactured to fit over the metal roofing at the side laps, provided the cover widths of both the metal roof sheets and the roof lights are manufactured to their correct widths.

(g) Penetrations through plastic sheeting to be avoided.

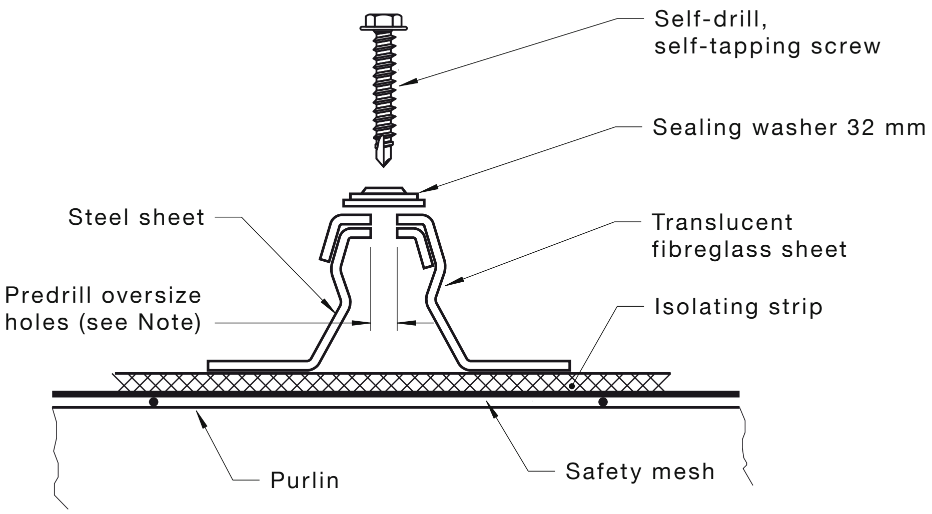
### 9.2.3 Fastening

#### 9.2.3.1 *Main fasteners*

All roof lights have to be pierce-fastened including those manufactured to match concealed-fastened profiles. The fasteners are to be located through the rib or crest of the sheet, generally in the same position as for the adjacent metal roofing fixing. Concealed-fastened profiles to be fastened at every rib at every purlin. The fasteners are to be 10 mm longer than those used for the metal roofing to accommodate the thickness of safety mesh and isolating strip and are to be complete with a 32 mm diameter sealing washer (see Figure 9.2.3.1).




(a) General layout




(b) Fastening detail

NOTE: Oversize hole size:

(a) Sheet length <8 m: Fastener diameter +3 mm.

(b) Sheet length >8 m: Fastener diameter +5 mm.

FIGURE 9.2.3.1 FASTENING

### 9.2.3.2 *Side lap fastening*

Side lap fastening may, on occasion, be necessary (manufacturer's specification to be referred to). It is necessary to ensure a tight weatherproof side lap, which is to be located at a maximum of 600 mm centres. Side lap fasteners are to be the same as those used on the adjacent metal roofing, and fixing holes in the overlapping sheet are to be pre-drilled to allow for thermal movement (see Figure 9.2.3.2).

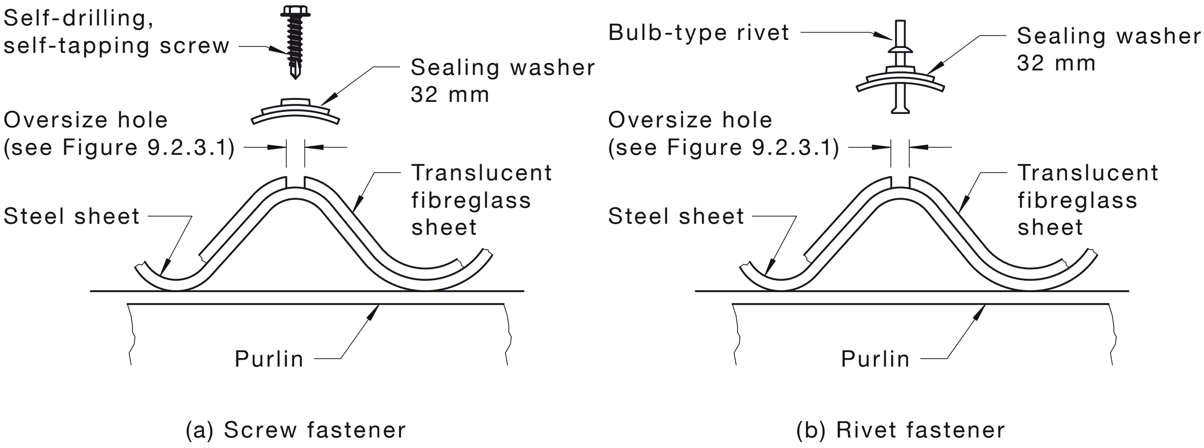



FIGURE 9.2.3.2 SIDE LAP FASTENING

**9.2.3.3** *Fastening holes*

Because fibreglass and polycarbonate roof lights have a greater rate of thermal expansion than steel, oversized fastener holes need to be pre-drilled to allow for thermal movement in the sheet (see Clause 9.2.3.4).

**9.2.3.4** *End lapping*

End lapping of sheets occurs on a purlin or girt. End laps for roofs are to be 300 mm.

**9.2.3.5** *End laps*

End laps are to be sealed with a silicone sealant or compressed foam strip after sealant. At all end laps between plastic and dissimilar material sheeting, two lines of compressive foam strip or suitable flexible sealant is to be placed across the full width of the lap approximately 150 mm apart and with one line 25 mm from the end of the plastic sheeting (see AS/NZS 1562.3).

NOTE: Translucent sheeting is to be only lapped over steel profiles. Metal overlapping translucent sheeting is to be discouraged. Also, it is impossible to insert multiple abutting translucent sheets under or over steel profile sheeting.

**9.2.3.6** *Profile closers*

Plastic roof sheeting is not easily reshaped after manufacture and will require special weather seals to be provided at all flashing and capping points. A foam closure strip matching the profile of the plastic sheeting is to be provided to seal the corrugations or pans of the profile under each flashing [see Figure 9.2.3.6(B)]. The foam closure is to be fitted to the pans or valleys of the plastic sheeting with a flexible waterproof sealant and to be continuous over the entire width of the plastic sheeting. Ideally, it should be fitted at least 100 mm behind the turn-down of the flashing.

NOTE: Some sealants are not compatible with certain types of plastic materials and can lead to stress cracking if incorrectly used. The manufacturer's instructions are to be adhered to.

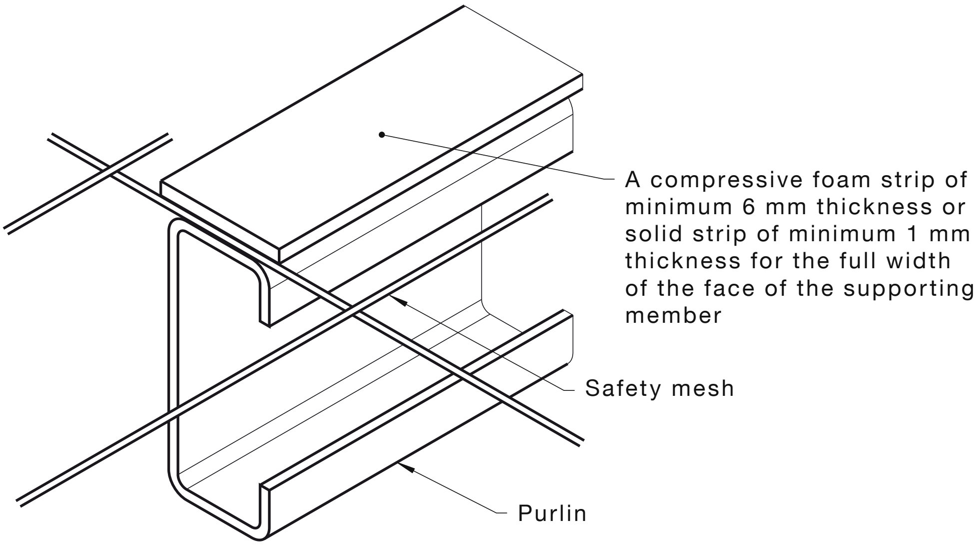
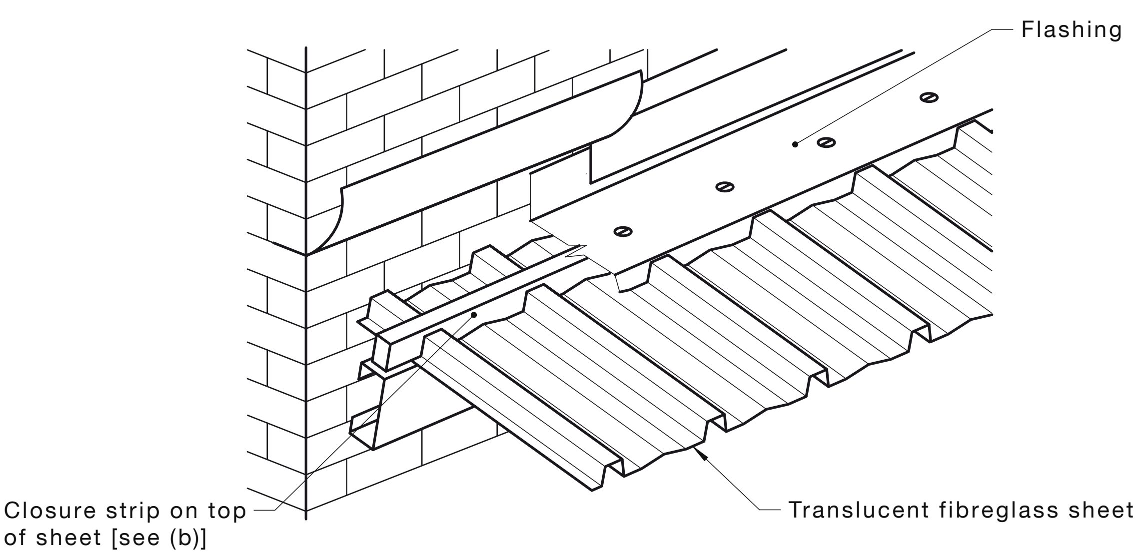
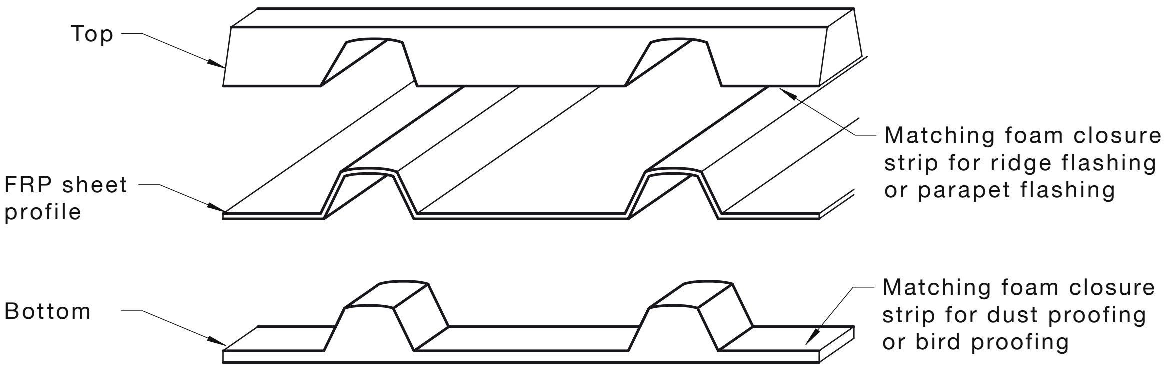


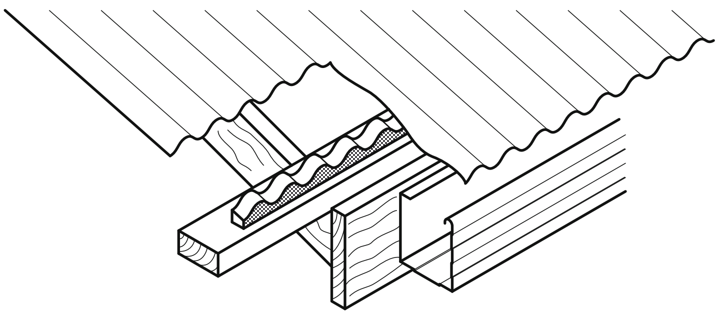

FIGURE 9.2.3.6(A) SHEET PROTECTION




(a) Parapet flashing




(b) Closure strip



(c) Eaves closure

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FIGURE 9.2.3.6(B) CORRUGATION CLOSERS

**9.2.3.7** *Fasteners*

The main fasteners to be 10 mm longer than those used for the metal roofing. The recommended screw lengths for the most commonly used profiles are shown in Table 9.2.3.7 (see Clause 9.2.3.1).

**TABLE 9.2.3.7**

**FASTENERS FOR TRANSLUCENT SHEETING**

| Profile | Fastener |
|---|---|
| Trapezoidal | 12 × 65 mm SD fasteners |
| Corrugated | 12 × 45 mm SD fasteners |
| Close pitch trapezoidal | 12 × 65 mm SD fasteners |
| Wide rib concealed-fastened decks | 12 × 65 mm SD fasteners |
| Side lap fasteners | 10 × 20 mm stitching screws<br>*or*<br>10 × 25 mm Type 17 |

LEGEND:
SD = self-drilling self-tapping screws

## 9.2.4 General handling

**9.2.4.1** *On-site storage*

Particular attention is to be paid to the following:

(a) Fibreglass or polycarbonate sheeting is normally delivered to the site in strapped bundles if intended for direct lifting on to a roof.

(b) Bundles of sheets being stored on site should be stacked clear of the ground and protected from rain with waterproof covers.

(c) Moisture can penetrate between the surface of stacked sheets by capillary action or wind and can cause deterioration of the surface coating, resulting in poor appearance or a shortened life expectancy.

**9.2.4.2** *Lifting to roof*

A spreader bar with a fabric sling is to be used when craning long lengths from truck to roof. Otherwise, sheets may be unloaded and passed up to the roof singly. Gloves are to be worn to protect the hands.

**9.2.4.3** *Roof traffic*

Roof lights are never to be walked on.

# SECTION 10 CEILING COMPONENTS AND ROOF EXTRAS

## 10.1 SCOPE OF SECTION

This Section provides information on ceiling components and roof extras in a building structure in both internal and external applications.

## 10.2 LOADBEARING AND NON-LOADBEARING SUPPORT SYSTEMS

Any cutting into the metal, timber or composite material of a wall frame system will require a design check and approval because of possible impairment of structural capacity of the frame and appearance of the lining.

Ceilings in domestic, commercial or industrial buildings may be direct-fixed to the roof or upper floor support system using truss chords, battens, beams and the like.

## 10.3 EXTERNAL APPLICATIONS

Ceiling installations in external applications (including awnings, eaves and parapets) are to be installed so that there is no water penetration from rain or wind-driven rain and the structure will withstand wind speeds that have been determined from AS/NZS 1170.2 or AS 4055 for the location and comply with fire rating where applicable. All access doors, penetrations, vents and the like are to be installed so that the ceiling system is reinstated structurally and aesthetically and complies with the appropriate fire ratings. Components in external applications include the following:

(a) *Trims, mouldings and cappings* These products in external wall or ceiling installations typically function to finish the installation, provide a means to prevent water and vermin penetration and, in some cases, provide a measure of fire retardation where fire ratings apply.

(b) *Access doors and channels* Access doors and channels supporting access panels are often required to permit access to below roof or ceiling areas.

(c) *Smoke vents* Smoke vents are often required to permit designed exhausting of smoke from under the roof area.

(d) *Explosion vents* Explosion vents are often required to permit designed pressure relief from building areas, including below roof, to prevent or substantially reduce structural damage in the event of an explosion within the building.

(e) *Smoke and fire curtains* Smoke and fire curtains are often required to permit designed prevention of the spread of fire or smoke between areas within a building, including below the roof area, in the event of fire and may form part of the fire rating for the building.

# APPENDIX A

# ATMOSPHERIC ENVIRONMENTS

## A1 GENERAL

This Appendix classifies atmospheric zones in Australia, based on their effect on the corrosion of steel and the life of a paint system. When selecting an appropriate protective coating system the overall atmospheric conditions in the location of the intended structure require consideration. A structure situated in an aggressive environment will require a much higher standard of corrosion protection than one in a benign environment. The environment can affect both the steel and the paint system. Of prime importance is the effect the environment has on the corrosion of steel. This Appendix describes the major factors affecting atmospheric corrosion and gives examples of corrosion rates of steel within seven atmospheric classifications, for various locations.

The effect of the environment on the life of the paint system is also highly important. The corrosive environments described do not necessarily affect bare steel in the same way as they affect coatings.

Environments that would not be considered particularly corrosive to steel, such as hot dry climates with a high amount of ultraviolet (UV) radiation, can cause early breakdown of some coatings. Tropical environments, with high humidity and rainfall, which promote mould and fungal growth, are far more aggressive to organic coatings than the corrosion rate would suggest. Furthermore, the colour of the paint may influence its performance in some environments.

In addition to climate effects, the local environmental effects (or microclimate) produced by the erection of a structure of installation of equipment need to be taken into account. Such on-site factors require additional consideration because a mildly corrosive atmosphere can be converted into an aggressive environment by microclimatic effects. A significant acceleration of corrosion rate can occur in the following circumstances:

(a) At locations where the metal surface remains damp for an extended period, such as where surfaces are not freely drained or are shaded from sunlight.

(b) On unwashed surfaces, i.e. surfaces exposed to atmospheric contaminants, notably coastal salts, but protected from cleansing rain.

Other microclimatic effects that may accelerate the corrosion of the substrate or the deterioration of its protective coating include acidic or alkaline fallout, industrial chemicals and solvents, airborne fertilisers and chemicals, prevailing winds that transport contamination, hot or cold surfaces and surfaces exposed to abrasion and impact. These effects can outweigh those of the macroclimatic zones described in Paragraph A2.

As a result of microclimatic effects it is very difficult, if not impossible, to predict accurately the aggressiveness of a given environment and a certain amount of educated judgement is required to assess its influence on the coating life.

The atmospheric classification outlined in this Appendix align with those in AS/NZS 2312, which classifies atmospheres in relation to corrosivity of bare steel. However, atmospheres as they apply to steel do not necessarily apply to prefinished metal products or to products that have aluminium as the base metal.

On-site factors require additional consideration because a mildly corrosive atmosphere can be converted into an aggressive environment by rain, dew or humid conditions and can thus affect the choice of a particular type of prefinished metal product.

Typical on-site factors or types of microclimates that may cause a breakdown of protective coatings include the following:

(i) Concentrations of industry.

(ii) Contamination from agricultural fertilisers or insecticides.

(iii) Damp locations not dried out by sunlight.

(iv) Exposure to sea breezes.

(v) Alkaline or acidic fallout.

(vi) Hot or cold surfaces.

(vii) Abrasion or impact.

(viii) Animal enclosures.

Prevailing winds, which can transport contamination from one location to another, also require consideration.

High levels of ultraviolet light and humidity have adverse effects on the performance of organic coatings and thus reduce their durability. Because of the effects of high humidity, the designer of a building should take special care to ensure that the effects of condensation are minimised.

It should also be borne in mind that stresses resulting from bending or roll-forming operations may cause cracking of paint films. In moderate exposure conditions, cracking does not adversely affect the expected life of a product; however, reduced service life can be expected in severe environments.

## A2 ATMOSPHERIC CLASSIFICATIONS (APPLICABLE TO STEEL)

A classification of the corrosivity of an atmosphere into one of five categories is given in ISO 9223. A number of surveys to determine the corrosion rate at various sites in Australia have made it possible to place each of these sites into one of the five ISO categories. However, the ISO categories are considered to be too broad to accurately indicate the performance of coatings and consequently it has been necessary to expand them into seven classifications to cover overall environmental aggressiveness. It is stressed that the micro environmental effects described in Paragraph El also require consideration. The seven atmospheric classifications make reference to one-year corrosion rate figures for mild steel. Figures are not given for copper and zinc which have much lower rates of corrosion than steel, or for aluminium which is highly resistant to atmospheric corrosion. The classifications are as follows:

(a) *Mild (ISO Category 1-2)* A mild environment will corrode mild steel at a rate of 1.3 g per year. The category includes all areas remote from the coast, industrial activity and the tropics. Sparsely settled regions, such as outback Australia, are typical examples, but it also includes rural communities other than those on the coast. Corrosion protection required for this category is minimal.

(b) *Moderate (ISO Category 2)* A moderate environment will cause a first year corrosion rate of mild steel of 10 g. The category includes areas with light industrial pollution or very light marine influence, or both. Typical areas are suburbs of cities on sheltered bays such as Melbourne, Adelaide, Hobart (except those areas near the coast) and most inland cities. Corrosion protection requirements are moderate and do not call for special measures.

(c) *Tropical (ISO Category 2)* A tropical environment includes coastal areas of north Queensland, Northern Territory, north-west Western Australia, Papua New Guinea and the Pacific Islands, except where directly affected by salt spray. This is the only category that cannot be delineated by corrosion rate and measurements would put these areas into ISO Category 2. Although corrosivity is generally low in tropical regions, the aggressiveness of the environment to organic coatings means special protection is required.

(d) *Industrial (ISO Category 3-4)* An industrial environment will cause a first year corrosion rate of mild steel to be greater than 25 g and is usually within ISO Category 3, but may extend into ISO Category 4 (greater than 50 g per year). The only areas within this category are around major industrial complexes inland from the sea. Examples occur around smelters in Port Pirie and Newcastle. There are only a few sites within this category in Australia. The pollution in these areas requires that coating systems be resistant to mild acid.

(e) *Marine (ISO Category 3)* A marine environment will cause a first year corrosion rate of mild steel of 25 g to 50 g. The category includes areas influenced to a moderate extent by coastal salts. The extent of the area varies considerably, depending on factors such as winds, topography and vegetation. For sheltered areas, such as occur around Port Phillip Bay, it extends from the coastline to about 100 m from the beach, but for most ocean-front areas, such as occur along the south-western corner of Western Australia, the south-eastern coast of South Australia, and the New South Wales and Queensland coasts, it generally extends from about 200 m from the coast to about 5 km inland, depending on the conditions. Large areas of Perth, Wollongong, Sydney and Newcastle are in this Zone. A high-performance coating system is required for long-term protection.

(f) *Severe marine (ISO Category 4)* A severe marine environment has high corrosivity and will cause a one year corrosion rate of steel to be in the range of 50 g to 80 g. In Australia, such regions are found in areas of continual breaking surf and near the coast. The extent to which such conditions extend inland depends on prevailing wind and geography, but is generally between 100 m from the beach front to about 300 m inland along the ocean coast. In high wind areas, this region may extend further inland.

(g) *Very severe (ISO Category 5)* This classification applies to marine environments and industrial and geothermal environments as follows:

(i) *Marine* Very severe marine environments have high corrosivity and will cause a one year corrosion rate of steel to be in excess of 80 g. In Australia such regions are found offshore and up to 100 m from the water line of areas of breaking surf. An appropriate prepainted coating system or a specialised high performance coating systems is required in this region.

(ii) *Industrial and geothermal* Areas of special corrosivity with high to very high corrosion rates occur underground, underwater, in splash zones and in chemical plants. For these areas, specific protection for the aggressive conditions is essential.

# APPENDIX B

# INSULATED PANELS

## B1 INTRODUCTION

### B1.1 Sandwich panels

An insulated panel is any laminated panel that is manufactured from different materials, permanently bonded together so that they act as a single structural element.

For the purposes of this Appendix, the description '*insulated panel*' refers to building cladding panels in most cases having metal facings to both surfaces, with an insulation core that fills the space between the two facings and is permanently bonded to both. In some instances, insulated panels will have one face of steel and the other face vinyl, foil, paper, plywood, aluminium and fibre cement.

These products are sometimes also referred to as '*sandwich panels*' or '*composite panels*'.

Site-assembled systems using profiled insulation, loose fill insulation or insulation batts are not insulated panels by this definition. These products are a '*built-up-metal roofing*' system and are not covered by this Appendix.

## B2 MATERIALS

### B2.1 The insulation core

The core material provides the thermal insulation, and contributes to the panel strength by providing most of the resistance to shear forces. There are various core materials, most commonly:

(a) Expanded polystyrene.

(b) Phenolic modified expanded polystyrene.

(c) Polyurethane.

(d) Polyisocyanurate.

(e) Mineral fibre.

Core thicknesses range from 20 mm to 300 mm with R values ranging from 1.4 to over 9 $m^2$.K/W for the range of insulated cores mentioned.

## B3 PANEL PERFORMANCE

### B3.1 Structural

Insulated panels are very rigid units in which the two facings act together with the insulation in resisting wind loads and foot traffic. Their combined strength and stiffness is far greater than the sum of the built-up parts of a traditional roof system. This is analogous to a lattice girder in which light angle sections are combined to create a stiff section that can bridge larger spans. Therefore, insulated panels have much larger spanning capability than traditional built up systems.

The strength and stiffness of panels is determined by both the core material and the thickness of the product. Profiling of one or both faces can further increase the strength.

All panel manufacturers publish load/span data, which should be consulted.

When fully fixed, insulated panels can support normal foot traffic without damage. The foam core provides continuous support to the external facing to resist indentation.

## B3.2 Thermal

### B3.2.1 *Thermal properties*

Insulated panels are extremely effective in thermal insulation as they can lock together and form an airtight seal. In a profiled roof panel, the insulation thickness varies depending on the regulations, codes of practice and guidance in each State and Territory, which stipulate the minimum 'R' value required for a specific building in a specific location. The specific R value for insulated panels will vary, based on core type and panel thickness. The panel manufacturers should be consulted regarding their specific products' thermal performance.

There is minimal thermal bridging when using insulated panels, whereas cold bridging only slightly occurs through the main fixings of the panel system and into the secondary steelwork below. It has been shown that this is unlikely to decrease the total 'R' value by any more than 1%–2%.

## B3.3 Fire

### B3.3.1 *Fire rating*

Fire regulations in buildings are generally directed at reducing the risk of death or injury to occupants and the public. This is achieved by the careful selection of materials with which the building envelope is constructed.

### B3.3.2 *Fire resistance*

Fire-resisting construction, conforming to the NCC, is possible with steel-skinned insulated panels provided that appropriate core materials and panel detailing are used.

### B3.3.3 *Fire resistance construction*

With regards to specific fire-resisting construction requirements, the panel manufacturer should be consulted as each product may have to be installed differently to a normal (non-fire resisting) installation. For example, the panel spans may be reduced, and internal joints may require stitching from the inside.

## B3.4 Supporting structure

### B3.4.1 *Support*

The panels are normally supported on purlins, battens, girts and frames, which are to be of adequate strength and stiffness. The structural engineer designs these supports. The greater inherent strength of insulated panels means that they require less supporting steel structure than other forms of roofing. Manufacturers' guidance should be sought on the specific span capabilities of their panels during the design process, due to the potential of the panels to reduce the amount of steel used to support the structure.

### B3.4.2 *Tolerance*

Purlins/girts are to be erected to close tolerances. Because the panels have high inherent stiffness, they cannot easily be deformed to follow uneven structures. The normal guidance for tolerance in the setting out of purlins/girts is that they should not be out of plane by more than one six hundredth of their spacing (e.g. ±3 mm for a 1.8 m purlin spacing). This tolerance applies to the actual fixing of the panels.

### B3.4.3 *Alignment*

The alignment of purlins/girts becomes critically important when panels are to be end-lapped. If the purlins are displaced, or if they wander in their length, the lap may not coincide with the support.

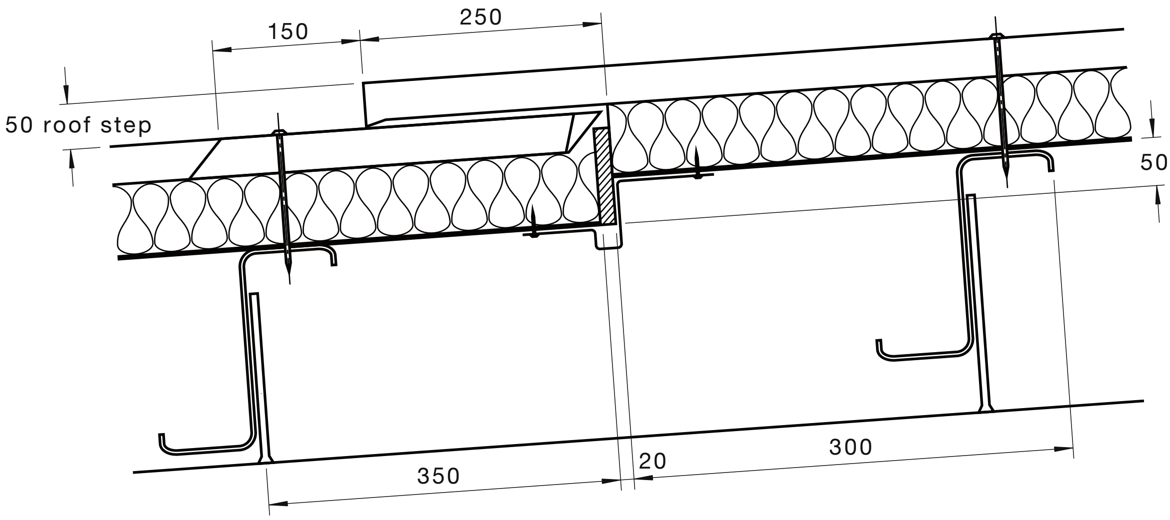
 

**B3.4.4** *Mid-roof jointing/lapping*

Long runs of roof panel often require mid-roof jointing or lapping. There are two distinct methods for this procedure as follows:

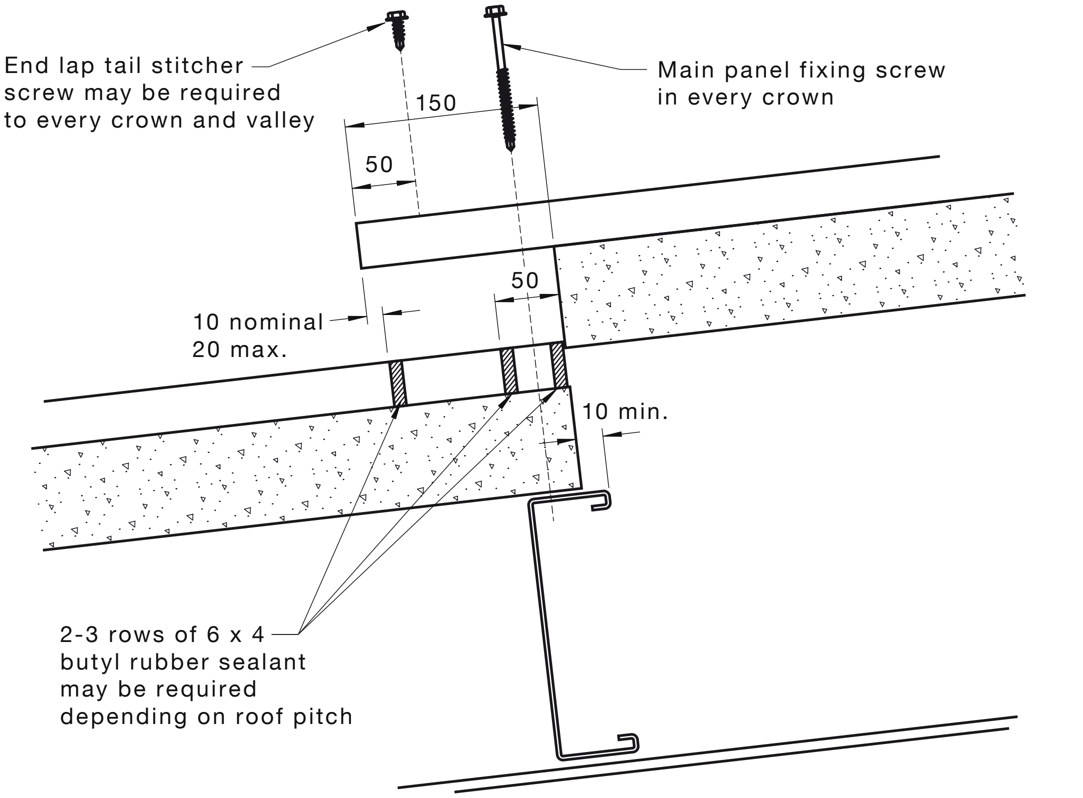
(a) *System A—Expansion step* This method allows true thermal movement and is the preferred method for Australian conditions. Upper and lower roofs are connected to purlins independently, therefore allowing for expansion and contraction. The expansion step joint is flashed utilising baffle flashing techniques proven in Australian conditions for over 25 years. This type of sheeting connection requires purlin/purlin cleats to be stepped. Expansion steps are the preferred method to connect long roof runs, offering good corrosion resistance and better control of thermal movement. This system requires the lower level panels to be installed first after core ‘cut back’ and sheet ‘turn up’ is completed. The baffle flashing is then notched over the lower level panels, ready for the upper level sheets to be installed.

(b) *System B—End lap joint* This method connects the upper and lower level roof runs in the same plane. This system requires the core material to be removed completely so sealant can form an effective, waterproof bond to the connecting steel skins. Unlike System A, this end-lapping process physically connects the upper and lower level panels. As both sides of the end lap require support, there is to be room for the fixing at one side; therefore, it is essential that the purlins be at least 60 mm wide and perfectly in line at each end lap position on the roof structure. When narrower purlins are used, it is possible to attach a structural angle or plate to extend the width at this position. In cases where the steelwork is not in line or parallel (more seen on roof slopes over 30 m long), it is sometimes preferable to use a double purlin at the end lap positions. The double purlin should be installed back to back.




DIMENSIONS IN MILLIMETRES

FIGURE B1 TYPICAL EXPANSION STEP DETAIL




DIMENSIONS IN MILLIMETRES

FIGURE B2 TYPICAL END LAP DETAIL

**B3.4.5** *Bracing*

Due to their inherent strength and mode of construction, insulated panels have bracing and diaphragm action capabilities that may be sourced from manufacturers; however, during installation, temporary bracing may be required to hold purlins/girts in the correct locations.

**B3.4.6** *Rigidity*

Most insulated panels have rigidity, which is not shared by single skin profiled sheets. They are able to accommodate relatively large openings without the need for additional structural support. As a guide only, holes up to 300 mm square or diameter suitably flashed off/ encapsulated should not need structural support or trimmers; however, advice on these scenarios should be sourced from the panel manufacturer.

**B3.4.7** *Span*

When considering span performance, other issues such as penetrations, trafficability, deflection limits in accordance with Australian Standards, distributed and concentrated imposed loads both short and long term, temperature differentials and thermal effects, number and type of fasteners, loadbearing and non-loadbearing configurations, etc., consult with the panel manufacturer during the building design phase. Most manufacturers have data based on testing and experience under Australian conditions.

# B4 ROOF APPLICATIONS

## B4.1 General

### B4.1.1 *Length*

The maximum length of insulated panels for roofing depends on the core material, profile and manufacturer. A range between 13–21 m is likely but should be confirmed by the manufacturer. Insulated panels due to their manufacturing nature are unable to be roll-formed on site.

### B4.1.2 *Pitch*

The minimum roof pitch for insulated panels varies depending on the manufacturer and profile; however, it is in the region of 2°–5°.

### B4.1.3 *End lap jointing*

At the end lap position, the internal lining and insulation is butt jointed over the purlin, and the overlap is formed in the external weather skin only. Manufacturers provide specific recommendations [usually as per Clause 7.15(b)]; the minimum recommended end lap for roof sheeting is 200 mm for roof slopes 1 in 12 (5°) and 1 in 4 (15°) and 150 mm for slopes above 1 in 4 with two or three runs of sealant, depending on roof pitch. Advice should be sourced from the panel manufacturer as to a recommended sealant.

### B4.1.4 *Side lapping*

At the side lap position, a single strip of sealant on the rib/crest of the panel may be required. Manufacturers provide specific recommendations (usually a 6 × 4 mm bead of sealant). Advice should be sourced from the panel manufacturer as to a recommended sealant. Side laps are usually stitched at 450–600 mm centres .

### B4.1.5 *Fixing*

Care should be taken when fixing insulated panels, to ensure that they are not over-tightened, which can cause a shallow dent around the fixing head. Similar issues can result from over-fixing. Fixing depth locators are generally recommended by the panel and fixing manufacturer to allow for the correct compression of the washer.

### B4.1.6 *Surface protection*

In high risk applications such as swimming pools, food processing and chill stores, refer to the IPCA *Code of Practice* (for installing insulated panels) from the Insulated Panel Council of Australasia (IPCA). Additional seals are usually required at the internal side lap lining and on the top of the purlins prior to panel installation. In such internal environments, a special internal coating may be required. Manufacturers' guidance should be sought on any special requirements for these applications.

### B4.1.7 *Laying sequence*

The laying sequence of insulated panels is sometimes different to traditional 'built-up roofing systems' and the panel manufacturer should be consulted prior to installation. Incorrect laying sequence may have an impact on the manufacturer's warranty offer on the product as well as possible waterproofing concerns.

# B5 WALL CLADDING

## B5.1 General

### B5.1.1 *Cladding*

Most insulated panels that are used for roofing may also be used for wall cladding; however, there are purpose-made products that are produced specifically for use on walls and can be installed vertical or horizontal. The maximum length of insulated panels for walling is generally around 7–16 m; however, sheets can be run up to 21 m (check with manufacturers for details).

NOTE: Size can be limited due to handling and transport considerations.

## B5.2 Finishes for wall cladding

### B5.2.1 *Coatings*

Wall panels may have the same paint coating systems as roof panels. In some applications, coatings specific to walling applications may be used.

### B5.2.2 *Coating protection*

Wall panel coatings are relatively thin, and can be easily damaged by careless handling and installation. The appearance of a large area of high gloss panels can be badly marred by scratches and scruffs (e.g. from moving ladders against finished areas). For this reason, manufacturers usually provide a protective strippable film to the external face of the panel. This film is stripped off, once the panels have been fixed into position and are no longer at risk. The cost of this film is very small when set against the cost of replacing panels. In general these strippable films should not be left on the panel as they become difficult to remove when exposed to UV over a period of time. The panel manufacturer usually states the 'maximum period of time' for the film to be left on the panel when exposed outdoors.

## B5.3 Purpose-made wall panels

### B5.3.1 *Strength*

Many designers, as well as building owners, prefer smaller profiled or flat wall panels to the high rib products usually associated with roof panels. Insulated panels are ideally suited to the manufacture of such products as they develop their strength from the sandwich of skins and insulation, and do not need high profiling to contribute to strength or stiffness.

### B5.3.2 *Length*

Continuously produced panels are usually made with a tongue and groove jointing system, which may incorporate concealed fixings. This is an aesthetically pleasing arrangement. The maximum length of insulated panels for walling is generally around 7–16 m; however, sheets can be run up to 21 m (check with manufacturers for details).

NOTE: Size can be limited due to handling and transport considerations.

### B5.3.3 *Profiles*

Each panel manufacturer has different profiles for the outer and inner skins. Manufacturers should be contacted for their range of profiles.

### B5.3.4 *Ancillaries*

Such products generally come with a whole range of purpose-made ancillaries including cranked corners, base channels and top hat sections, etc. Manufacturers provide technical literature giving guidance on the design and fixing of their systems. Users should always take account of these published data.

## B6 SITE WORK AND HANDLING

### B6.1 Site storage

Insulated panels merit special care during storage, prior to fixing. Since panels are produced to order, any accidental damage to either skin can lead to the whole panel needing to be replaced, which can be expensive and may incur extended delays.

### B6.2 Site handling

**B6.2.1** *Handling*

Packs may be lifted by forklift or crane (with spreader beams or wide forks depending on panel length). For roofing products, the packs may be either lifted onto a level surface at ground level, which reduces the risk of working at height, or be hoisted onto the roof structure, where they are less vulnerable to damage from other site activities. The project structural engineer should be consulted as to safe placement positions, based on the maximum pack weight if loading directly onto the roof structure. The panel manufacturer should be consulted for guidance.

**B6.2.2** *Lifting*

Individual panels may be moved by hand, but should never be lifted by the end lap end or carried at the side lap end, as this can cause delamination of the outer sheet.

**B6.2.3** *Surface protection*

Panels should be lifted cleanly off the stack. Sliding of the panel can damage the external or internal linings.

**B6.2.4** *Mechanical lifting*

Becoming increasingly popular with insulated panel installations is the use of purpose-made mechanical handling systems. These systems virtually eliminate the need for any manhandling, and make the speed of build time considerably quicker and easier over manhandling of product. Mechanical handling systems are normally attached to a crane which is normally required for the duration of the panel installation. The panel manufacturer should be consulted for approved lifting devices.

### B6.3 Health and safety

**B6.3.1** *Safety mesh*

Mesh is not normally required with insulated panel roof systems; however, other normal safety procedures and recommendations for building and roofing applications should be followed as per each State or Territory requirements.

**B6.3.2** *Personal protection*

All operators should wear gloves to protect their hands when handling metal sheets. Material Safety Data Sheets (MSDS) documentation should be sourced from the manufacturer and any guidelines strictly followed.

**B6.3.3** *Installation safety*

Insulated panel roofs contribute to site safety because, once fixed, they provide a safe working platform. They are trafficable for construction and maintenance activities (refer to manufacturer's guidelines/signage) at all practical spans; however, foot traffic on unfixed panels should be prohibited.

**B6.3.4** *Roof lights*

Roof lights are much lighter than insulated panels and from a 'trafficable point' should be treated in the same way as a built-up construction.

**B6.3.5** *Lap support*

Special care is required at end laps to ensure that both panels have sufficient width of bearing on the purlin flange.

**B6.3.6** *Roof safety considerations*

The installation of roof and wall panels on any building is to be planned carefully to ensure the work can proceed in safety.

The roofing and cladding contractor is to carry out the necessary 'site-specific risk assessments' and prepare all necessary 'method statements' to suit each particular project. Each Safe Work Method Statement is to take into account the detailed fixing recommendations contained in the installation guide.

The detailed Safe Work Method should include the roles and responsibilities of all persons involved in the project. This should indicate, but not necessarily be limited to, the details of the Site Supervisor, Safety Officer and what safety equipment will be used for each stage of the work. The sequence of the works should also be detailed to ensure compliance with all local regulatory authorities, OHS Acts and Regulations.

Compliance with OHS Acts and Regulations will require work to proceed in accordance with all relevant 'codes of practice' and any Australian Standards that have been adopted by the regulatory authorities as 'best industry practice'.

## B6.4 Fixing

**B6.4.1** *General*

The standard criteria for fixing metal cladding apply to insulated panels, although some additional criteria are given in Paragraphs B6.4.2 to B6.4.7.

**B6.4.2** *Set out*

Setting out of eaves and end laps are to be very accurate, as discrepancies cannot be accommodated by varying laps.

**B6.4.3** *Over-tightening*

Only fixings approved by the panel manufacturer are to be used with insulated panels, and they are not to be over-tightened or they will dimple the skin of the panels, which could result in weatherproofing problems. Correctly adjusted screw guns and fixing adapters should be used.

**B6.4.4** *Lap stitching*

Tail stitching may be required to the ends of an end lap (refer to panel manufacturer for current information on fixing requirements).

**B6.4.5** *Humidity*

In buildings where high humidity is anticipated, some manufacturers have internal liner side laps and both sides of the purlins bedded in a butyl mastic tape. These seals are to be continuous to be effective.

**B6.4.6** *Protective film*

Where a protective film is used by the manufacturer on delivery, it is to be removed as early as possible to prevent it from strong adherence, due to UV and high temperatures. The manufacturer's advice is to be followed.

**B6.4.7** *Rib/crest fixing*

Rib/crest fixings require extra care to ensure that they do not miss the purlin and are perpendicular to the surface to ensure the correct sealing of the washer.

## B6.5 Repairs to damaged panels

### B6.5.1 *Surface damage*

Badly damaged panels should be replaced and the manufacturer consulted for the best way to remove and replace the panels. Minor damage can often be repaired in situ.

### B6.5.2 *Sheet damage*

A dent, split or puncture in the external skin should not be left untreated as it will accelerate corrosion and, in some insulated products, absorb water into the insulation core. On a roof, it is sometimes possible to repair by over-skinning with a matching profiled skin. All edges are to be sealed to prevent moisture getting between the two skins. On a wall, it is not usually possible to over-skin and in this instance replacement of panel is generally the preferred option. The panel manufacturer should be consulted as to the preferred option.

### B6.5.3 *Surface scratches*

Rectification of scratches to the external metal skin is dependent on the length and depth of each scratch. In instances where the scratch has not exposed the bare metal, it is recommended to leave as is. In instances where the scratch has exposed base metal, remedial work will be required. The paint coating manufacturer's strict guidelines should be used in this instance.

## B6.6 Inspection and maintenance

The panel manufacturer should be consulted on specific inspection and maintenance procedures. These are important and, in most cases, are to be adhered to for warranty purposes.

NOTE: Insulated sandwich panel foam cores can be subject to termite and ant infestation and are often manufactured with an additive to resist infestation or, where installed in susceptible areas, treated with a suitable proprietary treatment such as disodium octaborate tetrahydrate or equivalent to complement total insect management systems.

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